

WDW-4

Classical Matter Occupancy and Joint-Constraint Closure under AASC

Scalar-clock regularity and torsion-free Standard Model carrier occupancy

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Exact scope. The integrated theorem occupies the regular Einstein–Klein–Gordon scalar-clock branch and a distinct torsion-free tetrad/spin/gauge Standard Model classical carrier on their declared common-domain interfaces. It proves the stated constraint and regularity closures without promoting determinant-line, bordism, regulated quantum-constraint, physical-inner-product, or global-clock claims.

Abstract

WDW-1–WDW-3 construct the covariant-to-canonical source bridge, proper spatial reduction, and the full off-shell hypersurface-deformation architecture, but leave two classical matter burdens deliberately open: occupation of the regular Einstein–Klein–Gordon scalar branch, and extension to a torsion-free tetrad/spin/gauge Standard Model carrier on one joint constrained domain. This integrated WDW-4 publication edition closes both burdens while preserving their distinct theorem objects and their one-way dependency.

Module 4A constructs an exact nonvacuum Einstein–Klein–Gordon datum on a closed hyperbolic three-manifold. The scalar momentum balances the Hamiltonian constraint, eliminates the lapse component of every generalized Killing initial datum, and supplies a timelike-gradient local clock. The scalar first-order column is retained in the coupled principal symbol. A fixed balanced normal operator yields a compatible loss-two smooth-tame right inverse, populating all clauses of ScalarReg3 and thereby promoting the WDW-3 conditional characteristic and finite refoliation results on a nonempty local chart.

Module 4B takes the accepted 4A chart as an analytic interface, not as a surrogate Standard Model proof. It constructs the torsion-free Levi–Civita tetrad/spin lift; freezes the three-generation minimal chiral Standard Model on the declared $(SU(3)_c \times SU(2)_L \times U(1)_Y)/\mathbb{Z}_6$, trivial-bundle branch; derives the complete graded Dirac–Bergmann carrier and all local Lorentz, internal Gauss, spatial, and Hamiltonian constraints; and proves their exact common-domain algebra. A small nonzero gauge–Higgs configuration with trivial joint internal stabilizer is corrected through the 4A tame split to an exact total-constraint zero. The full linearization retains the nonzero gauge/Higgs momentum column and admits an exact upper-triangular compatible tame right inverse. All twelve SMReg4B clauses are occupied, while the local scalar clock persists by openness.

The integration theorem records the precise interface: an occupied 4A datum and its tame chart are lifted by an even parametric correction to an occupied 4B datum; the corrected gravity/scalar projection is not misidentified as a separate pure 4A constraint zero. The resulting classical handoff retains the two constraint maps, the lifting relation, the joint quotient chart, local clock and failure data, target and topology guards, and the complete anomaly-status ledger. Perturbative Standard Model anomaly cancellation, the original Witten parity test, and $\mathbb{Z}_6$ representation descent are established at their stated levels. Determinant-line, bordism, dynamical-gravity, regulated quantum constraint, operator-domain, ordering, inner-product, and global-clock burdens remain explicitly outside the result.

Keywords: AASC; ADM; Einstein–Klein–Gordon; scalar clock; refoliation; tetrad gravity; chiral Standard Model; Dirac bracket; constraint algebra; tame inverse; Wheeler–DeWitt.

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Part I

Integrated publication architecture and theorem interface

1. The WDW-4 endpoint

The classical matter stage of the Wheeler–DeWitt arc has two logically different tasks. The first is to populate the scalar regularity antecedent left conditional by WDW-3. The second is to construct the larger tetrad/spin/gauge/fermion carrier and prove that its full classical constraint system is occupied and regular. Collapsing those tasks would obscure which hypotheses belong to Einstein–Klein–Gordon theory and which are introduced only by the Standard Model target.

Integrated WDW-4 question. Can the occupied scalar/clock branch and the occupied torsion-free Standard Model branch be presented as one classical theorem package while preserving their exact dependency, constraint-map, quotient, anomaly, topology, and failure boundaries?

The answer established below is affirmative. The edition is integrated at the level of source, notation, dependency, interface, certificate, and handoff. It does not identify the two phase carriers or replace either module’s proof by an appeal to the other.

2. Accepted module roles

The integration uses the final independently accepted WDW-4A and WDW-4B results. Their distinct mathematical roles and interfaces are preserved throughout the integrated theorem.

Module	Mathematical role	Preserved interface	Accepted result
WDW-4A	Einstein–Klein–Gordon scalar and local-clock occupancy	Regular scalar constraint chart, tame inverse, clock domain, and branch guard	W4.1–W4.8 with an occupied scalar/clock branch.
WDW-4B	Tetrad/spin/gauge/Higgs/fermion classical occupancy	Typed lift from the scalar module, joint constraints, quotient, anomaly ledger, and target guard	W4B.1–W4B.12 with an occupied torsion-free Standard Model branch.

The integrated source retains the theorem labels and exact typed interface needed to distinguish the two carriers and their zero loci.

3. Module map and reading order

Module	New mathematical burden	Controlling certificate	Final theorem
4A	Exact nonvacuum scalar constraint datum; generalized-KID elimination; coupled tame inverse; local clock	ScalarReg3, seven occupied rows	W4.8
4B	Tetrad/spin lift; chiral graded carrier; complete joint constraints and algebra; nonzero gauge–Higgs occupant; joint tame inverse	SMReg4B, twelve occupied rows	W4B.12
Integration	Typed 4A-to-4B lift, joint dependency certificate, consolidated classical handoff, and publication-level nonclaims	W4Reg, eight interface rows	W4.I.1

The theorem labels preserve the accepted numbering. “W4.*k*” denotes the scalar/clock module; “W4B.*k*” denotes the Standard Model module; and “W4.I.*k*” denotes an integration theorem. The ordinary numbered theorems, propositions, and lemmas remain local proof components.

4. The typed interface

Fix an occupied WDW-4A datum z_0 in its regular chart $\mathcal{V}_0$. The interface exported to WDW-4B is

$$\mathfrak{I}_{4A \rightarrow 4B} = (z_0, \mathcal{V}_0, \Phi_{4A}, \{R_z\}_{z \in \mathcal{V}_0}, \mu_{4A} = 2, \mathcal{U}_{\text{clk}}, \mathfrak{F}_{\text{clk}}, \mathfrak{G}_{\text{branch}}). \quad (1)$$

Its coordinates are, respectively, the exact scalar constraint zero, the occupied regular chart, the gravity–scalar constraint map, its compatible tame right-inverse family, the common coefficient loss, the local clock domain, the clock-failure data, and the branch guard. The last coordinate is exactly the already-defined $\mathfrak{G}_{\text{branch}}$ exported by the WDW-4A handoff (64); the integration introduces no renamed or implicit guard.

The WDW-4B construction adds an even gauge–Higgs parameter $(\mathcal{A}_\delta, H_\delta)$ and solves

$$\Phi_{4A}(z_0 + u_\delta) + \mathcal{S}_{\text{GH}}(z_0 + u_\delta; \mathcal{A}_\delta, H_\delta) = 0, \quad u_\delta \in \text{im } R_{z_0}, \quad u_\delta = O(\delta^2). \quad (2)$$

Write

$$\mathcal{L}_\delta(\mathfrak{I}_{4A \rightarrow 4B}) = w_\delta$$

for the full coframe/spin/gauge/Higgs/fermion datum assembled from that solution. Equation (2) is a lifting relation between two constraint problems. It does not assert $\Phi_{4A}(z_0 + u_\delta) = 0$ when $\delta \neq 0$; the gauge–Higgs energy is part of the total WDW-4B constraint.

5. No-backflow and proof-labor discipline

The integrated dependency order is

$$\begin{aligned} \text{WDW-1} &\longrightarrow \text{WDW-2} \longrightarrow \text{WDW-3} \longrightarrow (\text{W4.1} \longrightarrow \cdots \longrightarrow \text{W4.8}) \\ &\longrightarrow (\text{W4B.1} \longrightarrow \cdots \longrightarrow \text{W4B.12}) \longrightarrow \text{W4.I.1} \longrightarrow \text{WDW-5}. \end{aligned}$$

The arrows authorize downstream use only. Integration cannot repair a defect in either module, and the quantum handoff cannot establish a classical premise retroactively.

AASC supplies the necessary governance of determinate nondegenerate objects: Admissibility, Standing, Reference, and Irreversibility are presupposed by the fixed carriers, comparisons, transformations,

records, and endpoints used throughout. Conventional geometry, constrained Hamiltonian theory, elliptic analysis, hyperbolic PDE, and anomaly calculation still perform the positive construction. Kernel necessity does not replace any displayed constraint calculation, inverse estimate, or anomaly sum.

6. Integration fidelity

The module text below preserves the final theorem objects while aligning their headings, local cross-references, and the 4A/4B separation wording with the present two-module publication. No equation, theorem label, theorem hypothesis, theorem conclusion, calculation, proof dependency, certificate row, or nonclaim is removed. The integrated source preserves the declared interface and proof dependencies.

Nonclaim 6.1. An integrated edition is not a proof that WDW-4A and WDW-4B have the same phase carrier, the same constraint map, or the same zero locus. It is a proof package containing both theorem objects and their exact interface.

MODULE 4A

Scalar/Clock Matter Occupancy

Part II

WDW-4A: Frozen boundary, import discipline, and matter-target separation

7. Central question and exact endpoint

WDW-3 ended with a deliberately branch-typed statement. The vacuum compact core was occupied and regular. For a minimally coupled massive real scalar, the off-shell hypersurface-deformation algebra was proved, but the regular characteristic and finite refoliation conclusions were conditional on a six-coordinate certificate. The present question is therefore not whether a scalar term can be appended formally to the ADM Hamiltonian. It is:

WDW-4A question. Does the exact scalar model inherited from WDW-1 admit a nonvacuum, source-attached, licensed Einstein–Klein–Gordon datum that occupies every coordinate of `ScalarReg3`, thereby populates the WDW-3 conditional local quotient and refoliation theorems, and carries a certified local physical-clock use without being promoted to a global or preferred clock?

The endpoint is the following single theorem object.

Definition 7.1 (Covariant-to-canonical scalar/clock matter occupancy closure). On the accepted compact, smooth, boundaryless, minimal metric branch, the target is a nonempty class of minimally coupled real Klein–Gordon data that:

1. obey the complete coupled Hamiltonian and momentum constraints;
2. occupy every clause of `ScalarReg3`;
3. inherit the accepted WDW-2 proper spatial quotient and WDW-3 regular characteristic/refoliation quotient;
4. carry a local time-oriented timelike-gradient scalar-clock patch;
5. retain source, attachment, witness, boundary, relational, branch, singular, and failure coordinates;
6. define a classical, branch-typed handoff to WDW-5; and
7. remain strictly distinct from the tetrad/spin/gauge/fermion enrichment of WDW-4B.

How to read Module 4A

Layer	Status on entry	Work performed here
WDW-1 source/canonical scalar	Accepted	Identity-checked and specialized; not re-proved from nothing.
WDW-2 spatial quotient	Accepted	Imported on its own spatial constraint surface.
WDW-3 scalar HDA	Unconditional	Imported without changing signs, density types, or coefficient functions.
WDW-3 scalar regular quotient	Conditional on ScalarReg3	Promoted only after the six clauses are independently populated in Theorem W4.4.
New WDW-4A construction	Open on entry	Exact nonvacuum constraint datum, generalized-KID elimination, coupled symbol, fixed normal problem, tame Green family, and licensed history.
Clock result	Open on entry	One local timelike-gradient quotient use with explicit failure boundaries.
WDW-4B and WDW-5 quantum data	Not available	Retained as separately gated burdens; never used to repair WDW-4A.

The positive physical construction is conventional Einstein–Klein–Gordon mathematics. The AASC kernel is upstream governance: determinate carrier identity, admissibility, standing-bearing comparison, reference, and irreversible fixation are necessary to state and transport any of these objects. It does not numerically choose m , ε , Γ , or a gauge for the Cauchy theorem. Conversely, success of the downstream occupant does not authorize the kernel retroactively.

8. No-backflow dependency order

The dependency order is

non-degenerate determinate construction
 $\implies$ AASC kernel necessity
 $\implies$ realized physical interior and fixed target
 $\implies$ standing-preserving connection type
 $\implies$ minimal metric-branch Levi–Civita closure
 $\implies$ Module-A source attachment
 $\implies$ WDW-1 canonical source-faithful projection
 $\implies$ WDW-2 proper spatial redescription
 $\implies$ WDW-3 normal-deformation/refoliation closure
 $\implies$ WDW-4A scalar occupancy and local clock.

Every implication is one-way for proof authorization. A local clock cannot prove a source attachment that was missing before the split; a canonical Hamiltonian cannot supply its own covariant ancestor; a status ledger cannot replace the theorem it records; and WDW-4B cannot repair a WDW-4A scalar defect.

9. Imported theorem interfaces

The following sources supply the declared WDW-4A interfaces:

ID	Source	Licensed WDW-4A use
IMP-W1	Final WDW-1 publication edition	Scalar action, phase carrier, Legendre map, constraints, Dirac ledger, source map, and exact pointed lift.
IMP-W2	Final WDW-2 publication edition	Proper spatial generator, quotient, characteristic equality, and source descent.
IMP-W3	Final WDW-3 publication edition	Scalar off-shell HDA, ScalarReg3 schema, conditional regular/refoliation theorems, and branch-typed handoff.
IMP-W3R	WDW-3 handoff theorem	Authorization boundary for WDW-4 with no scalar status promotion.
IMP-ARC	WDW ARC dependency contract	WDW-4A/4B split and first-capstone target.
IMP-COV	Covariant Differentiation	Connection normal form and minimal metric Levi–Civita branch.
IMP-MA	Metric-Source Representation, Module A	Attachment, witness, tensor representation, Ward closure, and scalar witness.
IMP-CLK	Physical Clock as a Quotient	Local clock-domain tuple, standing, failure typing, and overlap guards.

The exact imported WDW-1 clauses are:

1. the minimally coupled action and its unique Module-A tensor representative;
2. the scalar phase carrier $C^\infty(\Sigma) \times \Gamma^\infty(\text{Dens } \Sigma)$;
3. the regular scalar Legendre map for positive lapse;
4. the two scalar deformation densities and the additive total constraints;
5. the complete scalar Dirac ledger;
6. the Hamilton-equation reconstruction of $\square_g \phi - m^2 \phi = 0$;
7. the on-shell Ward identity and compact-core boundary match; and
8. the map Src_ϕ from complete canonical histories to structured Module-A source classes, including one exact pointed occupant.

These are the scalar family and interface results of WDW-1, not a claim that every off-shell scalar point is already a closed source history [1].

The exact imported WDW-2 clauses are the proper spatial moment map, action-groupoid descent, regular spatial characteristic/orbit/vertical equality, and source/observable descent on the spatial constraint surface. No Hamiltonian self-bracket or normal regularity is attributed to WDW-2 [2].

The exact imported WDW-3 clauses are the massive scalar off-shell hypersurface-deformation algebra, the six-coordinate ScalarReg3 antecedent, and the implications from that antecedent to a split total constraint locus, full characteristic equality, the coefficient-sensitive conormal object, direct guarded path/homotopy evaluation, the Nash–Moser local equivalence, and local same-arrow closure. On entry the algebra is PROVED; the regular scalar quotient is CONDITIONAL. Module 4A changes the latter status only through Theorem W4.4.

10. Fixed conventions and common domains

The spacetime signature is $(-+++)$, the future unit normal satisfies $g(n, n) = -1$, and

$$K_{ij} = +\frac{1}{2}\mathcal{L}_n h_{ij}, \quad \pi^{ij} = \frac{\sqrt{h}}{2\kappa}(K^{ij} - Kh^{ij}).$$

The Gauss–Codazzi constraints, with $\Lambda = 0$, are

$$R(h) + K^2 - K_{ij}K^{ij} = 2\kappa\rho, \quad (3)$$

$$D_j(K^j_i - \delta^j_i K) = -\kappa j_i. \quad (4)$$

The scalar potential is $V(\phi) = m^2 \phi^2/2$, with $m^2 \geq 0$. The scalar momentum

$$p_\phi = \sqrt{h}\mathcal{L}_n \phi$$

is a weight-one scalar density. Consequently

$$\mathcal{H}_\perp^\phi = \frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} D_i \phi D^i \phi + \frac{\sqrt{h}}{2} m^2 \phi^2, \quad \mathcal{H}_i^\phi = p_\phi D_i \phi.$$

No sign or density convention in these equations is reselected here.

The smooth coupled carrier is

$$\mathcal{P}_{g+\phi}^\infty = \text{Met}^\infty(\Sigma) \times \Gamma^\infty(S^2 T\Sigma \otimes \text{Dens } \Sigma) \times C^\infty(\Sigma) \times \Gamma^\infty(\text{Dens } \Sigma),$$

with the smooth inverse-limit topology of the shifted Sobolev levels used in section 24. Functionals belong to the inherited algebra of smooth regular local functionals whose variational derivatives have the displayed tensor and density types. The carrier Σ is compact, oriented, connected, and boundaryless. Lapse is positive when it represents a foliation; lapse–shift descriptors in the constraint algebra need not be positive because they are infinitesimal linear variables.

11. WDW-4A/WDW-4B interface boundary

WDW-4A contains one real scalar, its covariant action, its canonical pair, and the common gravitational deformation constraints. It contains no internal gauge group, principal bundle, tetrad, spin structure, spin connection, local Lorentz constraint, fermionic primary or second-class constraint, anomaly calculation, or transported CPT theorem.

WDW-4B is the downstream theorem object that adds and reconciles precisely those carriers. In particular, a full Standard Model target requires at least:

1. tetrad and spin/tangential structures sufficient for fermions;
2. local Lorentz, internal gauge, and diffeomorphism constraints on one common carrier;
3. primary, secondary, second-class, graded, and reducibility ledgers;
4. global-form, line-sector, boundary, and large-transformation data;
5. curved-spacetime Poincaré/CPT specialization rather than unlicensed flat-space transport; and
6. classical and quantum anomaly burdens stated at their actual levels.

Absence of these objects is not a defect in WDW-4A; identifying them as already present would be a category error.

Theorem W4.1 (Exact inheritance, target separation, and acyclicity). The object defined in definition 7.1 inherits exactly the registered WDW-1, WDW-2, and WDW-3 scalar interfaces, with all sign, density, topology, source, boundary, relational, and branch coordinates fixed. WDW-4A and WDW-4B are distinct theorem objects. No WDW-4B or WDW-5 conclusion is a premise of W4.1–W4.8, and the dependency graph

$$\text{W4.1} \rightarrow \text{W4.2} \rightarrow \text{W4.3} \rightarrow \text{W4.4} \rightarrow \text{W4.5} \rightarrow \text{W4.6} \rightarrow \text{W4.7} \rightarrow \text{W4.8}$$

is acyclic.

Proof. The artifact hashes fix the imported sources, and the clause list above fixes the specializations rather than relying on titles. The common-domain equations fix every sign and density used downstream. The WDW-3 scalar regular status is explicitly conditional, so W4.1 does not promote it. The carrier comparison shows that WDW-4B has additional primitive variables and constraints not present in WDW-4A. Finally, each theorem in the displayed graph has only earlier W4 gates and accepted imports as premises; this is also machine-checked from `audit/theorem_dependency.csv`. Hence there is no backflow, cross-target rescue, or cycle. $\square$

Nonclaim 11.1 (No status by ancestry alone). Exact inheritance establishes identity and authorization. It does not prove that a ScalarReg3 occupant exists. That new burden is W4.3–W4.4.

Part III

WDW-4A: Covariant source identity and the exact scalar canonical package

12. Derivative and connection ancestry

For a real scalar, the first derivative is the exterior derivative:

$$\nabla_\mu \phi = \partial_\mu \phi = (d\phi)_\mu.$$

No affine connection is needed to define this first jet. The second derivative and the tensor divergence do require connection data:

$$\square_g \phi = g^{\mu\nu} \nabla_\mu \nabla_\nu \phi, \quad \nabla_\mu T_\phi^{\mu\nu}.$$

Their connection is not chosen inside the matter model. It is inherited from the accepted standing-preserving connection classification and its minimal, torsion-free, metric-compatible metric-branch specialization. Thus it is the Levi–Civita connection ∇^g , with identity principal symbol on the relevant tensor carriers [3]. This ancestry excludes a silent independent torsion, nonmetricity, Palatini, teleparallel, or spin-connection variable from WDW-4A.

Proposition 12.1 (Connection-typed scalar equation). *On the frozen branch, variation of the scalar action produces*

$$\square_g \phi - m^2 \phi = 0$$

with $\square_g$ formed from the inherited Levi–Civita connection. The stress-divergence identity is evaluated using the same connection. No connection degree of freedom belongs to the scalar phase pair.

Proof. The scalar first jet is $d\phi$. Integration by parts in the metric volume form converts $g^{-1}(d\phi, d\delta\phi)$ to $-(\nabla_\mu \nabla^\mu \phi) \delta\phi$; metric compatibility is exactly what preserves the volume and contracts the two derivative indices without an additional standing-changing term. The frozen connection theorem makes this the Levi–Civita operator. The phase carrier introduced below has only (ϕ, p_ϕ) , so no independent connection component can be generated by the Legendre map. $\square$

13. Action, source attachment, and stress representation

The downstream matter occupant is the action

$$S_\phi[g, \phi] = -\frac{1}{2} \int_M \sqrt{-g} \left(g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + m^2 \phi^2 \right) d^4x. \quad (5)$$

It is selected as the WDW-4A target; it is not claimed to be numerically generated by AASC. For each smooth covariant scalar solution ϕ , Module A supplies the structured source class

$$\hat{x}[\phi] = [\mathbf{p}_\phi, \text{Att}_\phi, W_\phi]_{\sim_{\text{src}}}, \quad (6)$$

where $\mathbf{p}_\phi$ is the operational response presentation, Att_ϕ is the attachment graph to the metric source slot, and W_ϕ is the complete smooth witness. The representation map has the unique tensor value

$$q_g(\hat{x}[\phi]) = T^\phi[g, \phi].$$

All four coordinates in this sentence are retained. A bare tensor field is not silently promoted to a structured source class.

Metric variation of equation (5) gives

$$T_{\mu\nu}^\phi = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\partial_\alpha \phi \partial^\alpha \phi + m^2 \phi^2). \quad (7)$$

This is both the operational-Hilbert representative and the tensor used in the coupled Einstein equation. Thus source representation precedes the 3 + 1 split.

Lemma 13.1 (Ward identity and closed-sector exchange). *For every smooth scalar,*

$$\nabla_\mu T_\phi^{\mu\nu} = (\square_g \phi - m^2 \phi) \nabla^\nu \phi. \quad (8)$$

Hence a solution of the scalar equation occupies the zero-external-exchange Ward coordinate $\nabla_\mu T_\phi^{\mu\nu} = 0$.

Proof. Apply the Leibniz rule to equation (7). Since second derivatives commute on a scalar, the terms $\nabla_\mu \phi \nabla^\mu \nabla^\nu \phi$ cancel the derivative of $\frac{1}{2} \nabla_\alpha \phi \nabla^\alpha \phi$; the potential derivative contributes $-m^2 \phi \nabla^\nu \phi$. The remaining factor is equation (8). No converse from a degenerate gradient is used. $\square$

14. ADM decomposition and regular Legendre map

Write

$$g = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt), \quad N > 0.$$

Then equation (5) becomes

$$\mathcal{L}_\phi = N\sqrt{h} \left[\frac{1}{2N^2} (\dot{\phi} - N^i D_i \phi)^2 - \frac{1}{2} D_i \phi D^i \phi - \frac{1}{2} m^2 \phi^2 \right].$$

The momentum and inverse velocity are

$$p_\phi = \frac{\partial \mathcal{L}_\phi}{\partial \dot{\phi}} = \frac{\sqrt{h}}{N} (\dot{\phi} - N^i D_i \phi), \quad \dot{\phi} = \frac{N}{\sqrt{h}} p_\phi + N^i D_i \phi. \quad (9)$$

The velocity Hessian is $\sqrt{h}/N > 0$. Therefore the scalar Legendre map is a smooth fibrewise isomorphism, with no scalar primary constraint.

The smooth phase carrier and two-form are

$$\begin{aligned} \mathcal{P}_\phi^\infty &= C^\infty(\Sigma) \times \Gamma^\infty(\text{Dens } \Sigma), \\ \Omega_\phi((\delta\phi_1, \delta p_1), (\delta\phi_2, \delta p_2)) &= \int_\Sigma (\delta p_1 \delta\phi_2 - \delta p_2 \delta\phi_1). \end{aligned} \quad (10)$$

Together with the gravitational pair this gives

$$\Theta_{g+\phi} = \int_\Sigma (\pi^{ij} \delta h_{ij} + p_\phi \delta\phi) d^3x, \quad \Omega_{g+\phi} = \delta\Theta_{g+\phi} = \Omega_g \oplus \Omega_\phi$$

under the inherited selected-variable diffeomorphism.

The Legendre transform gives

$$\mathcal{H}_\perp^\phi = \frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} D_i \phi D^i \phi + \frac{\sqrt{h}}{2} m^2 \phi^2, \quad (11)$$

$$\mathcal{H}_i^\phi = p_\phi D_i \phi. \quad (12)$$

The total densities are additive:

$$\mathcal{H}_\perp^{g+\phi} = \mathcal{H}_\perp^g + \mathcal{H}_\perp^\phi, \quad \mathcal{H}_i^{g+\phi} = \mathcal{H}_i^g + \mathcal{H}_i^\phi.$$

There is no gravitational-scalar velocity cross-term.

15. Stress projections and positivity

Let

$$\rho_\phi = T_{\mu\nu}^\phi n^\mu n^\nu, \quad j_i^\phi = -h_i^\mu T_{\mu\nu}^\phi n^\nu, \quad S_{ij}^\phi = h_i^\mu h_j^\nu T_{\mu\nu}^\phi.$$

Using $p_\phi = \sqrt{h} \mathcal{L}_n \phi$,

$$\rho_\phi = \frac{1}{2} \left(\frac{p_\phi}{\sqrt{h}} \right)^2 + \frac{1}{2} D_i \phi D^i \phi + \frac{1}{2} m^2 \phi^2, \quad (13)$$

$$j_i^\phi = -\frac{p_\phi}{\sqrt{h}} D_i \phi, \quad (14)$$

$$S_{ij}^\phi = D_i \phi D_j \phi + h_{ij} \left[\frac{1}{2} \left(\frac{p_\phi}{\sqrt{h}} \right)^2 - \frac{1}{2} D_k \phi D^k \phi - \frac{1}{2} m^2 \phi^2 \right]. \quad (15)$$

Consequently

$$\mathcal{H}_\perp^\phi = \sqrt{h} \rho_\phi, \quad \mathcal{H}_i^\phi = -\sqrt{h} j_i^\phi.$$

For $m^2 \geq 0$, each term in equation (13) is nonnegative, so $\rho_\phi \geq 0$ pointwise. Equality holds exactly where $p_\phi = 0$, $D\phi = 0$, and $m\phi = 0$.

16. Hamilton equations and complete scalar Dirac ledger

The matter Hamilton equations are

$$\dot{\phi} = \frac{N}{\sqrt{h}} p_\phi + N^i D_i \phi, \quad (16)$$

$$\dot{p}_\phi = D_i (N \sqrt{h} D^i \phi) + D_i (N^i p_\phi) - N \sqrt{h} m^2 \phi. \quad (17)$$

Their conjunction is $\square_g \phi - m^2 \phi = 0$, so lemma 13.1 supplies the same closed Ward coordinate as the covariant source. Compactness of Σ removes spatial surface terms; fixed endpoint scalar data close the time-boundary variation.

Sector	Scalar status	Reason
Primary scalar constraints	none	Hessian $\sqrt{h}/N$ is invertible.
Internal/Gauss constraints	none	Real scalar has no internal gauge carrier.
Second-class constraints	none	No scalar Dirac-bracket reduction.
Reducibility relations	none	No internal constraint complex.
Lapse/shift primaries	inherited gravitational P_N, P_i	Multipliers, not scalar canonical variables.
Deformation secondaries	total $\mathcal{H}_\perp^{g+\phi}, \mathcal{H}_i^{g+\phi}$	Their full algebra is the accepted WDW-3 result.
Boundary improvement	zero on compact core	Reopens only after a boundary branch is admitted.

17. History-to-source identity

Let $\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi)$ denote the WDW-1 domain of complete smooth coupled canonical histories that obey equations (16) and (17), the total constraints and retained ADM evolution equations, and reconstruct a smooth covariant pair (g_γ, ϕ_γ) . The exact source map is

$$\begin{aligned} \text{Src}_\phi : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi) &\longrightarrow \widehat{Q}_{\text{sm},0}^{\text{src}}, \\ [\gamma] &\longmapsto [\mathfrak{p}_{\phi_\gamma}, \text{Att}_{\phi_\gamma}, W_{\phi_\gamma}]_{\sim_{\text{src}}}, \end{aligned} \quad (18)$$

with

$$q_g(\text{Src}_\phi[\gamma]) = T^\phi[g_\gamma, \phi_\gamma].$$

It is constant under foliation-presentation equivalence because the reconstructed covariant pair and its attachment graph are unchanged. It is not constant on the entire off-shell phase carrier, and no such collapse is claimed.

Theorem W4.2 (Scalar source/canonical identity). The WDW-4A scalar action, connection ancestry, structured Module-A source class, stress tensor, phase carrier, symplectic form, regular Legendre map, constraint densities, Hamilton equations, Ward coordinate, compact-core boundary condition, and complete Dirac classification are exactly the scalar object accepted in WDW-1. The identities

$$\sqrt{h} \rho_\phi = \mathcal{H}_\perp^\phi, \quad -\sqrt{h} j_i^\phi = \mathcal{H}_i^\phi$$

make its covariant and canonical source projections coincide, and $\rho_\phi \geq 0$ for $m^2 \geq 0$.

Proof. Proposition 12.1 fixes the derivative and connection types. Equations (5), (7) and (8) supply the action, unique tensor representative, and closed Ward implication. The regular Legendre calculation equation (9) gives precisely the WDW-1 scalar carrier and proves the absence of scalar primary constraints. Direct Legendre transformation gives equations (11) and (12); direct projection of the already-defined tensor gives equations (13) to (15), establishing equality without inferring the tensor backward from ADM. The two Hamilton equations reconstruct the covariant equation, and equation (18) retains the complete attachment and witness coordinates. The Dirac table has no blank scalar sector. These are the exact identities licensed by IMP-COV, IMP-MA, and IMP-W1. $\square$

Nonclaim 17.1 (No universal matter lift). Theorem W4.2 proves one model-family interface. Gauge fields, fermions, higher-derivative matter, open systems, and actionless effective sources have different Legendre, constraint, and attachment burdens and are not consequences of this theorem.

Part IV

WDW-4A: Exact nonvacuum scalar occupancy and licensed source history

18. The compact hyperbolic carrier

Choose a torsion-free cocompact lattice $\Gamma < \text{Isom}^+(\mathbb{H}^3)$ and set

$$\Sigma = \mathbb{H}^3/\Gamma.$$

Such closed oriented hyperbolic three-manifolds exist [4]. Let h be the sectional-curvature -1 metric. Then

$$\text{Ric}(h) = -2h, \quad R(h) = -6, \quad Dh = 0.$$

This is the exact carrier used by the accepted WDW-3 vacuum occupant; neither its topology nor its orientation is changed.

Lemma 18.1 (No continuous hyperbolic isometry). *The compact metric (Σ, h) admits no nonzero Killing vector field.*

Proof. For a Killing field X , the integrated Bochner identity gives

$$\int_\Sigma |DX|_h^2 d\mu_h = \int_\Sigma \text{Ric}(X, X) d\mu_h = -2 \int_\Sigma |X|_h^2 d\mu_h.$$

The left side is nonnegative and the right side nonpositive, so both vanish and $X = 0$. Equivalently, the isometry group of a closed hyperbolic manifold is finite. This excludes continuous Killing fields, not discrete isometries; finite large-isometry data remain in the global-sector ledger. $\square$

19. The one-parameter coupled datum

Fix $\kappa > 0$ and $\varepsilon > 0$. Define

$$c_\varepsilon = \sqrt{1 + \frac{\kappa\varepsilon^2}{6}}, \quad (19)$$

and set

$$\phi_\varepsilon = 0, \quad p_{\phi,\varepsilon} = \varepsilon\sqrt{h}, \quad K_{ij}^\varepsilon = c_\varepsilon h_{ij}. \quad (20)$$

The canonical momentum determined by the frozen convention is

$$\begin{aligned} \pi_\varepsilon^{ij} &= \frac{\sqrt{h}}{2\kappa} (K_\varepsilon^{ij} - K_\varepsilon h^{ij}) \\ &= \frac{\sqrt{h}}{2\kappa} (c_\varepsilon - 3c_\varepsilon) h^{ij} = -\frac{c_\varepsilon\sqrt{h}}{\kappa} h^{ij}. \end{aligned} \quad (21)$$

Thus the phase-space datum is

$$z_\varepsilon = \left(h, -\frac{c_\varepsilon\sqrt{h}}{\kappa} h^{-1}; 0, \varepsilon\sqrt{h} \right) \in \mathcal{P}_{g+\phi}^\infty. \quad (22)$$

All components are smooth, and the map $\varepsilon \mapsto z_\varepsilon$ is smooth on $(0, \infty)$ and extends smoothly as an even/odd component family through $\varepsilon = 0$ once the sign of p_ϕ is retained.

20. Direct constraint calculation

At equation (20),

$$D_i \phi_\varepsilon = 0, \quad \frac{p_{\phi,\varepsilon}}{\sqrt{h}} = \varepsilon.$$

Therefore equations (13) and (14) give

$$\rho_{\phi,\varepsilon} = \frac{\varepsilon^2}{2}, \quad j_i^{\phi,\varepsilon} = 0. \quad (23)$$

The geometric traces are

$$K_\varepsilon = 3c_\varepsilon, \quad K_{ij}^\varepsilon K_\varepsilon^{ij} = 3c_\varepsilon^2.$$

Hence

$$\begin{aligned} R + K_\varepsilon^2 - K_{ij}^\varepsilon K_\varepsilon^{ij} &= -6 + 9c_\varepsilon^2 - 3c_\varepsilon^2 \\ &= -6 + 6c_\varepsilon^2 = \kappa\varepsilon^2 = 2\kappa\rho_{\phi,\varepsilon}. \end{aligned} \quad (24)$$

Since c_ε is spatially constant and $Dh = 0$,

$$D_j \left((K_\varepsilon)^j_i - \delta_i^j K_\varepsilon \right) = D_j (-2c_\varepsilon \delta_i^j) = 0 = -\kappa j_i^{\phi,\varepsilon}. \quad (25)$$

Both coupled constraints hold exactly. No small-error term, perturbative truncation, or implicit-function existence claim appears in this computation.

Proposition 20.1 (Nonvacuum status and vacuum limit). *For every $\varepsilon > 0$, z_ε is genuinely nonvacuum. As $\varepsilon \rightarrow 0$,*

$$z_\varepsilon \longrightarrow \left(h, -\frac{\sqrt{h}}{\kappa} h^{-1}; 0, 0 \right),$$

whose gravitational component is the accepted WDW-3 hyperbolic vacuum datum.

Proof. At positive ε ,

$$p_{\phi,\varepsilon} = \varepsilon\sqrt{h} \neq 0, \quad \rho_{\phi,\varepsilon} = \varepsilon^2/2 > 0.$$

Thus the matter phase coordinate and stress tensor are nonzero even though the slice value ϕ_ε vanishes. The limit follows from $c_\varepsilon \rightarrow 1$. None of equations (24) and (25) for positive ε is inferred from the limiting vacuum equation. $\square$

21. Einstein–Klein–Gordon development

The data $(h, K^\varepsilon; \phi_\varepsilon, \varepsilon)$ are smooth and satisfy the Einstein–Klein–Gordon constraints. In harmonic coordinates the reduced Einstein equations form a quasilinear wave system for the metric components, while the Klein–Gordon equation is a wave equation with smooth semilinear mass term and coefficients supplied by the metric. The standard local Cauchy theorem therefore gives a smooth development on a neighborhood of $\{0\} \times \Sigma$, and constraint propagation plus geometric uniqueness make it a development of the unreduced system [5, 6].

The Choquet–Bruhat–Geroch maximal-development theorem then applies to the geometrically unique Cauchy system: there is a maximal globally hyperbolic development, unique up to an initial-surface-fixing isometry [7]. Denote one representative by

$$(M_\varepsilon, g_\varepsilon, \phi_\varepsilon^{\text{dev}}, \iota_\varepsilon).$$

Here $\iota_\varepsilon : \Sigma \hookrightarrow M_\varepsilon$ induces precisely equation (20).

Remark 21.1 (Exact import boundary). “Maximal globally hyperbolic” means maximal among globally hyperbolic developments of these data. It does not mean geodesically complete, singularity-free, globally scalar-monotone, or inextendible in every category. No such conclusion is used below.

Choose any admissible local foliation through $\iota_\varepsilon(\Sigma)$. Its canonical split defines a history

$$\gamma_\varepsilon = \left[t \mapsto \left(h_\varepsilon(t), \pi_\varepsilon(t); \phi_\varepsilon^{\text{dev}}|_{\Sigma_t}, \sqrt{h_\varepsilon(t)} \mathcal{L}_{n_\varepsilon} \phi_\varepsilon^{\text{dev}}; N_\varepsilon, N_\varepsilon^i \right) \right]_{\mathcal{F}}.$$

The field equations imply

$$\gamma_\varepsilon \in \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi).$$

Use the WDW-1 map equation (18) to define

$$\hat{x}_\varepsilon = \text{Src}_\phi(\gamma_\varepsilon) = [\mathfrak{p}_{\phi_\varepsilon^{\text{dev}}}, \text{Att}_{\phi_\varepsilon^{\text{dev}}}, W_{\phi_\varepsilon^{\text{dev}}}]_{\sim_{\text{src}}}. \quad (26)$$

Its tensor representative is

$$qg(\hat{x}_\varepsilon) = T^\phi[g_\varepsilon, \phi_\varepsilon^{\text{dev}}],$$

and lemma 13.1 places it on the closed Ward locus. This source class is assigned from the covariant development; it is not inferred from a bare instantaneous ADM tensor.

Proposition 21.2 (Exact pointed source lift). *The tuple*

$$\lambda_\varepsilon = (\mathfrak{M}_\phi, \text{Src}_\phi, \gamma_\varepsilon, \text{Src}_\phi(\gamma_\varepsilon) = \hat{x}_\varepsilon)$$

is an exact WDW-1 pointed source lift. It retains the response presentation, attachment graph, smooth witness, unique tensor representative, Ward coordinate, compact-core boundary coordinate, and reconstructed history.

Proof. The local and maximal Cauchy theorems make γ_ε a complete history on its declared regular interval. WDW-1 proves that the canonical split of any such covariant Einstein–Klein–Gordon history belongs to the domain of Src_ϕ , and that the map returns the Module-A source class of the reconstructed covariant scalar. Applying that theorem to the representative above gives equation (26) and all listed coordinates. $\square$

Theorem W4.3 (Explicit coupled constraint occupancy). For every $\varepsilon > 0$, the smooth datum z_ε of equation (22) satisfies

$$\Phi_{g+\phi}(z_\varepsilon) = 0.$$

It is nonvacuum, depends smoothly on ε , has the exact canonical momentum equation (21), and belongs to the exact source class $\hat{x}_\varepsilon = \text{Src}_\phi(\gamma_\varepsilon)$ of a licensed smooth Einstein–Klein–Gordon development.

Proof. Smooth carrier membership and canonical momentum are equations (20) to (22). The Hamiltonian and momentum constraints are the identities equations (24) and (25). Nonvacuum status is proposition 20.1. The licensed history and exact source assignment are section 21 and proposition 21.2. These witnesses are all attached to the same datum and not assembled from unrelated examples. $\square$

Part V

WDW-4A: Generalized KIDs and the complete coupled weighted symbol

22. The coupled adjoint system

Let $z = (h, K; \phi, p_\phi)$ satisfy the coupled constraints and write

$$\varpi = \frac{p_\phi}{\sqrt{h}} = \mathcal{L}_n \phi.$$

A lapse-shift pair (N, X) belongs to the formal adjoint kernel of the coupled constraint map exactly when the corresponding infinitesimal spacetime deformation preserves all initial data. In the frozen convention the complete generalized-KID system is:

$$\mathcal{L}_X h_{ij} + 2NK_{ij} = 0, \quad (27)$$

$$D_i D_j N + \mathcal{L}_X K_{ij} - N \left[R_{ij} + K K_{ij} - 2K_i^k K_{kj} - \Lambda h_{ij} - \kappa (S_{ij}^\phi - \tfrac{1}{2} h_{ij} (S_\phi - \rho_\phi)) \right] = 0, \quad (28)$$

$$\mathcal{L}_X \phi + N\varpi = 0, \quad (29)$$

$$D_i (N\sqrt{h} D^i \phi) + D_i (X^i p_\phi) - N\sqrt{h} m^2 \phi = 0. \quad (30)$$

Equation (27) preserves the induced metric, equation (28) preserves the second fundamental form using the coupled Einstein evolution equation, and equations (29) and (30) preserve the scalar canonical pair. Thus no matter equation is hidden in the phrase “generalized KID.”

Proposition 22.1 (Adjoint-kernel/KID identity). *After the fixed density and Legendre identifications,*

$$\ker D\Phi_{g+\phi}(z)^\# = \{(N, X) : \text{equations (27) to (30) hold}\}.$$

Proof. Pair $D\Phi_{g+\phi}(z)$ with a smooth lapse-shift and integrate by parts on compact Σ . The metric and extrinsic-curvature coefficients are the two geometric preservation equations, including the matter stress term from the variation of ρ_ϕ, j_i^ϕ . The coefficient of the scalar momentum variation is $\mathcal{L}_X \phi + N\varpi$. The coefficient of the scalar configuration variation is the negative of equation (30). Hence the formal adjoint vanishes precisely on the displayed system. The canonical Legendre map is an isomorphism, so the same kernel is obtained in $(h, \pi; \phi, p_\phi)$ variables. $\square$

23. Direct generalized-KID elimination

At z_ε ,

$$\phi = 0, \quad D\phi = 0, \quad \varpi = \varepsilon, \quad K = c_\varepsilon h.$$

Equation equation (29) becomes

$$N\varepsilon = 0.$$

Since $\varepsilon > 0$, $N = 0$. Equation equation (27) then becomes

$$\mathcal{L}_X h = 0.$$

Lemma 18.1 gives $X = 0$. With $N = X = 0$, the remaining equations (28) and (30) hold only trivially.

Theorem 23.1 (No generalized KIDs at the scalar occupant). *For every $\varepsilon > 0$,*

$$\ker D\Phi_{g+\phi}(z_\varepsilon)^\# = 0.$$

Proof. The scalar-preservation equation eliminates the lapse at the nonzero scalar datum, after which the metric-preservation equation and compact hyperbolicity eliminate the shift. Apply proposition 22.1. $\square$

Remark 23.2 (Stronger than a vacuum import). At the vacuum point WDW-3 had to eliminate a conformal Killing pair. Here the matter equation first forces the lapse to zero. Thus the no-KID result is proved at the coupled point itself and does not follow by continuity from vacuum or by a genericity theorem.

24. The coupled shifted tower

Fix $r > \frac{3}{2}$ and set

$$\begin{aligned} E_r^\phi &= H^{r+2}(S^2T^*\Sigma) \oplus H^{r+1}(S^2T^*\Sigma) \\ &\quad \oplus H^{r+1}(\Sigma) \oplus H^r(\text{Dens } \Sigma), \end{aligned} \quad (31)$$

$$F_r = H^r(\text{Dens } \Sigma) \oplus H^r(T^*\Sigma \otimes \text{Dens } \Sigma), \quad (32)$$

$$\hat{Z}_r = H^{r+2}(\Sigma) \oplus H^{r+2}(T\Sigma). \quad (33)$$

The components of E_r^ϕ are (γ, k, φ, q) , respectively variations of (h, K, ϕ, p_ϕ) . The target retains the equal-row grading inherited from WDW-3. The smooth carrier is the inverse limit over the nested levels. The Legendre differential transports the first two components to the (h, π) tower.

Set

$$A_z = D\Phi_{g+\phi}(z) : E_r^\phi \longrightarrow F_r.$$

It is useful to work with the scalar-valued constraint expressions

$$\mathcal{H} = R + K^2 - K_{ij}K^{ij} - 2\Lambda - 2\kappa\rho_\phi, \quad \mathcal{M}_i = D_j(K^j_i - \delta_i^j K) + \kappa j_i^\phi$$

before restoring the fixed density maps. Their linearizations have the following complete differential-order table.

Row	$\gamma = \delta h$	$k = \delta K$	$\varphi = \delta \phi$	$q = \delta p_\phi$	Target
$\delta \mathcal{H}$	2	0	1	0	$H^r(\text{Dens})$
$\delta \mathcal{M}$	1	1	1	0	$H^r(T^*\Sigma \otimes \text{Dens})$

More explicitly, up to the fixed density identification,

$$\begin{aligned} D\mathcal{H}(\gamma, k, \varphi, q) &= -D^i D^j \gamma_{ij} + D^2 \text{tr } \gamma - \langle \text{Ric}, \gamma \rangle \\ &\quad + 2K \text{tr } k - 2\langle K, k \rangle + \mathcal{Z}_H(\gamma; K) \\ &\quad - 2\kappa \left[\varpi \left(\frac{q}{\sqrt{h}} - \frac{1}{2} \varpi \text{tr } \gamma \right) + D^i \phi D_i \varphi - \frac{1}{2} \gamma^{ij} D_i \phi D_j \phi + m^2 \phi \varphi \right], \end{aligned} \quad (34)$$

$$\begin{aligned} D\mathcal{M}_i(\gamma, k, \varphi, q) &= D_j k^j_i - D_i \text{tr } k + \mathcal{Z}_{M,i}(D\gamma; K) + \mathcal{Z}'_{M,i}(\gamma; DK) \\ &\quad - \kappa \left[\left(\frac{q}{\sqrt{h}} - \frac{1}{2} \varpi \text{tr } \gamma \right) D_i \phi + \varpi D_i \varphi \right]. \end{aligned} \quad (35)$$

Here $\mathcal{Z}_H$ is algebraic in γ , while $\mathcal{Z}_M$ is first order in γ . The formulas retain every scalar term.

At the exact point z_ε , the scalar principal contribution simplifies to

$$D\mathcal{H}_{\text{scalar}} = -2\kappa\varepsilon q/\sqrt{h} + \text{metric zeroth order}, \quad (36)$$

$$D\mathcal{M}_{i,\text{scalar}} = -\kappa\varepsilon D_i\varphi. \quad (37)$$

Thus the q -column is order zero, while the φ -column in the momentum row is genuinely first order. It is not discarded as “lower order.”

25. Formal adjoint types

On $\widehat{Z}_r$, the differential formal adjoint has the component regularities

$$A_z^\# : \widehat{Z}_r \longrightarrow \widetilde{E}_r^\phi,$$

where

$$\widetilde{E}_r^\phi = H^r(S^2T^*\Sigma) \oplus H^{r+1}(S^2T^*\Sigma) \oplus H^{r+1}(\Sigma) \oplus H^r(\text{Dens } \Sigma).$$

The last component is in fact smoother when $z = z_\varepsilon$, but it is embedded in the declared H^r slot. The component-order table is:

Adjoint component	N	X	Leading origin
γ	order 2	order 1	curvature and momentum connection terms
k	order 0	order 1	K -quadratic and divergence terms
φ	order 1	order 1	$D\phi$ -Hamiltonian and ϖX -momentum terms
q	order 0	order 0	scalar preservation $N\varpi + \mathcal{L}_X\phi$

This formal differential adjoint is not a Sobolev-level Hilbert adjoint. The Banach transpose and each Hilbert adjoint require separate duality/Riesz maps. No minimum-norm solution selected by a changing Hilbert product is used below.

26. Complete principal symbol

Assign column orders 2, 1, 1, 0 to (γ, k, φ, q) , respectively, and row order zero to both equations. For $\zeta \neq 0$, write

$$\mathbf{h}_2(\zeta)\gamma = -\zeta^i\zeta^j\gamma_{ij} + |\zeta|^2 \text{tr } \gamma, \quad (38)$$

$$\mathbf{m}_1(\zeta)k = \zeta_j k^j_i - \zeta_i \text{tr } k, \quad (39)$$

$$\mathbf{s}_H(z, \zeta)\varphi = -2\kappa \langle D\phi, \zeta \rangle \varphi, \quad (40)$$

$$\mathbf{s}_M(z, \zeta)\varphi = -\kappa \varpi \zeta_i \varphi. \quad (41)$$

Ignoring the conventional factor i in the symbol map,

$$a_{z,0}^\phi(\zeta)(\gamma, k, \varphi, q) = (\mathbf{h}_2(\zeta)\gamma + \mathbf{s}_H(z, \zeta)\varphi, \mathbf{m}_1(\zeta)k + \mathbf{s}_M(z, \zeta)\varphi). \quad (42)$$

The zero-order q -column is displayed in equations (34) and (35) but does not enter the selected principal symbol. At z_ε , $\mathbf{s}_H = 0$ and $\mathbf{s}_M = -\kappa\varepsilon\zeta_i$.

Lemma 26.1 (Coupled symbol surjectivity). *For every z in a sufficiently small coefficient neighborhood of z_ε and every $\zeta \neq 0$, $a_{z,0}^\phi(\zeta)$ is surjective onto the scalar-plus-covector equation fibre. Its gravitational columns alone are surjective.*

Proof. Given a scalar target, choose the trace/transverse part of γ so that $\mathbf{h}_2(\zeta)\gamma$ attains it. Given a covector target, choose the longitudinal components of k so that $\mathbf{m}_1(\zeta)k$ attains it. Set $\varphi = 0$. This is the accepted WDW-3 gravitational symbol argument. The scalar column in equation (42) adds an allowed image direction and cannot destroy surjectivity. Local uniformity follows by compactness of the unit cosphere and shrinking the coefficient chart. $\square$

27. One fixed scalar-coupled row balancer

Fix the WDW-3 auxiliary metric h_* , its density and bundle metrics, and the positive spectral operator $\Lambda_{*,S^2} = (1 + \Delta_{*,S^2})^{1/2}$. Let U_z^ϕ be the pointwise unitary transport from the geometric variation pairing to these fixed pairings. Define

$$\mathcal{B}_{z,*}^\phi = (U_z^\phi)^{-1} \text{diag} \left(\Lambda_{*,S^2}^{-2}, I, I, I \right) U_z^\phi : \tilde{E}_r^\phi \longrightarrow E_r^\phi. \quad (43)$$

It is positive, self-adjoint, injective, smooth-tame in z , and independent of r . The same pseudodifferential expression acts on smooth sections and restricts to every Sobolev level.

Define the square balanced normal operator

$$L_z^\phi = A_z \mathcal{B}_{z,*}^\phi A_z^\# : \hat{Z}_r \longrightarrow F_r. \quad (44)$$

The order -2 metric balancer converts the Hamiltonian metric-normal contribution from order four to order two. The k and φ identity blocks leave their order-one columns with order-two normal contributions. The q block is lower order.

At principal order,

$$\sigma_2(L_z^\phi)(\zeta) = a_{z,0}^\phi(\zeta) b_*(\zeta) a_{z,0}^\phi(\zeta)^*, \quad (45)$$

where b_* is positive and acts by $|\zeta|_*^{-2}$ on the metric column and by the identity on the k and scalar principal columns. In particular the scalar contribution is

$$\begin{pmatrix} \mathbf{s}_H \\ \mathbf{s}_M \end{pmatrix} \begin{pmatrix} \mathbf{s}_H \\ \mathbf{s}_M \end{pmatrix}^* \geq 0.$$

At z_ε it adds the positive semidefinite momentum block $\kappa^2 \varepsilon^2 \zeta \otimes \zeta$.

Proposition 27.1 (Positive elliptic coupled normal symbol). *After shrinking to a coefficient neighborhood $\mathcal{U}_\varepsilon$, there is $c'_\varepsilon > 0$ such that*

$$\left\langle u, \sigma_2(L_z^\phi)(\zeta) u \right\rangle \geq c'_\varepsilon |\zeta|_*^2 \left(|u_N|^2 + |u_X|^2 \right)$$

for $z \in \mathcal{U}_\varepsilon$, $u = (u_N, u_X)$, and $\zeta \neq 0$.

Proof. Lemma 26.1 makes the adjoint symbol injective. Positivity and injectivity of b_* therefore make $\langle a^*u, b_*a^*u \rangle$ positive for $u \neq 0$. Homogeneity supplies the factor $|\zeta|_*^2$; compactness of the unit cosphere and local uniformity give one positive constant. The scalar column is retained in $a_{z,0}^\phi$ and contributes a positive semidefinite term, so it cannot weaken the gravitational lower bound. $\square$

Part VI

WDW-4A: The tame Green family and populated ScalarReg3

28. Self-adjoint Fredholm realization

Use the fixed equation/multiplier density pairing inherited from WDW-3 to identify the Hamiltonian density with a lapse multiplier and the momentum density with a shift multiplier. On smooth sections,

$$\langle u, L_z^\phi v \rangle = \langle A_z^\# u, \mathcal{B}_{z,*}^\phi A_z^\# v \rangle.$$

Because $\mathcal{B}_{z,*}^\phi$ is self-adjoint, L_z^ϕ is formally self-adjoint. Proposition 27.1 makes it an elliptic classical pseudodifferential operator of order two on the complete lapse–shift bundle, with unbounded L_z^2 domain $H^2(\Sigma) \oplus H^2(T\Sigma)$. On compact Σ , the elliptic parametrix theorem gives Fredholmness and elliptic regularity [8]. Its self-adjoint realization has index zero.

Lemma 28.1 (Energy identity and kernel). *For every smooth multiplier $u = (N, X)$,*

$$\langle u, L_z^\phi u \rangle_{L_z^2} = \langle A_z^\# u, \mathcal{B}_{z,*}^\phi A_z^\# u \rangle_{L_z^2}. \quad (46)$$

Consequently

$$\ker L_z^\phi = \ker A_z^\#.$$

Proof. Equation (46) is the formal-adjoint identity for $A_z \mathcal{B}_{z,*}^\phi A_z^\#$. Positivity of $\mathcal{B}_{z,*}^\phi$ gives one inclusion. Its injectivity gives the reverse implication: if the energy is zero, its positive square root annihilates $A_z^\# u$, hence $A_z^\# u = 0$. $\square$

Corollary 28.2 (Invertibility at the occupant). *For every $\varepsilon > 0$,*

$$L_{z_\varepsilon}^\phi : \widehat{Z}_r \longrightarrow F_r$$

is bijective at every sufficiently high Sobolev level.

Proof. Theorem 23.1 and lemma 28.1 give zero kernel. Index zero gives zero cokernel. Elliptic regularity identifies the solutions obtained at different regularities with the same smooth distributional solution. $\square$

29. Uniform parameter estimates

Fix $q_0 > \frac{3}{2} + 4$. The coefficients of A_z use at most two derivatives of h , one derivative of K , one derivative of ϕ , and no derivative of p_ϕ . The transports in $\mathcal{B}_{z,*}^\phi$ are order zero and the auxiliary spectral operator is fixed. Relative to the shifted domain equation (31), the controlling coefficient loss is therefore

$$\mu_\phi = 2. \quad (47)$$

The scalar column does not increase the WDW-3 metric loss: its highest coefficient derivative is one, whereas curvature already requires two.

Shrink the coefficient neighborhood of z_ε so that the elliptic symbol bound and low-level invertibility are uniform. For $q \geq q_0$, the parameter-elliptic estimate is

$$\|u\|_{\widehat{Z}_q} \leq C_q \left(\|L_z^\phi u\|_{F_q} + (1 + \|z\|_{E_{q+\mu_\phi}^\phi}) \|u\|_{\widehat{Z}_{q_0}} \right). \quad (48)$$

At z_ε , the low-norm term is removed by invertibility. A Neumann argument at level q_0 , followed by equation (48), gives a possibly smaller open neighborhood $\mathcal{V}_\varepsilon$ on which every L_z^ϕ is invertible.

Define

$$G_z^\phi = (L_z^\phi)^{-1} : F_q \longrightarrow \widehat{Z}_q, \quad Q_z^\phi = \mathcal{B}_{z,*}^\phi A_z^\# G_z^\phi : F_q \longrightarrow E_q^\phi. \quad (49)$$

Then

$$A_z Q_z^\phi = A_z \mathcal{B}_{z,*}^\phi A_z^\# G_z^\phi = L_z^\phi G_z^\phi = \text{id}_{F_q}. \quad (50)$$

Theorem 29.1 (Smooth-tame Green and right-inverse families). *After shrinking $\mathcal{V}_\varepsilon$, the following statements hold for every $q \geq q_0$:*

$$\|G_z^\phi f\|_{\widehat{Z}_q} \leq C_q \left[\|f\|_{F_q} + (1 + \|z\|_{E_{q+\mu_\phi}^\phi}) \|f\|_{F_{q_0}} \right], \quad (51)$$

$$\|Q_z^\phi f\|_{E_q^\phi} \leq C_q \left[\|f\|_{F_q} + (1 + \|z\|_{E_{q+\mu_\phi}^\phi}) \|f\|_{F_{q_0}} \right]. \quad (52)$$

Every parameter derivative exists and obeys a one-high-norm tame estimate. More precisely, for $m \geq 1$,

$$D^m G_z^\phi[\dot{z}_1, \dots, \dot{z}_m] = \sum_{\mathcal{P}=(B_1, \dots, B_k)} (-1)^k G_z^\phi D^{|B_1|} L_z^\phi[\dot{z}_{B_1}] G_z^\phi \dots D^{|B_k|} L_z^\phi[\dot{z}_{B_k}] G_z^\phi, \quad (53)$$

where the sum is over ordered partitions of $\{1, \dots, m\}$. With $u_0 = f$, it satisfies

$$\left\| D^m G_z^\phi[\dot{z}_1, \dots, \dot{z}_m] f \right\|_{\widehat{Z}_q} \leq C_{q,m} (1 + \|z\|_{E_{q+\mu_\phi}^\phi}) \sum_{j=0}^m \|u_j\|_{\mathcal{X}_j(q)} \prod_{k \neq j} \|u_k\|_{\mathcal{X}_k(q_0 + \mu_\phi)}, \quad (54)$$

where

$$\mathcal{X}_0(q) = F_q, \quad \mathcal{X}_j(q) = E_{q+\mu_\phi}^\phi \quad (j \geq 1).$$

Every derivative of $Q_z^\phi = \mathcal{B}_{z,*}^\phi A_z^\# G_z^\phi$ obeys the corresponding Leibniz-expanded estimate with the same fixed coefficient loss.

Proof. The uniform symbol inequality of proposition 27.1, compact pseudodifferential estimates, Sobolev multiplication, and Moser commutator estimates give equation (48). The kernel is absent at z_ε ; continuity gives invertibility at the low level on $\mathcal{V}_\varepsilon$, and the elliptic estimate lifts it to all higher levels. This proves equation (51).

The coefficient audit preceding equation (47) shows that every $D^j L_z^\phi$ has a one-high-norm estimate with loss two. Differentiating $L_z^\phi G_z^\phi = 1$ once gives

$$D G_z^\phi[\dot{z}] = -G_z^\phi(D L_z^\phi[\dot{z}]) G_z^\phi.$$

Induction gives the ordered-partition identity equation (53). Applying the uniform elliptic estimate to each factor and using the tame product inequality gives equation (54). Finally, $A_z^\#$ is a differential operator with the component orders displayed in section 22, while $\mathcal{B}_{z,*}^\phi$ has fixed component orders $(-2, 0, 0, 0)$. Their composition maps to E_q^ϕ ; Leibniz differentiation of that composition proves equation (52) and all derivative estimates for Q_z^ϕ . $\square$

30. Sobolev-level compatibility

Lemma 30.1 (One equation inverse on all levels). *If $f \in F_{r+1}$, then*

$$\iota_{\widehat{Z}} G_z^{\phi, (r+1)} f = G_z^{\phi, (r)} \iota_F f, \quad \iota_E Q_z^{\phi, (r+1)} f = Q_z^{\phi, (r)} \iota_F f.$$

Proof. Both multiplier expressions solve the same distributional equation

$$L_z^\phi u = f$$

defined by the single pseudodifferential expression equation (44). Injectivity makes the distributional solution unique, and elliptic regularity promotes it to $\widehat{Z}_{r+1}$. Applying the same fixed $\mathcal{B}_{z,*}^\phi A_z^\#$ proves compatibility of Q_z^ϕ . No Sobolev-dependent Riesz map, Hilbert adjoint, or minimum-norm complement appears. $\square$

31. Split constraint locus

Equation (50) gives the explicit splitting

$$E_q^\phi = \ker A_z \oplus \operatorname{im} Q_z^\phi. \quad (55)$$

Indeed,

$$v = (v - Q_z^\phi A_z v) + Q_z^\phi A_z v,$$

and applying A_z shows that the intersection is zero. The smooth-tame submersion theorem therefore gives a split smooth-tame zero locus

$$\mathcal{C}_{g+\phi}^{\operatorname{reg}} = \Phi_{g+\phi}^{-1}(0) \cap \mathcal{V}_\varepsilon$$

whose Hilbert-level charts agree under inclusion. This is a local regular chart; points with generalized KIDs remain outside it as singular strata.

32. Clause-by-clause ScalarReg3 occupancy

Certificate 32.1 (ScalarReg3). For a scalar-branch point z , set

$$\operatorname{ScalarReg3}(z) \iff \begin{cases} \Phi_{g+\phi}(z) = 0, \\ \ker D\Phi_{g+\phi}(z)^\# = 0, \\ \operatorname{im} D\Phi_{g+\phi}(z) \text{ is closed,} \\ D\Phi_{g+\phi}(z) \text{ has a compatible tame right inverse,} \\ \text{the pointed source lift is exact,} \\ \text{the Einstein–Klein–Gordon history is licensed.} \end{cases}$$

Certificate coordinate	Witness	Reason
Constraint zero	<i>equations</i> (24) and (25)	Direct hyperbolic/pure-trace algebra.
Adjoint kernel zero	<i>theorem</i> 23.1	Scalar preservation gives $N = 0$; hyperbolicity gives $X = 0$.
Closed range	<i>equation</i> (50)	Range is all of F_q , hence closed.
Compatible tame right inverse	Q_z^ϕ , <i>theorem</i> 29.1 and <i>lemma</i> 30.1	Fixed balanced normal equation; loss $\mu_\phi = 2$.
Exact pointed source lift	λ_ε , <i>proposition</i> 21.2	WDW-1 source map applied to the same coupled development.
Licensed history	$(M_\varepsilon, g_\varepsilon, \phi_\varepsilon^{\operatorname{dev}})$	Local EKG Cauchy theorem plus maximal globally hyperbolic development.

Theorem W4.4 (ScalarReg3 analytic occupancy). For every fixed $\varepsilon > 0$,

$$\operatorname{ScalarReg3}(z_\varepsilon).$$

Moreover, after shrinking around z_ε , there is a nonempty split smooth-tame regular constraint chart on which all six coordinates remain occupied for the licensed local Einstein–Klein–Gordon developments. The compatible right inverse is

$$Q_z^\phi = \mathcal{B}_{z,*}^\phi A_z^\# (L_z^\phi)^{-1}, \quad \mu_\phi = 2.$$

Proof. The table above gives an independent theorem and witness for each coordinate. Openness of invertibility follows from the uniform low-level Neumann argument; openness of the no-KID condition follows from the same energy estimate. Nearby smooth constraint data have local geometrically unique Einstein–Klein–Gordon developments by the same Cauchy theorem, and the WDW-1 source construction applies to each resulting history. Thus the passing set is nonempty and contains z_ε . Generalized-KID points are not removed from the global object; they remain singular strata outside this chart. $\square$

Nonclaim 32.2 (No use of the corrected scalar genericity statement). Neither a generic no-KID slogan nor the corrected Saraykar–Joshi claim is load-bearing. The proof uses one explicit coupled point, one displayed coupled symbol, and one fixed normal equation. Comparison results may indicate abundance but cannot substitute for these witnesses.

Part VII

WDW-4A: Occupied scalar characteristic and finite-refoliation closure

33. Promotion rule

WDW-3 proves implications of the form

$$\text{ScalarReg3}(z) \implies \mathcal{R}_j(z),$$

where $\mathcal{R}_j$ ranges over split regularity, characteristic equality, conormal closure, evaluation reconstruction, and local same-arrow closure. It does not assert the antecedent. Theorem W4.4 now supplies it at z_ε and on the smaller regular chart. This section performs only that licensed modus ponens.

Audit condition 33.1 (No ledger-only promotion). The scalar branch changes from CONDITIONAL to PROVED/OCCUPIED ON THE WDW-4A CHART only after Theorem W4.4. The machine dependency graph makes W4.4 a direct predecessor of W4.5.

34. Split scalar constraint surface

Let

$$\mathcal{C}_{g+\phi}^{\text{reg}} = \Phi_{g+\phi}^{-1}(0) \cap \mathcal{V}_\varepsilon.$$

The right inverse equation (49) and splitting equation (55) give

$$T_z \mathcal{P}_{g+\phi}^\infty = \ker D\Phi_{g+\phi}(z) \oplus \text{im } Q_z^\phi.$$

Thus the scalar total constraint surface is a split smooth-tame submanifold on the chart. Its intersection with the accepted WDW-2 spatial constraint surface is exact:

$$\mathcal{C}_{g+\phi}^{\text{reg}} \subset \mathcal{C}_{\text{si},g+\phi},$$

and the WDW-2 spatial characteristic/orbit/vertical equality is retained on its own tangent space. It is not re-proved by setting lapse zero in a theorem whose base is $\mathcal{C}_{g+\phi}^{\text{reg}}$.

35. Conormal Lie–Rinehart object

Let $\mathcal{A}$ be the accepted algebra of smooth regular local functionals and $\mathcal{I} \subset \mathcal{A}$ the ideal generated by the total scalar deformation constraints on the regular chart. WDW-3 proves that

$$\mathfrak{A}_{\text{con},g+\phi}^{\text{reg}} = \mathcal{I} / \mathcal{I}^2$$

carries the coefficient-sensitive conormal Lie–Rinehart/algebroid bracket induced by the off-shell hypersurface-deformation algebra. In particular, a coefficient (N, X) is not frozen as a constant token: its phase dependence contributes the Leibniz and anchor terms required by the quotient.

The occupied scalar chart supplies an actual base for this object. The bracket remains

$$\begin{aligned}\{C(0, X), C(0, Y)\} &= C(0, [X, Y]), \\ \{C(0, X), C(N, 0)\} &= C(\mathcal{L}_X N, 0), \\ \{C(N, 0), C(M, 0)\} &= C\left(0, h^{-1}(N dM - M dN)\right),\end{aligned}\tag{56}$$

with the same sign and density convention as WDW-3. Matter terms cancel in the total off-shell bracket exactly as already proved there; WDW-4A does not recalculate or alter equation (56).

36. Full characteristic equality

Let $\mathfrak{R}_{g+\phi}$ denote the accepted lapse–shift deformation anchor. At a passing regular point WDW-3 proves

$$\ker\left(\Omega_{g+\phi}|_{\mathcal{C}_{g+\phi}^{\text{reg}}}\right) = \text{im } \mathfrak{R}_{g+\phi}.\tag{57}$$

The forward inclusion follows from the first-class constraints. The reverse inclusion in WDW-3 uses a distributional multiplier, the fixed balanced normal problem, elliptic regularity, and weak nondegeneracy; it is conditional exactly on the right-inverse and history coordinates now supplied by W4.4. Therefore equation (57) is occupied on $\mathcal{V}_\varepsilon$.

Remark 36.1 (Singular complement). If $\ker A_z^\# \neq 0$, the normal operator is not invertible and the local result above does not apply. Such KID points remain singular groupoid objects with isotropy and possible second-order obstruction data. They are neither deleted nor identified with regular objects.

37. Direct finite path functor

Fix one licensed maximal development of a passing source fibre. Let $\mathcal{O}_{\text{emb}}$ be the guarded local orbit of admissible Cauchy embeddings and $\mathcal{P}_{\text{char}}$ the corresponding canonical characteristic plaque. Evaluation is the already-defined map

$$\text{Eval}(i) = \text{canonical data induced by } i.$$

On compatible shifted path spaces, WDW-3 defines directly

$$\text{Eval}^{\text{path}} : \Pi_1^{\text{guard}}(\mathcal{O}_{\text{emb}}) \longrightarrow \Pi_1^{\text{guard}}(\mathcal{P}_{\text{char}}), \quad [i_\bullet] \longmapsto [\text{Eval}(i_\bullet)].$$

The proof in WDW-3 checks path differentiation, characteristic image, homotopy preservation, representative independence, source and target, units, inversion, composition, restriction, presentation augmentation, every standing guard, and compatibility with the level inclusions. No infinite-dimensional Lie-II theorem is imported.

The guards retained on the WDW-4A scalar chart include:

1. fixed covariant development and pointed source fibre;
2. unchanged Module-A attachment and tensor representative;
3. compact-core boundary status;
4. proper, identity-connected embedding changes only;
5. no transition through a generalized-KID stratum;
6. unchanged relational record/readout coordinate for the use under comparison; and
7. unchanged WDW-4A branch target.

38. Nash–Moser local evaluation equivalence

On the smooth shifted tower, WDW-3 constructs a two-sided inverse to differentiated evaluation from the same constraint right inverse. In the present scalar chart, the replacement of the vacuum Q_z by the coupled Q_z^ϕ is already licensed by the conditional scalar theorem. Its smooth-tame estimates are theorem 29.1. Hence the projected evaluation map satisfies the hypotheses of Hamilton’s Nash–Moser inverse theorem [9].

After shrinking the embedding chart:

1. evaluation is a smooth-tame diffeomorphism from the embedding orbit to a characteristic plaque;
2. the plaque is a graph in a split constraint-surface chart;
3. both charts are contractible;
4. no generalized KID or finite local isotropy occurs; and
5. all source, attachment, boundary, relational, and branch guards are constant.

Every characteristic path in the plaque consequently has a unique local embedding-path lift. Fixed-endpoint source paths are homotopic relative endpoints; evaluation maps the homotopy to a guarded characteristic homotopy. The local monodromy groupoids reduce to pair groupoids and local foliation holonomy is trivial.

Proposition 38.1 (Local scalar same-arrow closure). *Within the declared regular contractible plaque, any two proper scalar refoliation routes with the same guarded endpoints determine the same canonical arrow.*

Proof. Lift both characteristic paths uniquely through the local evaluation diffeomorphism. Their embedding paths lie in a contractible guarded chart and are homotopic relative endpoints. Functoriality of $\text{Eval}^{\text{path}}$ identifies the two canonical path classes. $\square$

Nonclaim 38.2 (No global same-endpoint theorem). Proposition 38.1 does not identify routes in different maximal developments, through singular strata, across mapping-class or large sectors, across boundary-charge changes, or between relationally distinct records. It is a local pair-groupoid statement.

39. Source and observable descent

Proper paths in the chart change only slicing presentation. Therefore

$$\text{Src}_\phi(\gamma) = \hat{x}_\varepsilon$$

is constant along the guarded route; the attachment graph, witness, tensor representative, Ward coordinate, and reconstruction ancestry remain fixed. WDW-2 observable descent and WDW-3 relational coherence then imply that any admitted observable constant on the complete characteristic distribution descends to the local quotient. Gauge-variant instantaneous coordinates do not acquire primitive standing merely because the chart is regular.

Theorem W4.5 (Inherited scalar refoliation promotion). On the nonempty WDW-4A regular chart supplied by Theorem W4.4:

1. the total scalar constraint locus is split smooth-tame;
2. its characteristic distribution equals the complete lapse–shift distribution;
3. the coefficient-sensitive conormal Lie–Rinehart/algebroid object is populated;
4. the WDW-2 spatial quotient and WDW-3 total quotient agree on their exact registered intersection;
5. the direct guarded path/homotopy evaluation functor is defined;

6. evaluation is a local smooth-tame equivalence of embedding and characteristic plaques;
7. proper common-endpoint routes determine the same local canonical arrow; and
8. source, attachment, boundary, relational, singular, and branch guards are preserved.

The complement containing generalized-KID points remains singular, and no global smooth superspace or global refoliation group action is asserted.

Proof. Theorem W4.4 proves the complete antecedent of each accepted WDW-3 conditional scalar theorem. The conclusions in the first seven clauses are those exact implications, with the coupled tame inverse supplied by theorem 29.1. Source and observable preservation use the accepted WDW-2/WDW-3 descent theorems on the fixed pointed fibre. The locality and singular-stratum boundaries are premises of the imported results and are retained verbatim. $\square$

Part VIII

WDW-4A: A local failure-typed scalar clock

40. Timelike-gradient certificate

On the initial leaf of the licensed development,

$$D_i\phi_\varepsilon = 0, \quad \mathcal{L}_n\phi_\varepsilon = \frac{p_{\phi,\varepsilon}}{\sqrt{h}} = \varepsilon.$$

Therefore

$$g_\varepsilon^{\mu\nu}\nabla_\mu\phi_\varepsilon^{\text{dev}}\nabla_\nu\phi_\varepsilon^{\text{dev}} = -(\mathcal{L}_n\phi)^2 + |D\phi|_h^2 = -\varepsilon^2 < 0 \quad (58)$$

on $\iota_\varepsilon(\Sigma)$. Choose time orientation so that the inherited unit normal n is future-directed. Since $d\phi(n) = \mathcal{L}_n\phi = \varepsilon > 0$ in signature $(-+++)$, the timelike vector $\nabla\phi$ is past-directed there.

The scalar

$$\chi = g_\varepsilon^{-1}(d\phi_\varepsilon^{\text{dev}}, d\phi_\varepsilon^{\text{dev}})$$

is continuous. Since the initial leaf is compact and $\chi = -\varepsilon^2$ uniformly, there is an open spacetime neighborhood $\mathcal{U}_{\text{clk},\varepsilon}$ of that leaf on which

$$\chi < -\frac{\varepsilon^2}{2} < 0 \quad (59)$$

after shrinking if necessary. The gradient remains timelike and consistently oriented. The level sets of ϕ are therefore smooth spacelike hypersurfaces on this neighborhood.

Define

$$V_\phi = \frac{\nabla\phi}{g^{-1}(d\phi, d\phi)}. \quad (60)$$

Then

$$V_\phi(\phi) = d\phi(V_\phi) = \frac{g(\nabla\phi, \nabla\phi)}{g^{-1}(d\phi, d\phi)} = 1.$$

Because the denominator is negative and $\nabla\phi$ is past-directed, V_ϕ is future-directed. Along its integral curves, ϕ is strictly increasing and is a valid local parameter. Compactness gives one common short interval

$$I_\varepsilon = (-\delta_\varepsilon, \delta_\varepsilon)$$

whose ϕ -level sets cover a smaller neighborhood of the initial leaf.

41. Characteristic transversality

For a deformation descriptor (N, X) , the canonical action on the scalar is

$$\delta_{(N,X)}\phi = \mathcal{L}_X\phi + N \frac{p_\phi}{\sqrt{h}}. \quad (61)$$

At z_ε ,

$$\delta_{(N,0)}\phi = N\varepsilon.$$

Hence the gauge condition

$$\chi_\tau(z) = \phi - \tau = 0$$

is transverse to the normal part of the characteristic distribution whenever the admitted lapse does not vanish. Spatial diffeomorphisms tangent to a ϕ -level set are already handled by the WDW-2 proper quotient; they are not falsely fixed by the scalar clock.

On the neighborhood equation (59), the timelike gradient is nowhere zero. The local flow-box theorem for V_ϕ identifies the neighborhood with a short interval of level sets. In the regular canonical chart, the implicit-function theorem and equation (61) give a unique nearby intersection of each normal characteristic route with $\chi_\tau = 0$. Thus the clock is transverse in exactly the local, quotient-relative sense needed below.

42. Clock-domain tuple

The complete clock object is

$$\mathfrak{D}_{\text{clk}}^{(\varepsilon)} = \left(\mathcal{C}_\phi, \mathcal{U}_\phi, \mathcal{A}_\phi, \mathcal{Q}_\phi, \mathcal{R}_\phi, \mathcal{K}_\phi, \mathfrak{F}_{\text{clk}}^{(\varepsilon)} \right), \quad (62)$$

with coordinates:

1. **candidate** $\mathcal{C}_\phi = \phi_\varepsilon^{\text{dev}}$;
2. **use** $\mathcal{U}_\phi$: local ordering and labeling of the spacelike level sets $\phi = \tau$, $\tau \in I_\varepsilon$;
3. **anchors** $\mathcal{A}_\phi = (\hat{x}_\varepsilon, \iota_\varepsilon(\Sigma), \tau_0 = 0, \text{time orientation})$;
4. **quotient profile** $\mathcal{Q}_\phi$: the WDW-2 proper spatial quotient followed by the occupied WDW-3 local characteristic quotient;
5. **record/readout** $\mathcal{R}_\phi$: the scalar value τ recorded at the unique local characteristic intersection, together with the chosen relational observable values;
6. **continuation** $\mathcal{K}_\phi$: the flow of V_ϕ , restricted to the licensed regular source fibre and the interval I_ε ; and
7. **failure** $\mathfrak{F}_{\text{clk}}^{(\varepsilon)}$: the explicit boundary in section 44.

This is the specialization of the accepted physical-clock quotient, not an assertion that a bare scalar coordinate is intrinsically a clock [10].

43. Relational observable descent

Let F be an admitted smooth local functional on the regular scalar constraint chart. For $\tau \in I_\varepsilon$, define

$$\mathcal{O}_{F|\phi}(\tau)([z]) = F(z_\tau), \quad (63)$$

where z_τ is the unique local representative on the guarded characteristic route through z satisfying $\phi = \tau$, after the WDW-2 spatial quotient. This definition is valid only in the clock domain equation (62).

Proposition 43.1 (Local quotient descent). *$\mathcal{O}_{F|\phi}(\tau)$ is independent of the chosen proper refoliation path within the contractible guarded chart and therefore descends to the occupied local characteristic quotient.*

Proof. Transversality gives a unique local gauge-surface intersection. If two proper paths connect the same guarded endpoints, proposition 38.1 identifies their canonical arrows. They therefore select the same z_τ in the spatially reduced chart. Equation (63) has the same value. Source, attachment, boundary, and branch coordinates remain constant by Theorem W4.5. $\square$

Remark 43.2 (Use-relative readout). If no operational record/readout is attached to a proposed F , the construction is restricted to relational conditioning and is not promoted to an observed clock record. The clock ledger distinguishes those two uses rather than assuming measurement completion.

44. Failure boundary

The certified clock domain ends at the first occurrence of any of the following:

1. $g^{-1}(d\phi, d\phi) = 0$ (null-gradient boundary);
2. $g^{-1}(d\phi, d\phi) > 0$ (spacelike gradient);
3. $d\phi = 0$, equivalently loss of the flow equation (60);
4. vanishing normal speed $p_\phi/\sqrt{h}$ in the selected foliation;
5. loss of transversality to the normal characteristic distribution;
6. more than one guarded intersection with the same $\phi = \tau$ surface;
7. failure of the selected record/readout attachment;
8. exit from the occupied ScalarReg3 chart or encounter with generalized-KID isotropy;
9. source, attachment, boundary-charge, relational-record, or branch change; or
10. termination of the licensed local development or of the common interval I_ϵ .

For $m > 0$, the evolution equation can produce turning points; these are expected failure events and are not concealed by the positive initial momentum. The theorem does not assert that $\mathfrak{F}_{\text{clk}}^{(\epsilon)}$ is never reached.

45. Compatible local clock atlases

Suppose a second clock candidate ψ is valid on an overlapping regular domain. A licensed overlap map must be a smooth strictly monotone reparametrization

$$\psi = f_{\psi\phi}(\phi)$$

on each connected overlap, preserve the registered time orientation, map guarded quotient classes to guarded quotient classes, and transport readout and failure coordinates. On triple overlaps it must satisfy the cocycle relation

$$f_{\chi\psi} \circ f_{\psi\phi} = f_{\chi\phi}.$$

No claim is made that all local clocks admit one global atlas. The overlap conditions state what would be required.

Proposition 45.1 (No preferred foliation). *The certified scalar clock does not select a primitive preferred foliation.*

Proof. The clock is defined only after fixing a use, a regular quotient chart, a source fibre, and a failure domain. Any lawful alternative satisfying the overlap requirements describes the same quotient relations on its overlap. The scalar level sets are therefore one admitted local presentation class, not a new primitive standing coordinate. Outside their failure boundary the theorem is silent rather than exclusive. $\square$

Theorem W4.6 (Local material-clock realization). The occupant z_ϵ carries a nonempty local clock domain $\mathfrak{D}_{\text{clk}}^{(\epsilon)}$ such that:

1. $d\phi$ is timelike and consistently oriented;
2. the level sets $\phi = \tau$ form local spacelike slices;
3. ϕ is strictly increasing along the future-directed V_ϕ ;
4. the scalar gauge is transverse to normal characteristic motion;
5. admitted relational observables descend by equation (63);
6. a record/readout role is retained or the use is explicitly restricted to relational conditioning;
7. the complete failure boundary is registered; and
8. no global clock or preferred foliation is produced.

Proof. Equations (58) to (60) prove timelikeness, openness, slicing, and monotonicity. Equation (61) and the local flow box prove transversality and unique local intersections. Proposition 43.1 proves quotient descent. The domain tuple equation (62) retains use and readout coordinates, while section 44 supplies the failure boundary. Proposition 45.1 prevents local success from becoming a primitive foliation claim. $\square$

Nonclaim 45.2 (No quantum problem-of-time solution). The theorem supplies no physical inner product, quantum observable algebra, quantum constraint operator, or quantum clock. It is a classical local quotient statement.

Part IX

WDW-4A: Branch-typed handoff and capstone closure

46. Final branch statuses

Object	Publication status	Boundary
Klein–Gordon covariant/canonical identity	IMPORTED/VERIFIED	Exact WDW-1 model and Module-A source interface.
Explicit coupled constraint datum	PROVED/OCCUPIED	$z_\varepsilon, \varepsilon > 0$.
ScalarReg3	PROVED/OCCUPIED	Nonempty local regular chart around z_ε .
Scalar characteristic/refoliation quotient	PROVED/OCCUPIED LOCALLY	Accepted WDW-3 conclusions on the populated chart.
Scalar material clock	PROVED/OCCUPIED LOCALLY	Timelike-gradient domain before $\mathfrak{F}_{\text{clk}}^{(\varepsilon)}$.
Global scalar clock	NOT CLAIMED	Turning points and other failures remain possible.
WDW-4B Standard Model enrichment	PROVED IN MODULE 4B	Not a WDW-4A premise; see Theorem W4B.12.
WDW-5 quantum construction	OUT OF SCOPE	No functional derivative, regulator, ordering, anomaly, inner product, or quantum clock.

47. The exact WDW-5 input

Define

$$\mathfrak{H}_{\text{W4A}} = \left(\mathcal{P}_{g+\phi}^\infty, \Omega_{g+\phi}, \Phi_{g+\phi}, \mathcal{C}_{g+\phi}^{\text{reg}} // \mathbb{C}_{g+\phi}^{\text{char}}, \mathfrak{A}_{\text{con}, g+\phi}^{\text{reg}}, \overline{\text{Src}}_\phi, \mathcal{U}_{\text{clk}}, \mathfrak{F}_{\text{clk}}^{(\varepsilon)}, \mathfrak{G}_{\text{branch}} \right). \quad (64)$$

The coordinates are:

1. the exact smooth coupled scalar–gravity phase carrier;
2. its weak canonical symplectic form;
3. the complete total deformation–constraint map;
4. the occupied local characteristic quotient, represented groupoid-faithfully;
5. the coefficient-sensitive conormal Lie–Rinehart/algebroid object;
6. the descended structured source map on the fixed pointed fibres;
7. the family of certified local clock domains;
8. their failure ledger; and
9. the branch guard retaining the singular complement, WDW-4B boundary, compact-core boundary status, and all global/large-sector exclusions.

WDW-5 may use this object as a classical configuration-space input. It must still construct and type, rather than assume:

1. a connection or differential calculus on the groupoid/local-atlas configuration object;
2. functional derivatives with their domains and covariance;
3. a measure, rigging map, or other inner-product structure;
4. regulator and renormalization data;
5. factor ordering;
6. a quantum constraint algebra and anomaly test;

7. domains and self-adjointness properties of any operator; and
8. the relation, if any, between classical local clock conditioning and a quantum clock.

Theorem W4.7 (Branch-typed WDW-5 handoff). $\mathfrak{H}_{W4A}$ is a nonempty, source-attached, branch-typed classical handoff. Every coordinate in equation (64) has standing on the WDW-4A regular scalar chart, and every declared failure, singular, boundary, and WDW-4B guard is retained. The handoff contains no unproved functional derivative, Wheeler–DeWitt operator, regulator, ordering, anomaly theorem, physical inner product, or quantum clock.

Proof. The phase, symplectic, constraint, and source coordinates are W4.2–W4.3. The regular quotient and conormal coordinates are W4.4–W4.5. The clock domain and failure ledger are W4.6. Nonemptiness follows from z_ε . The branch guard is explicit in W4.1, W4.5, and the final status table. None of the eight listed WDW-5 burdens is constructed in W4.1–W4.6, so none is inserted into $\mathfrak{H}_{W4A}$. $\square$

48. Capstone theorem

Theorem W4.8 (Core scalar/clock matter occupancy and target separation). On the frozen compact, smooth, boundaryless, minimal metric branch with $\Lambda = 0$, $\kappa > 0$, and a minimally coupled real scalar of $V(\phi) = m^2\phi^2/2$, $m^2 \geq 0$, there exists a nonempty class of exact source-attached canonical data such that:

1. the complete Einstein–Klein–Gordon constraints hold;
2. the data are genuinely nonvacuum;
3. all generalized KIDs vanish;
4. the complete coupled constraint map has a fixed Sobolev-compatible, smooth-tame right inverse with coefficient loss two;
5. every coordinate of ScalarReg3 is occupied;
6. the accepted WDW-2 spatial and WDW-3 characteristic, conormal, path-functor, Nash–Moser, and local same-arrow conclusions are occupied on a regular chart;
7. the same class supports a nonempty local time-oriented timelike-gradient scalar-clock quotient with relational descent and a complete failure boundary; and
8. the resulting object supplies the branch-typed classical handoff $\mathfrak{H}_{W4A}$ while remaining distinct from WDW-4B and every quantum Wheeler–DeWitt construction.

Proof. W4.1 fixes the target, exact imports, separation, and acyclic order. W4.2 identifies the covariant source and canonical scalar as one inherited model. W4.3 supplies the exact nonvacuum datum and its licensed source history. W4.4 proves all six ScalarReg3 coordinates. W4.5 applies the accepted conditional quotient and refoliation theorems only after that occupancy. W4.6 constructs the local clock domain and failure profile. W4.7 assembles exactly the certified classical handoff. Every capstone clause is therefore a direct descendant of an earlier proved gate, and the dependency graph contains no reverse edge. $\square$

Conventional mathematical result. The explicit compact-hyperbolic Einstein–Klein–Gordon initial datum solves both constraints, has no generalized KIDs, admits a fixed elliptic balanced normal inverse, and develops to a local and maximal globally hyperbolic source history.

AASC result. The physical object is retained with its complete admissibility, standing, reference, source, attachment, comparison, singular, boundary, relational, and failure coordinates. Proper local refoliation changes presentation without multiplying terminal standing; standing-bearing differences are not quotiented away.

ARC result. The missing WDW-3 scalar antecedent is populated, the scalar local refoliation branch is occupied, and WDW-5 receives a certified classical matter object. WDW-4B remains a separate enrichment rather than an interpretive appendage or an illicit premise.

49. Exact nonclaims

The capstone does not establish:

1. a global material clock or global scalar monotonicity;
2. a preferred foliation;
3. geodesic completeness or singularity avoidance;
4. regularity at generalized-KID points;
5. a global smooth superspace;
6. global same-endpoint path independence;
7. a global Hamiltonian group action of refoliations;
8. a deparametrized physical Hamiltonian;
9. a configuration-space connection or functional differential calculus;
10. a Wheeler–DeWitt operator;
11. factor ordering, regularization, or renormalization;
12. quantum constraint closure or anomaly freedom;
13. a physical inner product or quantum observable algebra;
14. a quantum clock theorem;
15. a full Standard Model matter target;
16. tetrad, spin-connection, local-Lorentz, or fermionic closure;
17. transport of flat-space Poincaré/CPT or anomaly results into dynamical gravity;
18. boundary or asymptotic charges beyond the accepted compact core; or
19. torsionful, metric-affine, Palatini, teleparallel, or higher-jet branches.

50. Conclusion

The WDW-4A burden was an occupancy problem. WDW-3 already supplied the conditional scalar algebraic and geometric machinery, but correctly refused to treat its antecedent as automatic. The data equation (20) close that gap with one exact, nonvacuum, source-attached construction. The nonzero normal scalar momentum simultaneously balances the Hamiltonian constraint, destroys the lapse component of every generalized KID, and makes the scalar a local timelike-gradient clock. The complete scalar first-order symbol is retained in the normal-operator proof, and one fixed equation inverse supplies compatibility across the smooth Sobolev tower.

The result is deliberately local where the mathematics is local and deliberately branch-typed where later choices remain. It occupies the scalar core required before configuration-space quantization can be posed faithfully. It does not prove WDW-4B by itself and does not pre-answer the quantum theorem object of WDW-5.

Part X

WDW-4A: Audit appendices

A. Sign and density derivations

A.1 Normal decomposition

With signature $(-+++)$,

$$\partial_t = Nn + N^i \partial_i, \quad \mathcal{L}_n \phi = \frac{1}{N}(\dot{\phi} - N^i D_i \phi).$$

The scalar kinetic contraction is

$$g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi = -(\mathcal{L}_n \phi)^2 + D_i \phi D^i \phi.$$

Substitution into the covariant action gives

$$\mathcal{L}_\phi = N\sqrt{h} \left[\frac{1}{2}(\mathcal{L}_n \phi)^2 - \frac{1}{2}D_i \phi D^i \phi - \frac{1}{2}m^2 \phi^2 \right],$$

and differentiation with respect to $\dot{\phi}$ gives the weight-one density

$$p_\phi = \sqrt{h} \mathcal{L}_n \phi.$$

The Legendre transform is

$$\begin{aligned} p_\phi \dot{\phi} - \mathcal{L}_\phi &= p_\phi \left(N \frac{p_\phi}{\sqrt{h}} + N^i D_i \phi \right) \\ &\quad - N\sqrt{h} \left[\frac{1}{2} \left(\frac{p_\phi}{\sqrt{h}} \right)^2 - \frac{1}{2}D_i \phi D^i \phi - \frac{1}{2}m^2 \phi^2 \right] \\ &= N \left[\frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} D_i \phi D^i \phi + \frac{\sqrt{h}}{2} m^2 \phi^2 \right] + N^i p_\phi D_i \phi. \end{aligned}$$

This verifies both the sign and the density weight of $\mathcal{H}_\perp^\phi, \mathcal{H}_i^\phi$.

A.2 Stress projections

Since

$$n^\mu \partial_\mu \phi = \frac{p_\phi}{\sqrt{h}},$$

direct contraction of equation (7) gives

$$T(n, n) = \frac{1}{2} \left(\frac{p_\phi}{\sqrt{h}} \right)^2 + \frac{1}{2} |D\phi|^2 + \frac{1}{2} m^2 \phi^2.$$

The mixed projection is

$$-h_i^\mu T_{\mu\nu} n^\nu = -\frac{p_\phi}{\sqrt{h}} D_i \phi.$$

Multiplication by $\sqrt{h}$ therefore gives

$$\sqrt{h} \rho_\phi = \mathcal{H}_\perp^\phi, \quad -\sqrt{h} j_i^\phi = \mathcal{H}_i^\phi.$$

At the occupant, $D\phi = 0$ and $p_\phi/\sqrt{h} = \varepsilon$, giving $\rho = \varepsilon^2/2$, $j = 0$.

A.3 Gravitational convention

The convention

$$K_{ij} = +\frac{1}{2}\mathcal{L}_n h_{ij}$$

implies

$$\pi^{ij} = \frac{\sqrt{h}}{2\kappa}(K^{ij} - Kh^{ij}).$$

For $K_{ij} = c_\varepsilon h_{ij}$, $K = 3c_\varepsilon$, so

$$\pi^{ij} = -\frac{c_\varepsilon \sqrt{h}}{\kappa} h^{ij}.$$

This is the WDW-1/WDW-3 sign and is not the opposite extrinsic-curvature convention often used in ADM references.

B. Module-A scalar attachment and Ward certificate

Coordinate	WDW-4A occupant	Closure
Response presentation	$\mathfrak{p}_{\phi_\varepsilon^{\text{dev}}}$	Metric response of the covariant action equation (5).
Attachment graph	$\text{Att}_{\phi_\varepsilon^{\text{dev}}}$	Same scalar and metric variables as the licensed development.
Complete witness	$W_{\phi_\varepsilon^{\text{dev}}}$	Smooth field, action variation, tensor, field equation, and regularity data.
Tensor representative	$q_g(\widehat{x}_\varepsilon) = T^\phi[g_\varepsilon, \phi_\varepsilon^{\text{dev}}]$	Operational-Hilbert uniqueness.
Canonical projections	$\mathcal{H}_\perp^\phi = \sqrt{h}\rho_\phi$, $\mathcal{H}_i^\phi = -\sqrt{h}j^\phi$	Direct tensor projection, not backward definition.
Ward coordinate	$\nabla_\mu T^{\mu\nu} = 0$	$\square_g \phi - m^2 \phi = 0$ and equation (8).
Boundary coordinate	Compact spatial core; fixed temporal endpoints	No spatial flux term in the variation.
History fibre	$\text{Src}_\phi(\gamma_\varepsilon) = \widehat{x}_\varepsilon$	Exact WDW-1 pointed lift.

The table prevents two shortcuts: a bare stress tensor is not a complete source class, and a canonical initial point is not a complete source history. The licensed development supplies the missing history coordinate.

C. Exact import excerpts and non-overclaim boundaries

ID	Imported clause	Specialization	Non-overclaim
IMP-W1	Strong scalar source/canonical lift, including action, regular Legendre map, scalar constraints, Dirac ledger, Ward match, and Src_ϕ .	Same real massive scalar and compact core.	No ScalarReg3, refoliation, local clock, or quantum operator.
IMP-W2	Proper spatial moment map, action-groupoid descent, regular spatial characteristic equality, and source/observable descent.	Apply to the scalar spatial constraint surface.	No Hamiltonian self-bracket or normal regularity.

ID	Imported clause	Specialization	Non-overclaim
IMP-W3	Massive scalar off-shell HDA and conditional implications from populated ScalarReg3 to regular characteristic and finite refoliation closure.	Antecedent supplied only by W4.4.	No global group action, global superspace, or scalar occupancy on entry.
IMP-COV	Identity-principal-symbol derivative classification and minimal metric Levi–Civita closure.	$d\phi$ is scalar first jet; $\square_g$ and ∇T use Levi–Civita.	No superspace connection or independent matter choice.
IMP-MA	Attachment-safe response representation, unique tensor, Ward/composite closure, and scalar witness.	Use complete $(\mathfrak{p}, \text{Att}, W)$ class.	No canonical split inferred from the tensor alone.
IMP-CLK	Clock-domain tuple, use-relative standing, local realization, failure typing, and overlap guards.	Timelike-gradient scalar patch only.	No global, preferred, or quantum clock.
EXT-HYP	Existence and geometry of closed hyperbolic three-manifolds.	Choose one torsion-free cocompact Γ .	No rigidity statement beyond the used curvature/no-Killing facts.
EXT-EKG	Local smooth well-posedness and geometric uniqueness for the Einstein–Klein–Gordon Cauchy system.	Smooth compact constraint data in harmonic gauge.	No long-time existence, completeness, or global monotonicity.
EXT-CBG	Existence and uniqueness of a maximal globally hyperbolic development.	Apply after local existence and geometric uniqueness.	No geodesic completeness or maximal extension outside the globally hyperbolic category.
EXT-PSIDO	Compact elliptic pseudodifferential parametrix and regularity.	Apply to the square balanced L_z^ϕ .	No claim that the raw overdetermined A_z is Fredholm.
EXT-NM	Hamilton smooth-tame inverse/submersion results.	Used only after the explicit tame inverse family exists.	The theorem does not supply missing estimates.

D. Complete generalized-KID audit

Component	Equation	At z_ε
Metric	$\mathcal{L}_X h_{ij} + 2N K_{ij} = 0$	After $N = 0$, $\mathcal{L}_X h = 0$.
Extrinsic curvature	<i>equation (28)</i>	Automatically zero after $N = X = 0$.
Scalar configuration	$\mathcal{L}_X \phi + N \varpi = 0$	$N \varepsilon = 0$, hence $N = 0$.
Scalar momentum	$D_i(N\sqrt{h}D^i\phi) + D_i(X^i p_\phi) - N\sqrt{h}m^2\phi = 0$	Automatically zero after $N = X = 0$.
Hyperbolic shift	$\mathcal{L}_X h = 0$	Bochner identity gives $X = 0$.

All four preservation equations are required to identify the full adjoint kernel. Only two are needed

sequentially to eliminate it at the occupant.

E. Weighted symbol and row-balancer audit

Constraint row	γ	k	φ	q	Exact scalar principal term at z_ε
Hamiltonian	2	0	1	0	φ -term vanishes; $-2\kappa\varepsilon q/\sqrt{h}$ is order zero.
Momentum	1	1	1	0	$-\kappa\varepsilon D_i\varphi$, genuinely order one.

The balancer has component orders

$$(-2, 0, 0, 0)$$

on (γ, k, φ, q) . Hence the principal normal contributions are:

$$h_2\Lambda_{*,S^2}^{-2}h_2^* \quad (\text{order } 2), \quad m_1m_1^* \quad (\text{order } 2), \quad s_\phi s_\phi^* \quad (\text{order } 2).$$

The last term is positive semidefinite. Surjectivity of the gravitational symbol already gives a strictly positive lower bound. The raw differential composition $A_z A_z^\#$, whose rows have unequal natural orders, is not inverted on the equal-row target.

F. Tame-estimate and derivative-loss ledger

Map	Level type	Loss/control	Reason
A_z	$E_q^\phi \rightarrow F_q$	Built into shifts 2, 1, 1, 0.	Curvature, divergence, scalar first jet, momentum density.
$A_z^\#$	$\widehat{Z}_q \rightarrow \widetilde{E}_q^\phi$	Formal differential orders 2, 1, 1, 0.	Complete adjoint table.
$\mathcal{B}_{z,*}^\phi$	$\widetilde{E}_q^\phi \rightarrow E_q^\phi$	Fixed component orders $(-2, 0, 0, 0)$.	Sobolev-index-independent auxiliary spectral operator.
L_z^ϕ	$\widehat{Z}_q \rightarrow F_q$	Order two; coefficient loss $\mu_\phi = 2$.	Metric curvature is the largest coefficient derivative.
G_z^ϕ	$F_q \rightarrow \widehat{Z}_q$	<i>equation</i> (51), loss two.	Uniform parameter-elliptic estimate.
$D^m G_z^\phi$	Multilinear parameter family	<i>equations</i> (53) and (54).	Ordered partitions from differentiating $LG = 1$.
Q_z^ϕ	$F_q \rightarrow E_q^\phi$	<i>equation</i> (52), loss two.	Composition $\mathcal{B}_{z,*}^\phi A_z^\# G_z^\phi$.
Level inclusions	$q + 1 \rightarrow q$	Exact commutation.	Uniqueness for one distributional equation and elliptic regularity.

G. Physical-clock certificate

Coordinate	Witness	Failure test
Candidate	$\phi_\varepsilon^{\text{dev}}$	$d\phi = 0$ or loss of source identity.
Use	Local slice label/order $\phi = \tau$	No unique local level-set use.
Anchor	$(\hat{x}_\varepsilon, \Sigma, \tau_0 = 0, \text{orientation})$	Source, initial surface, or orientation changes.
Timelikeness	$g^{-1}(d\phi, d\phi) < -\varepsilon^2/2$	Null or spacelike gradient.
Transversality	$\delta_{(N,0)}\phi = Np_\phi/\sqrt{h}$	Vanishing normal speed or rank loss.
Quotient profile	WDW-2 spatial then WDW-3 characteristic quotient	Exit from regular chart or guarded branch.
Readout	τ plus admitted relational values	Attachment or operational readout failure.
Continuation	Future-directed increasing- ϕ flow of V_ϕ on I_ε	Turning point, multiple intersection, or end of interval.
Overlap	Smooth orientation-preserving monotone map with cocycle	Noninvertible transition or cocycle failure.

H. WDW-3 status-promotion trace

WDW-3 scalar conclusion	Status on entry	WDW-4A antecedent	Publication status
Off-shell HDA	Proved	Exact import only	Proved
ScalarReg3	Conditional; unoccupied	W4.3–W4.4	Proved/occupied
Split total constraint locus	Conditional	Q_z^ϕ , W4.4	Proved locally
Full characteristic equality	Conditional	Populated ScalarReg3	Proved locally
Conormal Lie–Rinehart object	Conditional	Split regular base	Occupied locally
Direct path/homotopy functor	Conditional	Licensed history and regularity	Occupied locally
Nash–Moser evaluation equivalence	Conditional	Smooth-tame inverse family	Proved locally
Same canonical arrow	Conditional	Contractible plaque/no KID	Proved locally
Global quotient/group action	Not claimed	None	Not claimed

I. WDW-4B interface-separation matrix

Carrier/burden	WDW-4A	Disposition
Real scalar and density momentum	Present	Core theorem object.
Metric ADM pair	Present	Accepted WDW-1 branch.
Internal gauge bundle/group	Absent	WDW-4B.
Tetrad and spin structure	Absent	WDW-4B.
Spin connection/local Lorentz constraint	Absent	WDW-4B.
Fermionic canonical and second-class sectors	Absent	WDW-4B.
Nonabelian Gauss constraints/reducibility	Absent	WDW-4B.
Global form, line sectors, anomaly ledger	Absent	WDW-4B.
Curved-spacetime CPT transport	Absent	Separately proved or guarded in WDW-4B.
Quantum Wheeler–DeWitt data	Absent	WDW-5, not WDW-4B rescue.

J. Acceptance conditions

ID	Condition	Proof surface	Status
AC4A-01	Exact accepted artifact identities are recorded.	Section 9	PASS
AC4A-02	Every imported theorem has source, specialization, and boundary.	Appendix C	PASS
AC4A-03	WDW-1 scalar variables, signs, and densities are unchanged.	Section 10 and appendix A	PASS
AC4A-04	WDW-2 spatial conclusions are correctly attributed.	W4.1, W4.5	PASS
AC4A-05	WDW-3 scalar regular conclusions remain conditional before occupancy.	W4.1, W4.5	PASS
AC4A-06	Module A is used with complete attachment and witness coordinates.	Section 13 and appendix B	PASS
AC4A-07	Physical Clock results are specialized only to the local patch.	W4.6	PASS
AC4A-08	The scalar action is declared.	Equation (5)	PASS
AC4A-09	Connection ancestry is explicit.	Proposition 12.1	PASS
AC4A-10	Canonical variables and all constraint contributions are exact.	W4.2	PASS
AC4A-11	The complete scalar Dirac ledger is present.	W4.2	PASS
AC4A-12	Stress and source correspondence is exact.	Equations (6) and (7)	PASS
AC4A-13	The Ward identity follows from the scalar equation.	Lemma 13.1	PASS
AC4A-14	Scalar energy positivity holds for $m^2 \geq 0$.	Equation (13)	PASS
AC4A-15	The coupled hyperbolic datum solves both constraints.	Equations (24) and (25)	PASS
AC4A-16	The coupled datum is genuinely nonvacuum.	Proposition 20.1	PASS
AC4A-17	Every generalized-KID equation is displayed.	Equations (27) to (30)	PASS
AC4A-18	Scalar preservation forces lapse zero.	Theorem 23.1	PASS
AC4A-19	Compact hyperbolicity forces the shift/Killing field zero.	Lemma 18.1 and theorem 23.1	PASS
AC4A-20	The complete coupled principal symbol is displayed.	Equation (42)	PASS
AC4A-21	Scalar terms are correctly typed by order.	Equations (34) and (35) and appendix E	PASS
AC4A-22	The fixed row balancer is Sobolev-level compatible.	Equation (43) and lemma 30.1	PASS
AC4A-23	The normal operator is self-adjoint elliptic Fredholm of index zero.	Proposition 27.1 and corollary 28.2	PASS
AC4A-24	The energy identity identifies the kernel with generalized KIDs.	Lemma 28.1	PASS
AC4A-25	The Green family has declared tame estimates.	Equation (51)	PASS
AC4A-26	All parameter derivatives of the Green family are controlled.	Equations (53) and (54)	PASS
AC4A-27	Sobolev inclusion compatibility is proved.	Lemma 30.1	PASS
AC4A-28	The pointed source lift is exact.	Proposition 21.2	PASS
AC4A-29	The Einstein–Klein–Gordon history is licensed.	Section 21	PASS
AC4A-30	ScalarReg3 is populated on a nonempty class.	W4.4	PASS

ID	Condition	Proof surface	Status
AC4A-31	WDW-3 scalar promotion occurs only after AC4A-30.	W4.4 $\rightarrow$ W4.5	PASS
AC4A-32	Characteristic equality and conormal closure are inherited exactly.	W4.5	PASS
AC4A-33	Direct path functor and local same-arrow theorem are inherited exactly.	W4.5	PASS
AC4A-34	The scalar gradient is timelike on the initial leaf.	Equation (58)	PASS
AC4A-35	A nonempty local timelike-gradient patch is proved.	Equation (59)	PASS
AC4A-36	Clock transversality and quotient descent are proved.	Proposition 43.1	PASS
AC4A-37	The clock failure boundary is explicit.	Section 44	PASS
AC4A-38	No global or preferred clock is claimed.	Proposition 45.1	PASS
AC4A-39	WDW-4A and WDW-4B remain distinct theorem objects.	W4.1, Appendix I	PASS
AC4A-40	WDW-5 receives only the certified classical branch-typed object.	W4.7	PASS

K. Adversarial-map adjudication

ID	Hostile attack	Exact closure	Status
ADV4A-01	AASC generated the Klein–Gordon action.	False dependency reading; the action is a declared downstream occupant, W4.1–W4.2.	CLOSED
ADV4A-02	The scalar source is inferred backward from ADM.	Module-A class equation (6) and Src_ϕ precede/certify the split.	CLOSED
ADV4A-03	The stress tensor is assumed.	Metric variation and operational-Hilbert uniqueness, equation (7).	CLOSED
ADV4A-04	The scalar Hamiltonian has no covariant ancestor.	Action, Legendre transform, and stress projections in W4.2 and appendix A.	CLOSED
ADV4A-05	The scalar derivative imports an unexplained connection.	$d\phi$ is the first jet; $\square_g, \nabla T$ use the frozen Levi–Civita branch.	CLOSED
ADV4A-06	Constraint occupancy is perturbative hand-waving.	Exact identities equations (24) and (25).	CLOSED
ADV4A-07	The occupant is actually vacuum.	$p_\phi = \varepsilon\sqrt{h} \neq 0$, $\rho = \varepsilon^2/2 > 0$.	CLOSED
ADV4A-08	No-KID is imported from the vacuum point.	Scalar preservation gives $N = 0$ directly at z_ε , then $X = 0$.	CLOSED
ADV4A-09	Generic no-KID data do not prove occupancy.	One exact point is constructed; no genericity premise.	CLOSED
ADV4A-10	Saraykar–Joshi supplies the missing theorem.	Comparison only; not a dependency of W4.4.	CLOSED
ADV4A-11	Closed range follows only from no-KID.	$A_z Q_z^\phi = 1$, so the range is the complete closed target.	CLOSED
ADV4A-12	Overdetermined elliptic is called Fredholm.	Fredholmness is claimed only for square balanced L_z^ϕ .	CLOSED
ADV4A-13	The right inverse changes with Sobolev level.	One equation inverse and uniqueness, lemma 30.1.	CLOSED
ADV4A-14	Scalar terms destroy the WDW-3 symbol.	Full symbol equation (42); scalar normal term is positive semidefinite.	CLOSED
ADV4A-15	A ledger label promotes ScalarReg3.	Six independent witnesses in W4.4.	CLOSED
ADV4A-16	WDW-3 already proved scalar occupancy.	WDW-3 antecedent is recorded as conditional until W4.4.	CLOSED
ADV4A-17	The scalar clock is only a coordinate choice.	Timelikeness, use, transversality, quotient descent, readout, and failure are proved in W4.6.	CLOSED
ADV4A-18	Positive momentum proves a global clock.	Only $\mathcal{U}_{\text{clk},\varepsilon}$ is certified; turning points are failures.	CLOSED
ADV4A-19	The local clock selects a preferred foliation.	Use-relative quotient class and guarded overlap maps, proposition 45.1.	CLOSED
ADV4A-20	Clock success solves the quantum problem of time.	Explicit quantum nonclaim after W4.6 and in W4.7.	CLOSED
ADV4A-21	The scalar result is the Standard Model.	WDW-4B separation theorem and appendix I.	CLOSED
ADV4A-22	Flat-space CPT or anomaly closure transfers automatically.	No such import occurs in the dependency ledger.	CLOSED
ADV4A-23	The local quotient is a global superspace.	W4.5 retains groupoid/local-chart and global nonclaim boundaries.	CLOSED
ADV4A-24	KID points are deleted.	They remain singular strata outside $\mathcal{V}_\varepsilon$.	CLOSED

ID	Hostile attack	Exact closure	Status
ADV4A-25	The WDW-5 operator is determined.	W4.7 transfers only $\mathfrak{H}_{W4A}$ and lists every open quantum burden.	CLOSED

L. Publication-consistency correspondence

The theorem dependency, scalar regularity, Sobolev-loss, claim-status, acceptance, and adversarial surfaces above are read together. A status promotion is invalid if it lacks its theorem dependency, if the right-inverse construction reintroduces a Sobolev-dependent Hilbert adjoint, if the scalar principal column disappears, or if a local clock is reported globally.

MODULE 4B

**Torsion-Free Tetrad/Spin/Gauge Standard
Model Occupancy**

Part XI

WDW-4B: Frozen target, dependency order, and imports

1. The downstream WDW-4B theorem object

WDW-4A proves a scalar-matter theorem. It supplies a smooth compact hyperbolic Einstein–Klein–Gordon constraint datum z_0 , a nonempty regular chart $\mathcal{V}_0$, a compatible smooth-tame right inverse

$$R_z : F_q \longrightarrow E_q^{4A}, \quad D\Phi_{4A}(z)R_z = 1, \quad \mu_{4A} = 2,$$

and a local scalar clock [11]. It contains no Standard Model phase carrier. In particular it proves neither a coframe lift nor local Lorentz reduction, internal Gauss law, chiral fermionic second-class block, Standard Model anomaly statement, or Standard-Model-coupled quantum operator.

Module 4B is downstream of that accepted object. Its dependency order is

$$\begin{aligned} &\text{non-degenerate determinate construction} \implies \text{AASC kernel necessity} \implies \text{fixed realized target} \\ &\implies \text{WDW-1} \implies \text{WDW-2} \implies \text{WDW-3} \implies \text{WDW-4A} \implies \text{W4B.1} \implies \cdots \implies \text{W4B.12}. \end{aligned}$$

No implication is reversible merely because a downstream construction succeeds.

1.1 AASC and conventional proof labor

AASC is not an optional interpretation of determinate objects. Its Admissibility, Standing, Reference, and Irreversibility coordinates are the minimal necessity conditions used whenever a nondegenerate determinate carrier, comparison, record, or endpoint is asserted [12]. Here they govern target fidelity: global form, bundle sector, spin structure, representation, source attachment, stabilizer, anomaly status, clock failure, and singular-stratum data cannot be erased if changing them changes an admitted continuation or conclusion.

This necessity does not generate the Standard Model action. Conventional differential geometry, constrained Hamiltonian theory, elliptic analysis, hyperbolic PDE, and anomaly calculations carry every construction below. The covariant action is an independently declared physical occupant. A failure in any one of those conventional constructions is not repairable by invoking kernel necessity.

2. Exact frozen branch

The spacetime signature is $(-+++)$, $\epsilon_{0123} = +1$, and $\kappa = 8\pi G > 0$. The inherited Cauchy carrier is a fixed compact, connected, boundaryless, oriented hyperbolic three-manifold Σ . Every oriented three-manifold is parallelizable and spin; one particular spin structure s is frozen [13, 14]. The accepted WDW-4A Sobolev base satisfies

$$q_0 > \frac{3}{2} + 4,$$

so multiplication, composition, curvature, and the order-two coefficient estimates used below are tame.

Coordinate	Frozen value	Guard
Gravity	Second-order metric GR in a nondegenerate Lorentz-coframe presentation	No independent connection, torsion, Holst, Palatini, or Einstein–Cartan dynamics
Clock	Accepted real massive WDW-4A scalar	Local and nonpreferred
Internal group	$G_{\text{SM}}^{(6)} = (SU(3)_c \times SU(2)_L \times U(1)_Y)/\mathbb{Z}_6$	Declared global-form branch
Topology	Trivial principal bundle $P_0 \rightarrow \Sigma$; no external line insertion	Other bundles, monopoles, and line sectors are distinct targets
Matter	Three minimal chiral generations; no ν_R	Neutrino-mass and sterile sectors require target update
Higgs	One complex weak doublet, $Y = 1/2$	Represents the imported $H_{\text{br}/\text{fix}}$ role; potential data are declared
Parameters	Positive g_3, g_2, g_1 , fixed $Y_u, Y_d, Y_e, \lambda, v$	No numerical value is derived
Proof class	Classical $\mathbb{Z}_2$ -graded carrier; analytic theorem on its even body	No Banach theorem over untyped odd variables
Boundary	Compact and boundaryless	Surface/asymptotic data reopen on other targets

3. Import interfaces and failure modes

The accepted WDW line is used only through the interfaces stated above. The structural Standard Model spine and SM-1–SM-6 arc specify the representation-response inventory, carrier/skin statuses, the three-generation record, and the structural generation certificate [15]. They do not supply a curved spin action or canonical constraint proof. The Higgs arc supplies the structural bridge/fixation role, not a value of v, λ , or a pole mass. The symmetry arc supplies target-update and anomaly typing; it does not transport fixed-background Poincare or CPT closure into dynamical gravity.

The external inputs are also narrow:

1. spin geometry supplies the fixed spin lift and Levi–Civita spin connection;
2. the Dirac–Bergmann theorem supplies the reduction rule once the actual second-class matrix has been calculated [16, 17];
3. the hypersurface-deformation calculation supplies the geometric bracket template, but the target’s matter terms are calculated here [18];
4. the tame inverse theorem applies only after a smooth-tame map and right-inverse family have been typed [19];
5. hyperbolic Einstein–Yang–Mills and Yang–Mills–Higgs results license the local bosonic development after the constraints are solved [20–22]; and
6. the local and global anomaly sources apply only to their stated representation and tangential structures [23–25].

First-order tetrad/connection analyses are comparison sources only [26, 27]; their torsion equations and

induced four-fermion terms are not imported.

3.1 External-theorem specialization matrix

Interface	Source hypotheses	WDW-4B specialization	Failure boundary
EXT-SPIN	Oriented spin manifold, nondegenerate coframe, metric-compatible torsion-free connection	Fixed $s, e \in Q_e^q$, direct cotangent reduction and $\omega_{LC}(e)$ in W4B.2	Degenerate tetrad, changed spin structure, or independent connection
EXT-SM-ACTION	Spin Lorentz manifold and representations of the gauge group	$G_{SM}^{(6)}$ descent proved in W4B.3; one second-order curved action in W4B.4	Alternative global form, bundle, fermion, or torsion branch
EXT-DIRAC	Regular primary constraint matrix with inverse distribution	Full matrix (78) on the nondegenerate coframe branch	$\det e = 0$ or a changed first-order kinetic target
EXT-BOS-CAN	Differentiable canonical generators on one domain	Compact $\partial\Sigma = \emptyset$, complete currents, raw/improved generators, and direct sector bracket in W4B.6–W4B.7	Boundary terms, omitted currents, or incompatible domains
EXT-ANOM	Specified chiral representations and tangential/global structure	All-left-handed sums, Witten parity, ordinary spin target, and $\mathbb{Z}_6$ descent in W4B.8	Other tangential structures or the open determinant-line/bordism problem
EXT-IFT	Smooth-tame map, compatible inverse family, uniform loss	Accepted WDW-4A loss-two split, even gauge–Higgs parameter, and (109)	Odd directions, loss of uniform invertibility, or a formal-only expansion
EXT-EYMH	Smooth constraint data and hyperbolic coordinate/gauge reduction	Smooth compact corrected bosonic datum; Higgs and scalar are smooth semilinear lower-order extensions	No constraint zero, nonlicensed gauge, or a claim of global completeness

Theorem W4B.1 (Target freeze, separation, and no backflow). The object constructed in this module is the fixed local classical branch listed above. WDW-4A is an accepted predecessor, not a disguised proof of WDW-4B. The action and its parameters are declared physical target data; AASC governs their faithful use but does not generate them. Failure of a WDW-4B-only gate cannot invalidate WDW-4A, and success of WDW-4B cannot authorize any earlier premise.

Proof. The displayed dependency order is acyclic. Each accepted import has been restricted to an interface already proved in that ancestor, while every new object is introduced only in a later gate. The frozen table distinguishes same-target coordinates from richer-scope exits. Since none of the definitions of WDW-1–WDW-4A refers to the objects introduced here, no backward dependency exists. $\square$

Nonclaim 3.1. The theorem neither derives the Standard Model action from the kernel nor claims that the chosen $\mathbb{Z}_6$, trivial-bundle, torsion-free branch is the only physically possible target.

Part XII

WDW-4B: Torsion-free tetrad and spin lift

4. The Lorentz-redundant coframe carrier

Let $\eta_{IJ} = \text{diag}(-1, 1, 1, 1)$. Define

$$Q_e^q = \left\{ e_i^I \in H^{q+2}(T^*\Sigma \otimes \mathbb{R}^{1,3}) : h_{ij} := e_i^I e_j^J \eta_{IJ} > 0 \right\}.$$

The selected orientation fixes the future unit normal

$$n_I(e) = -\frac{1}{6\sqrt{h}} \epsilon_{IJKL} \epsilon^{ijk} e_i^J e_j^K e_k^L, \quad n_I n^I = -1, \quad n_I e_i^I = 0.$$

Thus e alone fixes the internal normal on the selected connected branch. Lapse and shift reconstruct a spacetime tetrad along the leaf by

$$E_0^I = Nn^I + N^i e_i^I, \quad E_i^I = e_i^I.$$

The connected proper orthochronous gauge group $\mathcal{L}^{q+2} = H^{q+2}(\Sigma, SO^+(1, 3))$ acts freely on Q_e^q . The map

$$\varrho : Q_e^q \longrightarrow \text{Met}^{q+2}(\Sigma), \quad \varrho(e) = h = e^\top \eta e,$$

has exactly the $\mathcal{L}^{q+2}$ -orbits as its fibres. Local sections are obtained by choosing an orthonormal frame for h . Consequently

$$Q_e^q / \mathcal{L}^{q+2} \cong \text{Met}^{q+2}(\Sigma)$$

on this fixed spin/orientation branch.

5. The derived spin connection

Let E_μ^I be the tetrad of a local development and $\Gamma_{\mu\nu}^\rho(g)$ its Levi-Civita connection. The unique metric-compatible connection satisfying the torsion-free tetrad postulate is

$$\omega_\mu^I{}_J(E) = E^I{}_\nu (\partial_\mu E^\nu{}_J + \Gamma_{\mu\rho}^\nu E^\rho{}_J), \quad dE^I + \omega^I{}_J(E) \wedge E^J = 0. \quad (65)$$

It is antisymmetric in I, J . Its spin lift is

$$\nabla_\mu^{\text{spin}} \psi = \partial_\mu \psi + \frac{1}{4} \omega_{\mu IJ}(E) \gamma^{IJ} \psi.$$

Equation (65) is a definition from E and its first derivative. No spatial ω_i^{IJ} and no conjugate momentum appear as an independent canonical pair. The temporal quantity used below is the arbitrary angular velocity of the moving Lorentz frame; it is a multiplier for the cotangent-lift moment map, not a Palatini connection.

Lemma 5.1 (Torsion-free target separation). *Varying the second-order action through E varies $\omega_{\text{LC}}(E)$ by the chain rule. It does not produce an independent Cartan equation, torsion constraint, or induced Einstein–Cartan four-fermion term.*

Proof. The configuration map is $E \mapsto (g(E), \omega_{\text{LC}}(E))$. Hence allowed tangent vectors satisfy $\delta\omega = D\omega_{\text{LC}}(E)[\delta E]$. An independent variation $\delta\omega$ is not in this tangent space. The Cartan equation arises only after enlarging the configuration space to independent (E, ω) , which is a different target. $\square$

6. Cotangent reduction to the accepted ADM pair

On $T^*Q_e^q$ use the canonical one-form

$$\Theta_e = \int_\Sigma P^i{}_I \delta e_i^I.$$

The cotangent-lifted Lorentz action has moment map

$$\mathcal{J}_{IJ}^e = 2P^i{}_{[I} e_{J]i}. \quad (66)$$

When spinors are present, the total moment map is

$$\mathcal{J}_{IJ} = \mathcal{J}_{IJ}^e + \mathcal{J}_{IJ}^\Psi, \quad \mathcal{J}_{IJ}^\Psi = \frac{i}{2} \sum_{\Psi, r} \Psi_r^\dagger \Sigma_{IJ} \Psi_r, \quad (67)$$

with the conventional chiral restriction of Σ_{IJ} . The half-density canonical transformation includes the finite momentum shift that removes δe -terms from the fermionic kinetic one-form; the same shift moves the spin term between P and $\mathcal{J}^\Psi$ without changing (67).

Lemma 6.1 (Pure gravitational symplectic reduction). *At zero fermion and $\mathcal{J}^e = 0$, each Lorentz orbit has a unique symmetric metric momentum π^{ij} determined by*

$$P^i{}_I = 2\pi^{ij}e_{jI}.$$

Moreover

$$\Theta_e = \int_{\Sigma} \pi^{ij} \delta h_{ij}, \quad T^*Q_e^q //_0 \mathcal{L}^{q+2} \cong T^*\text{Met}^{q+2}(\Sigma).$$

Proof. The boost part of $\mathcal{J}^e = 0$ removes the components of $P^i{}_I$ normal to the span of e_i^I ; the rotation part makes $P^i{}_I e_j^I$ symmetric. Hence the displayed symmetric π^{ij} exists uniquely. Since $\delta h_{ij} = 2e_{(i|I|} \delta e_{j)}^I$,

$$P^i{}_I \delta e_i^I = 2\pi^{ij} e_{jI} \delta e_i^I = \pi^{ij} \delta h_{ij}.$$

The Lorentz action is free on nondegenerate coframes, so the zero-level cotangent reduction is smooth and gives the asserted symplectomorphism. $\square$

7. Algebraic Lorentz split

The surjectivity used later can be displayed pointwise. Given $\lambda_{IJ} \in \mathfrak{so}(1,3)^*$, decompose

$$\lambda_{IJ} = 2n_{[I} b_{J]} + r_{IJ}, \quad n^I b_I = 0, \quad n^I r_{IJ} = 0.$$

With $b^i = e^{iI} b_I$ and $r^{ij} = e^{iI} e^{jJ} r_{IJ}$, set

$$(S_e \lambda)^i{}_I = n_I b^i + \frac{1}{2} e_{Ik} r^{ki}. \quad (68)$$

Direct contraction gives

$$2(S_e \lambda)^i{}_{[I} e_{J]i} = \lambda_{IJ}.$$

Thus S_e is an order-zero right inverse for the vertical momentum column of $D\mathcal{J}^e$. On a bounded nondegenerate coframe neighborhood,

$$\|S_e \lambda\|_{H^q} \leq C_q (1 + \|e\|_{H^{q+2}}) \|\lambda\|_{H^q}.$$

Theorem W4B.2 (Torsion-free tetrad/spin lift). The frozen metric branch has a Lorentz-redundant nondegenerate coframe and fixed spin lift. Its only spin connection is $\omega_{\text{LC}}(e)$. The local Lorentz moment map is (67); reduction returns the accepted ADM metric pair and adds no gravitational propagating degree of freedom. Its vertical momentum linearization has the compatible algebraic right inverse (68).

Proof. Existence and quotient identification are established by the construction of Q_e^q . Uniqueness of the torsion-free connection is (65); lemma 5.1 excludes an independent connection. The moment-map and symplectic assertions are lemma 6.1; (68) supplies the required split. $\square$

Part XIII

WDW-4B: Global form and representation ledger

8. Integral hypercharge convention

For the global-form calculation use the integral charge

$$y = 6Y \in \mathbb{Z}.$$

If $z = e^{i\theta} \in U(1)_Y$, a field of charge y transforms by z^y . The local covariant derivative retains the conventional hypercharge $Y = y/6$ and coupling g_1 ; this is only a normalization of the $U(1)$ connection.

Field	$SU(3)_c$	$SU(2)_L$	Y	$y = 6Y$
Q_L^r	3	2	1/6	1
u_R^r	3	1	2/3	4
d_R^r	3	1	-1/3	-2
L_L^r	1	2	-1/2	-3
e_R^r	1	1	-1	-6
H	1	2	1/2	3

Here $r = 1, 2, 3$. No right-handed neutrino is included. The conjugate $\tilde{H} = i\sigma^2 H^*$ has $Y = -1/2$.

9. Descent to the declared $\mathbb{Z}_6$ quotient

Let

$$\zeta = \left(e^{2\pi i/3} \mathbf{1}_3, -\mathbf{1}_2, e^{i\pi/3} \right).$$

It has order six in $SU(3) \times SU(2) \times U(1)$. Its action on the table is

$$\begin{aligned} Q_L : e^{2\pi i/3} (-1) e^{i\pi/3} &= 1, & u_R : e^{2\pi i/3} e^{4i\pi/3} &= 1, \\ d_R : e^{2\pi i/3} e^{-2i\pi/3} &= 1, & L_L : (-1) e^{-i\pi} &= 1, \\ e_R : e^{-2i\pi} &= 1, & H : (-1) e^{i\pi} &= 1. \end{aligned}$$

Therefore all matter and Higgs representations factor through

$$G_{\text{SM}}^{(6)} = (SU(3)_c \times SU(2)_L \times U(1)_Y) / \langle \zeta \rangle.$$

This verifies faithfulness to the chosen quotient; it does not select the quotient from local observations. The alternatives $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3$ have the same local Lie algebra but different allowed Wilson–t Hooft lines and theta identifications [25].

10. Bundle, generation, Higgs, and parameter statuses

Only the trivial principal $G_{\text{SM}}^{(6)}$ -bundle P_0 is occupied below. Connections are global Lie-algebra-valued one-forms in a selected trivialization; changing the trivialization is a proper gauge redescription. No external line is inserted. Nontrivial principal bundles, magnetic charge, monopoles, and alternative line spectra are not quotiented into this branch.

The three copies in the representation table are the imported generation roles. Anomaly cancellation below repeats generationwise and does not derive the number three. The fixed complex doublet is the conventional representative of the imported structural bridge/fixation carrier $H_{\text{br/fix}}$; the representative is not promoted into a structure-independent primitive identity.

The positive gauge couplings g_3, g_2, g_1 , complex Yukawa matrices Y_u, Y_d, Y_e , and Higgs parameters $\lambda > 0, v > 0$ are fixed target data. The potential is

$$V(H) = \lambda \left(H^\dagger H - \frac{v^2}{2} \right)^2, \quad (69)$$

with zero vacuum offset. Theta coordinates are registered and fixed to the local classical branch value. No numerical mass, mixing, CP, or renormalization-group result is inferred.

Theorem W4B.3 (Global form and representation closure). The six representation types, three generation copies, minimal no- ν_R fermion branch, Higgs doublet, symbolic parameters, trivial bundle, and no-line sector are fixed consistently. Every representation descends to the declared $\mathbb{Z}_6$ quotient. Alternative global forms, bundles, line sectors, Higgs realizations, neutrino sectors, and numerical parameter choices remain distinct target variants.

Proof. The table fixes every local representation and charge. The six explicit ζ -actions equal one, proving quotient descent. The remaining statements are target coordinates, not conclusions inferred from that calculation; their guards prevent the prohibited promotions. $\square$

Part XIV

WDW-4B: Covariant ancestor and source bridge

11. One target-matched action

Write the gauge potentials as

$$A_\mu = A_\mu^{(3)a} T_a, \quad W_\mu = W_\mu^A \tau_A, \quad B_\mu,$$

with Hermitian generators $\text{tr}(T_a T_b) = \text{tr}(\tau_A \tau_B) = \frac{1}{2} \delta$. For a field in representation (R_3, R_2, Y) ,

$$\mathcal{D}_\mu = \nabla_\mu^{\text{spin}} - i g_3 A_\mu^a T_a^{R_3} - i g_2 W_\mu^A \tau_A^{R_2} - i g_1 Y B_\mu. \quad (70)$$

For H , the spin term is absent. Every connection in (70) has now been identified: $\omega_{\text{LC}}(E)$ for spin and the three declared internal connections for representation response.

With $e = |\det E_\mu^I| = \sqrt{-g}$, define

$$\begin{aligned} S_{4B} = & \frac{1}{2\kappa} \int e (R - 2\Lambda) - \frac{1}{2} \int e \left(g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + m_\phi^2 \phi^2 \right) \\ & - \frac{1}{4} \int e \left(F_{\mu\nu}^a F^{a\mu\nu} + W_{\mu\nu}^A W^{A\mu\nu} + B_{\mu\nu} B^{\mu\nu} \right) \\ & - \int e \left((\mathcal{D}_\mu H)^\dagger \mathcal{D}^\mu H + V(H) \right) + S_{\text{Weyl}} + S_Y. \end{aligned} \quad (71)$$

The symmetrized chiral kinetic action is

$$S_{\text{Weyl}} = \frac{i}{2} \sum_{\Psi, r} \int e \left(\bar{\Psi}_r \gamma^\mu \mathcal{D}_\mu \Psi_r - (\mathcal{D}_\mu \bar{\Psi}_r) \gamma^\mu \Psi_r \right),$$

with the chiral projectors and representations from Theorem W4B.3. Finally

$$S_Y = - \int e [\bar{Q}_L Y_d H d_R + \bar{Q}_L Y_u \tilde{H} u_R + \bar{L}_L Y_e H e_R + \text{h.c.}]. \quad (72)$$

Generation indices are contracted by the displayed matrices; color and weak indices are contracted in their unique invariant channels.

12. Currents, stress tensor, and Ward identities

All sources are varied from (71):

$$T_{\mu\nu} = -\frac{2}{e} \frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}, \quad j_{3a}^\mu = -\frac{1}{e} \frac{\delta S_{\text{matter}}}{\delta A_\mu^a}, \quad j_{2A}^\mu = -\frac{1}{e} \frac{\delta S_{\text{matter}}}{\delta W_\mu^A}, \quad j_Y^\mu = -\frac{1}{e} \frac{\delta S_{\text{matter}}}{\delta B_\mu}.$$

Because the connection is derived, tetrad variation includes the Belinfante spin improvement and gives a symmetric tensor. Its gauge and Higgs parts are, for example,

$$T_{\mu\nu}^{\text{YM}} = \sum_s \left(F_{\mu\rho}^{(s)} F_\nu^{(s)\rho} - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma}^{(s)} F^{(s)\rho\sigma} \right), \quad (73)$$

$$T_{\mu\nu}^H = (\mathcal{D}_\mu H)^\dagger \mathcal{D}_\nu H + (\mathcal{D}_\nu H)^\dagger \mathcal{D}_\mu H - g_{\mu\nu} ((\mathcal{D}H)^\dagger \mathcal{D}H + V). \quad (74)$$

The scalar term is the accepted WDW-4A tensor. The fermionic tensor is the symmetrized tetrad variation of $S_{\text{Weyl}} + S_Y$, not the unsymmetrized canonical tensor.

Gauge invariance under a compactly supported infinitesimal parameter α gives

$$D_\mu j_\alpha^\mu = - \sum_\Phi \mathcal{E}_\Phi (\alpha \cdot \Phi),$$

where $\mathcal{E}_\Phi = e^{-1} \delta S / \delta \Phi$. Hence every internal current is covariantly conserved on the matter equations. Diffeomorphism invariance gives the off-shell identity

$$\nabla_\mu T^\mu{}_\nu = \sum_\Phi \mathcal{E}_\Phi \mathcal{D}_\nu \Phi + F_{\nu\mu}^a j_{3a}^\mu + W_{\nu\mu}^A j_{2A}^\mu + B_{\nu\mu} j_Y^\mu. \quad (75)$$

The gauge equations convert the last three terms into divergences of the gauge stress. Equivalently, sectorwise,

$$\nabla T_{\text{YM}} = -F \cdot (j_H + j_\Psi), \quad \nabla T_H = F \cdot j_H + \mathcal{X}_Y, \quad \nabla T_\Psi = F \cdot j_\Psi - \mathcal{X}_Y,$$

and the Yukawa exchange $\mathcal{X}_Y$ cancels. Together with the scalar equation, $\nabla_\mu T^\mu{}_\nu = 0$ on shell.

13. Normal and tangential source projections

Let n^μ be the future unit normal and $h^\mu{}_i$ the leaf projector. The source coordinates are

$$\rho = T_{\mu\nu} n^\mu n^\nu, \quad j_i = -T_{\mu\nu} n^\mu h^\nu{}_i, \quad S_{ij} = T_{\mu\nu} h^\mu{}_i h^\nu{}_j.$$

Legendre transformation of the same action gives

$$\mathcal{H}_\perp^{\text{matter}} = \sqrt{h} \rho, \quad \mathcal{H}_i^{\text{matter}} = \sqrt{h} j_i \quad (76)$$

up to the internal Gauss and Lorentz improvements made explicit in part XVI. This follows term by term: electric and magnetic energies for gauge fields, normal and spatial covariant derivatives for the Higgs, the first-order normal spinor term, the Yukawa scalar, and the accepted clock contributions.

The pointed source object attached to any licensed history is

$$\hat{x}_{4B} = [\mathbf{p}_{4B}, \text{Att}_{4B}, T_{4B}, (j_3, j_2, j_Y), \text{Ward status}]_{\sim_{\text{src}}}.$$

It is assigned from the covariant history and action. It is not reconstructed from an unattached instantaneous stress tensor.

Theorem W4B.4 (Covariant action and source attachment). The declared torsion-free target has the single covariant ancestor (71). Its derivatives, representations, currents, symmetric stress tensor, gauge Ward identities, sector exchange cancellations, and total on-shell conservation are fixed by variation of that action. The normal and tangential projections agree with the canonical matter constraint densities as in (76), and each licensed history carries an exact pointed source class.

Proof. Gauge and diffeomorphism variation yield the displayed Noether identities. The three sector exchange equations sum to (75) with the matter equations imposed. The Legendre identity $p\dot{q} - \mathcal{L} = N\sqrt{h}\rho + N^i\sqrt{h}j_i$ for each sector gives (76). Because every term comes from (71), the response, attachment, Ward, and canonical projections refer to the same occupant. $\square$

Part XV

WDW-4B: The complete graded canonical carrier

14. Even variables and primary multiplier pairs

The non-multiplier even carrier is

$$\mathcal{P}_0 = \{(e_i^I, P^i_I), (\phi, p_\phi), (A_i^a, E_a^i), (W_i^A, E_A^i), (B_i, E_Y^i), (H, \Pi_H), (H^\dagger, \Pi_{H^\dagger})\}.$$

The symplectic potential is

$$\Theta_0 = \int_\Sigma \left(P^i_I \delta e_i^I + p_\phi \delta \phi + E_a^i \delta A_i^a + E_A^i \delta W_i^A + E_Y^i \delta B_i + \Pi_H \delta H + \delta H^\dagger \Pi_{H^\dagger} \right).$$

The independent-complex-coordinate convention treats $H, H^\dagger$ as independent during variation and restores conjugacy on the real section.

The lapse N , shift N^i , Lorentz angular velocity λ_0^{IJ} , and internal temporal potentials (A_0^a, W_0^A, B_0) have primary momenta

$$p_N = 0, \quad p_i = 0, \quad p_{\lambda IJ} = 0, \quad p_{0a}^{(3)} = p_{0A}^{(2)} = p_0^Y = 0.$$

The Lorentz multiplier is not an independent spatial spin connection.

15. Odd carrier and the actual second-class matrix

Let the composite index

$$\mathcal{A} = (\Psi, r, a, c, m)$$

include species, generation r , two-component spin index a , and the applicable color and weak indices c, m . There are, per generation,

$$12 + 6 + 6 + 4 + 2 = 30$$

complex Weyl components: 12 in Q_L , 6 each in u_R, d_R , 4 in L_L , and 2 in e_R .

Before half-densitization, the normal kinetic term has the form

$$\frac{i}{2} \int_\Sigma \sqrt{h} (\psi_{\mathcal{A}}^\dagger \dot{\psi}_{\mathcal{A}} - \dot{\psi}_{\mathcal{A}}^\dagger \psi_{\mathcal{A}}) + \text{coframe-velocity terms}.$$

Set

$$\Psi_{\mathcal{A}} = h^{1/4} \psi_{\mathcal{A}}.$$

The coframe-velocity terms are absorbed by the finite canonical shift of P^i_I mentioned in equation (67). The odd kinetic one-form becomes

$$\Theta_1 = \frac{i}{2} \int_\Sigma \left(\Psi_{\mathcal{A}}^\dagger \delta \Psi_{\mathcal{A}} + \Psi_{\mathcal{A}} \delta \Psi_{\mathcal{A}}^\dagger \right),$$

with the sign determined by right differentiation of odd variables. Introducing unconstrained odd momenta gives the primary pairs

$$\chi_{\mathcal{A}} = \pi_{\mathcal{A}} - \frac{i}{2} \Psi_{\mathcal{A}}^\dagger = 0, \quad \bar{\chi}_{\mathcal{A}} = \bar{\pi}_{\mathcal{A}} - \frac{i}{2} \Psi_{\mathcal{A}} = 0. \quad (77)$$

With the even graded Poisson bracket, their only nonzero matrix entry is

$$C_{\mathcal{A}\mathcal{B}}(x, y) = \{\chi_{\mathcal{A}}(x), \bar{\chi}_{\mathcal{B}}(y)\} = -i \delta_{\Psi\Psi'} \delta_{rr'} \delta_{aa'} \delta_{cc'} \delta_{mm'} \delta_\Sigma(x, y). \quad (78)$$

Absent indices are omitted rather than summed. Before half-densitization, the same matrix is $-i\sqrt{h}$ times the corresponding identity. Thus nondegeneracy of e is exactly the required invertibility condition.

The inverse kernel is

$$(C^{-1})^{\mathcal{A}\mathcal{B}}(x, y) = i \delta^{\mathcal{A}\mathcal{B}} \delta_\Sigma(x, y).$$

The graded Dirac bracket

$$\{F, G\}_D = \{F, G\} - \{F, \chi_A\}(C^{-1})^{AB}\{\bar{\chi}_B, G\} - \{F, \bar{\chi}_A\}(C^{-1})^{AB}\{\chi_B, G\}$$

therefore imposes (77) strongly and gives

$$\{\Psi_A(x), \Psi_B^\dagger(y)\}_D = -i \delta_{AB} \delta_\Sigma(x, y). \quad (79)$$

All even brackets retain their canonical form. The formula displays every density, species, generation, chirality, spin, color, and weak coordinate; zero fermion excitation does not make the matrix disappear.

16. Legendre map and reduced carrier

The gauge momenta are the densitized electric fields

$$E_s^i = \frac{\sqrt{h}}{N} h^{ij} (F_{0j}^{(s)} - N^k F_{kj}^{(s)}), \quad s = 3, 2, Y,$$

and

$$\Pi_H = \frac{\sqrt{h}}{N} (\mathcal{D}_0 H - N^i \mathcal{D}_i H)^\dagger, \quad \Pi_{H^\dagger} = \frac{\sqrt{h}}{N} (\mathcal{D}_0 H - N^i \mathcal{D}_i H).$$

The first-order fermion Legendre map is exactly the constraint pair (77). After its Dirac reduction, the reduced odd carrier consists of the 90 complex half-density components Ψ_A and their adjoints with bracket (79).

Theorem W4B.5 (Graded canonical carrier and fermion reduction). The action (71) has the complete even and odd canonical carrier displayed above. All temporal multipliers have vanishing primary momenta. The first-order chiral kinetic term yields (77); its full graded matrix is (78), invertible on every nondegenerate coframe branch, and the reduced half-density bracket is (79). No analytic implicit-function argument has been used in an odd direction.

Proof. The even momenta follow by direct differentiation of the normal velocities in (71). Right differentiation of the symmetrized odd kinetic term gives (77); the fundamental graded brackets give (78) and its inverse. Substitution in the defining Dirac formula gives (79). All steps are algebraic in the odd variables. $\square$

Part XVI

WDW-4B: The complete constraint system

17. Local Lorentz and internal Gauss laws

The total Lorentz constraint is (67). It is distinct from the internal gauge constraints. With the representation table of Theorem W4B.3, define

$$\mathcal{G}_3^a = D_i E_3^{ia} + g_3 \sum_r (Q_{Lr}^\dagger T^a Q_{Lr} + u_{Rr}^\dagger T^a u_{Rr} + d_{Rr}^\dagger T^a d_{Rr}), \quad (80)$$

$$\begin{aligned} \mathcal{G}_2^A &= D_i E_2^{iA} + i g_2 (\Pi_H \tau^A H - H^\dagger \tau^A \Pi_{H^\dagger}) \\ &\quad + g_2 \sum_r (Q_{Lr}^\dagger \tau^A Q_{Lr} + L_{Lr}^\dagger \tau^A L_{Lr}), \end{aligned} \quad (81)$$

$$\mathcal{G}_Y = \partial_i E_Y^i + i g_1 Y_H (\Pi_H H - H^\dagger \Pi_{H^\dagger}) + g_1 \sum_{\Psi, r} Y_\Psi \Psi_r^\dagger \Psi_r. \quad (82)$$

The adjoint and sign convention is the one for which $\{A_i, \mathcal{G}[\alpha]\}_D = -D_i \alpha$ and $\{\Phi, \mathcal{G}[\alpha]\}_D = i g \alpha \cdot \Phi$. Equations (80)–(82) contain all generation and representation multiplicities.

18. Hamiltonian and spatial densities

The accepted gravity and clock densities are denoted $\mathcal{H}_\perp^g, \mathcal{H}_i^g, \mathcal{H}_\perp^\phi, \mathcal{H}_i^\phi$. For $s = 3, 2, Y$,

$$\mathcal{H}_\perp^{\text{YM}} = \sum_s \left[\frac{1}{2\sqrt{h}} E_s^i \cdot E_{si} + \frac{\sqrt{h}}{4} F_{sij} \cdot F_s^{ij} \right], \quad (83)$$

$$\mathcal{H}_i^{\text{YM}} = \sum_s E_s^j \cdot F_{sij}. \quad (84)$$

The Higgs contributions are

$$\mathcal{H}_\perp^H = \frac{1}{\sqrt{h}} \Pi_H \Pi_{H^\dagger} + \sqrt{h} \left(h^{ij} (\mathcal{D}_i H)^\dagger \mathcal{D}_j H + V(H) \right), \quad (85)$$

$$\mathcal{H}_i^H = \Pi_H \mathcal{D}_i H + (\mathcal{D}_i H)^\dagger \Pi_{H^\dagger}. \quad (86)$$

On the reduced half-density carrier,

$$\mathcal{H}_\perp^\Psi = -\frac{i}{2} \sum_{\Psi, r} \Psi_r^\dagger \alpha^i \overleftrightarrow{\mathcal{D}}_i \Psi_r + \mathcal{H}_\perp^Y, \quad (87)$$

$$\mathcal{H}_i^\Psi = -\frac{i}{2} \sum_{\Psi, r} \Psi_r^\dagger \overleftrightarrow{\mathcal{D}}_i \Psi_r, \quad (88)$$

where $\mathcal{H}_\perp^Y = \sqrt{h} \mathcal{L}_Y$ with the sign fixed by (72). The half-density spin connection includes the corresponding density connection, so (88) is a density of weight one.

The total normal constraint is

$$\mathcal{H}_\perp^{\text{tot}} = \mathcal{H}_\perp^g + \mathcal{H}_\perp^\phi + \mathcal{H}_\perp^{\text{YM}} + \mathcal{H}_\perp^H + \mathcal{H}_\perp^\Psi, \quad (89)$$

and the gauge/Lorentz-covariant raw spatial constraint is

$$\mathcal{V}_i = \mathcal{H}_i^g + \mathcal{H}_i^\phi + \mathcal{H}_i^{\text{YM}} + \mathcal{H}_i^H + \mathcal{H}_i^\Psi. \quad (90)$$

These are exactly the normal and tangential projections (76).

19. Raw and improved spatial generators

Let

$$\mathcal{A}_i = (A_i, W_i, B_i), \quad \omega_i = \omega_{\text{LC}, i}(e).$$

The raw generator $\mathcal{V}[\xi] = \int \xi^i \mathcal{V}_i$ acts by the gauge- and Lorentz-covariant Lie derivative. The generator of the ordinary spatial Lie derivative on every reduced variable is

$$\mathcal{D}[\xi] = \mathcal{V}[\xi] - \mathcal{G}[\iota_\xi \mathcal{A}] - \mathcal{J}[\iota_\xi \omega]. \quad (91)$$

For example,

$$\{A_i, \mathcal{V}[\xi]\}_D = \xi^j F_{ji}, \quad \{A_i, \mathcal{D}[\xi]\}_D = \xi^j F_{ji} + D_i(\xi^j A_j) = \mathcal{L}_\xi A_i.$$

The same Cartan identity applies to W, B ; the Lorentz completion changes the covariant transport of e, Ψ into their ordinary spatial Lie derivatives. Thus raw and improved generators are not interchangeable.

20. Full Dirac Hamiltonian and boundary differentiability

Before fermionic reduction the Dirac Hamiltonian is

$$\begin{aligned} H_D = & \int_\Sigma (N \mathcal{H}_\perp^{\text{tot}} + N^i \mathcal{V}_i + \lambda^{IJ} \mathcal{J}_{IJ} + A_0^a \mathcal{G}_{3a} + W_0^A \mathcal{G}_{2A} + B_0 \mathcal{G}_Y) \\ & + \int_\Sigma (u_N p_N + u^i p_i + u_\lambda^{IJ} p_{\lambda IJ} + u_3^a p_{0a}^{(3)} + u_2^A p_{0A}^{(2)} + u_Y p_0^Y) \\ & + \int_\Sigma \sum_{\mathcal{A}} (\nu^{\mathcal{A}} \chi_{\mathcal{A}} + \bar{\nu}^{\mathcal{A}} \bar{\chi}_{\mathcal{A}}). \end{aligned} \quad (92)$$

After Dirac reduction the last line is imposed strongly. Replacing $\mathcal{V}$ by $\mathcal{D}$ shifts the displayed Lorentz and internal multipliers by $\iota_N\omega$ and $\iota_N\mathcal{A}$; it does not change the constraint surface.

Every integration by parts used here has zero surface term because $\partial\Sigma = \emptyset$. This proves differentiability for the frozen branch only. A finite boundary or asymptotic end requires surface charges and a new target.

21. Degree-of-freedom certificate

Counts below are real phase dimensions per spatial point, excluding multiplier pairs unless stated otherwise. The redundant coframe has $12 + 12 = 24$ even dimensions; clock, gauge, and Higgs sectors contribute 2, 72, 8, respectively. Thus the non-multiplier even phase dimension is 106. Adding four lapse/shift, six Lorentz, and twelve internal multiplier pairs gives $106 + 44 = 150$.

There are 90 complex Weyl components in three generations. Treating fields, adjoints, and their unconstrained momenta independently gives 360 real odd dimensions before primary reduction. The 180 second-class conditions leave odd rank 180.

Stage	Even phase dim.	Odd rank	Explanation
Non-multiplier, before reduction	106	360	Full canonical momenta
After fermion second class	106	180	$\Psi, \Psi^\dagger$ Dirac pair
After local Lorentz	94	180	Six first-class generators remove 12
After internal Gauss	70	180	Twelve generators remove 24
After four spacetime constraints	62	180	Four generators remove 8

The final even count is 31 configuration degrees of freedom: two gravitational, one clock-scalar, 24 gauge polarizations, and four real Higgs components. On a free bosonic quotient the odd variables form an associated graded vector bundle of the same rank 180; subtracting the bosonic gauge dimensions again from its fibre would double count the group reduction.

Theorem W4B.6 (Full classical constraint inventory). Equations (67), (80)–(82), (89)–(90), and (92) give every primary, second-class, local Lorentz, internal Gauss, spatial, and normal constraint of the frozen target. The improved generator is (91); it is not conflated with the raw generator. The degree count returns the ADM gravitational propagating content and the expected Standard Model bosonic and graded fermionic carriers. No boundary completion is required on the closed branch.

Proof. All primary constraints follow from absent multiplier velocities and the first-order fermion kinetic term. Variation of the temporal gauge and Lorentz multipliers gives (67) and (80)–(82). The Legendre identity for lapse and shift gives (89)–(90). Cartan's formula proves (91). First- and second-class dimension subtraction gives the table. Compactness without boundary eliminates all integration surfaces. $\square$

Part XVII

WDW-4B: Exact common-domain joint algebra

22. Common reduced domain

Fix $q \geq q_0$. All brackets in this part are evaluated on the same fermion-Dirac-reduced Sobolev carrier:

$$\begin{aligned} e &\in H^{q+2}, & P &\in H^{q+1}, & \phi &\in H^{q+1}, & p_\phi &\in H^q, \\ \mathcal{A} &\in H^{q+1}, & E &\in H^q, & H &\in H^{q+1}, & \Pi_H &\in H^q, & \Psi &\in H^{q+1/2}. \end{aligned}$$

Smooth fields are a common invariant core. The displayed formulas extend continuously to the stated levels; no bracket is assembled from incompatible completions.

23. Moment-map brackets

The canonical even brackets and (79) give

$$\{\mathcal{J}[\lambda], \mathcal{J}[\mu]\}_D = \mathcal{J}[[\lambda, \mu]], \quad \{\mathcal{G}[\alpha], \mathcal{G}[\beta]\}_D = \mathcal{G}[[\alpha, \beta]], \quad (93)$$

$$\{\mathcal{J}[\lambda], \mathcal{G}[\alpha]\}_D = 0, \quad \{\mathcal{D}[\xi], \mathcal{J}[\lambda]\}_D = \mathcal{J}[\mathcal{L}_\xi \lambda], \quad (94)$$

$$\{\mathcal{D}[\xi], \mathcal{G}[\alpha]\}_D = \mathcal{G}[\mathcal{L}_\xi \alpha], \quad \{\mathcal{D}[\xi], \mathcal{D}[\eta]\}_D = \mathcal{D}[[\xi, \eta]]. \quad (95)$$

For example, the fermion contribution to the first row is the standard bilinear calculation

$$\{-\Psi^\dagger t_\lambda \Psi, -\Psi^\dagger t_\mu \Psi\}_D = -\Psi^\dagger [t_\lambda, t_\mu] \Psi,$$

and exactly matches the coframe or internal bosonic moment map. Thus the fermion spin and gauge currents are not spectators in the closure.

24. Normal brackets: sector calculation

Let

$$s^i = h^{ij}(N\partial_j M - M\partial_j N).$$

Only terms with one spatial derivative of a smearing survive antisymmetrization in N, M . The canonical calculation gives the following derivative kernels:

For the gauge block, suppressing the sector label,

$$\frac{\delta \mathcal{H}_{\text{YM}}[N]}{\delta E^i} = \frac{N}{\sqrt{h}} E_i, \quad \frac{\delta \mathcal{H}_{\text{YM}}[N]}{\delta A_i} = -D_j(N\sqrt{h} F^{ji}).$$

After one integration by parts,

$$\begin{aligned} \int_\Sigma \left[\frac{\delta \mathcal{H}_{\text{YM}}[N]}{\delta A_i} \frac{\delta \mathcal{H}_{\text{YM}}[M]}{\delta E^i} - (N \leftrightarrow M) \right] \\ = \int_\Sigma s^i E^j \cdot F_{ij} = \mathcal{V}_{\text{YM}}[s]. \end{aligned} \quad (96)$$

For the Higgs block,

$$\frac{\delta \mathcal{H}_H[N]}{\delta \Pi_H} = \frac{N}{\sqrt{h}} \Pi_{H^\dagger}, \quad \frac{\delta \mathcal{H}_H[N]}{\delta H^\dagger} = -\mathcal{D}_i(N\sqrt{h} \mathcal{D}^i H) + N\sqrt{h} \partial_{H^\dagger} V.$$

The potential contribution is symmetric in NM , while the derivative term gives

$$\{\mathcal{H}_H[N], \mathcal{H}_H[M]\}_D = \int_\Sigma s^i \left(\Pi_H \mathcal{D}_i H + (\mathcal{D}_i H)^\dagger \Pi_{H^\dagger} \right) = \mathcal{V}_H[s]. \quad (97)$$

The clock calculation is the identical real-scalar specialization. For the reduced Weyl block, (79) gives

$$\{\mathcal{H}_\Psi[N], \mathcal{H}_\Psi[M]\}_D|_{\partial N, \partial M} = -\frac{i}{2} \int_\Sigma s^i \Psi^\dagger \overset{\leftrightarrow}{\mathcal{D}}_i \Psi = \mathcal{V}_\Psi[s].$$

Curvature commutators in the remaining fermion terms cancel against the gauge–fermion and tetrad–fermion cross variations. Yukawa variations contain no derivative of a smearing and cancel under $N \leftrightarrow M$. The internal and Lorentz moment-map terms enter only when the resulting covariant raw generator is rewritten as the ordinary spatial generator through (91).

Sector	Derivative term after $N \leftrightarrow M$	Result
Gravity	curvature variation against π^{ij}	$\mathcal{V}_g[s]$
Clock	p_ϕ against $h^{ij}\partial_i\phi\partial_j\phi$	$\mathcal{V}_\phi[s]$
Yang–Mills	E^i against $F_{ij}F^{ij}$	$\mathcal{V}_{\text{YM}}[s]$
Higgs	Π_H against $(\mathcal{D}_i H)^\dagger \mathcal{D}^i H$	$\mathcal{V}_H[s]$
Fermions	reduced bracket of the two first-order spatial Dirac terms	$\mathcal{V}_\Psi[s]$, with curvature cross terms canceled by the dynamical connection/coframe blocks
Potential/Yukawa	ultralocal in canonical coordinates	zero after antisymmetrization

The gauge–Higgs and gauge–fermion cross terms combine with the covariant derivatives into the same internal moment maps already present in (91); the tetrad–fermion cross terms combine into $\mathcal{J}[\iota_s\omega]$. Therefore the raw result is

$$\{\mathcal{H}[N], \mathcal{H}[M]\}_D = \mathcal{V}[s] = \mathcal{D}[s] + \mathcal{G}[\iota_s\mathcal{A}] + \mathcal{J}[\iota_s\omega]. \quad (98)$$

Equation (91) is a change of spatial-constraint basis, not an off-shell deletion of the last two terms. Consequently the pure improved-generator formula is only the weak identity

$$\{\mathcal{H}[N], \mathcal{H}[M]\}_D \stackrel{\mathcal{G}=\mathcal{J}=0}{\approx} \mathcal{D}[s]. \quad (99)$$

The exact off-shell improved-basis formula remains (98); (99) records its restriction to the joint constraint surface. This distinction is the promised explicit “up to gauge” content.

Lemma 24.1 (Interaction cancellation). *The Higgs potential and Yukawa density contribute no independent derivative-of-smearing term to the normal-normal bracket. Their gauge and Lorentz variations cancel within the moment-map terms in (98).*

Proof. Both densities are ultralocal scalar densities of weight one. A bracket with a kinetic momentum term is proportional to NM ; antisymmetrization in N, M kills it. Brackets with covariant spatial kinetic terms organize by the gauge and Lorentz Ward identities into the generators of the corresponding representation action, already displayed in (98). $\square$

25. Complete first-class table

The remaining rows follow by direct generator action:

$$\{\mathcal{D}[\xi], \mathcal{H}[N]\}_D = \mathcal{H}[\mathcal{L}_\xi N], \quad (100)$$

$$\{\mathcal{G}[\alpha], \mathcal{H}[N]\}_D = 0, \quad \{\mathcal{J}[\lambda], \mathcal{H}[N]\}_D = 0. \quad (101)$$

The zeros in (101) are exact: each summand in (89) is an internal and Lorentz scalar after all currents and spin terms are included. Equations (93)–(101) also satisfy the Jacobi identities because they were calculated with one Dirac bracket; no bracket is imposed as an independent axiom.

Theorem W4B.7 (Exact off-shell joint constraint algebra). On the common reduced domain, the Lorentz, internal gauge, improved spatial, and normal constraints form the exact first-class algebra (93)–(101). Its structure function is the inherited inverse metric h^{-1} . The raw normal-normal bracket is $\mathcal{V}[s]$; in the improved spatial basis its exact off-shell expression is (98), including the displayed Gauss and Lorentz coefficients. The pure- $\mathcal{D}[s]$ expression is only the weak identity (99). No field-dependent Lorentz or gauge term is suppressed, and no appeal to covariance substitutes for the sector calculation.

Proof. The moment-map rows follow from the fundamental reduced brackets and the explicit coframe, Higgs, and fermion bilinears. The sector table derives every derivative-of-smearing contribution; lemma 24.1 closes the interactions. Their sum is $\mathcal{V}[s]$, and direct substitution of (91) gives the exact improved-basis expression (98) and its weak restriction (99). Direct action on a scalar density and the invariance of $\mathcal{H}_\perp^{\text{tot}}$ give (100)–(101). Compactness removes boundary remainders. $\square$

Part XVIII

WDW-4B: Target-indexed anomaly status

26. One-generation perturbative sums

For anomaly calculations convert every right-handed field to a left-handed charge conjugate:

$$Q : (\mathbf{3}, \mathbf{2})_{1/6}, \quad u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}, \quad d^c : (\bar{\mathbf{3}}, \mathbf{1})_{1/3}, \quad L : (\mathbf{1}, \mathbf{2})_{-1/2}, \quad e^c : (\mathbf{1}, \mathbf{1})_1.$$

Normalize $T(\mathbf{3}) = T(\mathbf{2}) = 1/2$ and $A(\mathbf{3}) = 1$, $A(\bar{\mathbf{3}}) = -1$. The one-generation sums are

$$SU(3)^3 : 2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 2 - 1 - 1 = 0, \quad (102)$$

$$SU(3)^2 Y : 2 \left(\frac{1}{6} \right) \frac{1}{2} + \left(-\frac{2}{3} \right) \frac{1}{2} + \left(\frac{1}{3} \right) \frac{1}{2} = 0, \quad (103)$$

$$SU(2)^2 Y : 3 \left(\frac{1}{6} \right) \frac{1}{2} + \left(-\frac{1}{2} \right) \frac{1}{2} = 0, \quad (104)$$

$$Y^3 : 6 \left(\frac{1}{6} \right)^3 + 3 \left(-\frac{2}{3} \right)^3 + 3 \left(\frac{1}{3} \right)^3 + 2 \left(-\frac{1}{2} \right)^3 + 1^3 = 0, \quad (105)$$

$$\text{grav}^2 Y : 6 \left(\frac{1}{6} \right) + 3 \left(-\frac{2}{3} \right) + 3 \left(\frac{1}{3} \right) + 2 \left(-\frac{1}{2} \right) + 1 = 0. \quad (106)$$

The pure $SU(2)^3$ local anomaly vanishes because the doublet is pseudoreal; mixed products with a single traceless nonabelian generator vanish. Three identical copies multiply every zero by three. Nothing in (102)–(106) selects the number of generations.

27. Original and newer $SU(2)$ diagnostics

After color multiplicity, one generation contains three quark weak doublets and one lepton weak doublet. Thus it contains four isospin-1/2 left-handed multiplets. Three generations contain twelve. Both are even, so the original mod-two Witten anomaly is absent [23].

The newer $SU(2)$ anomaly is sensitive to the tangential structure and to isospin $4k + 3/2$ representations [24]. The frozen target has an independently fixed spin structure and an honest $\text{Spin}(1, 3) \times G_{\text{SM}}^{(6)}$ lift, not a replacement spin- $SU(2)$ structure, and its weak fermions are only isospin 1/2. Accordingly the new diagnostic supplies no additional obstruction on this branch. This statement is target-indexed; it is not a bordism classification of every alternative tangential structure.

28. Global form and open rows

Theorem W4B.3 verifies representation descent to the chosen $\mathbb{Z}_6$ quotient and registers the corresponding line/theta data. It does not prove the triviality of every determinant line over the space of dynamical tetrads and gauge fields. The exact status ledger is:

ID	Target	Status and reason
AN4B-01	Classical gauge/Lorentz invariance	PROVED: action and first-class algebra
AN4B-02	One-generation local polynomial sums	PROVED: (102)–(106)
AN4B-03	Three-generation local sums	PROVED: three copies of zero
AN4B-04	Original Witten $SU(2)$ anomaly	PROVED ABSENT: 4 doublets/generation, 12 total
AN4B-05	New $SU(2)$ /tangential diagnostic	CHECKED: ordinary spin target; no $4k + 3/2$ multiplet
AN4B-06	Faithful $\mathbb{Z}_6$ global form	PROVED FOR REPRESENTATIONS: Theorem W4B.3
AN4B-07	Determinant-line, bordism, and large-diffeomorphism closure	OPEN FOR WDW-6
AN4B-08	Regulated quantum gravitational constraint algebra	OUT OF WDW-4B SCOPE
AN4B-09	CPT/discrete-symmetry recovery	SEPARATELY GATED; unused as a premise
AN4B-10	Boundary/inflow anomaly	INACTIVE WITH GUARD: $\partial\Sigma = \emptyset$

Theorem W4B.8 (Anomaly-status closure at the frozen scope). The declared chiral representation is free of the displayed local perturbative anomalies generation by generation and of the original Witten $SU(2)$ anomaly. It descends to the chosen $\mathbb{Z}_6$ global form and passes the frozen spin/tangential diagnostic. Determinant-line, bordism, large-diffeomorphism, dynamical-gravity, and regulated quantum-constraint anomaly closure are not proved and retain the statuses shown above. No CPT theorem is a premise of W4B.1–W4B.12.

Proof. The local claims are the explicit sums (102)–(106). The mod-two count is 4 per generation. Quotient descent is the six-field calculation in Theorem W4B.3; the tangential statement follows from the frozen ordinary spin lift and representation inventory. None of those finite calculations implies a theorem about the open rows, which are therefore retained. $\square$

Part XIX

WDW-4B: Explicit nonzero gauge–Higgs occupation

29. The accepted base point

Fix one accepted WDW-4A datum

$$z_0 = (h_0, \pi_0, \phi_0, p_{\phi,0}) \in \mathcal{V}_0$$

on the compact hyperbolic Σ . It satisfies

$$\Phi_{4A}(z_0) = 0, \quad \ker D\Phi_{4A}(z_0)^\# = 0, \quad D\Phi_{4A}(z)R_z = 1$$

on a neighborhood $\mathcal{V}_0$, with coefficient loss two. Choose a coframe lift e_0 in the fixed spin branch. Set

$$H_{\text{vac}} = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix}.$$

At zero gauge connection, constant H_{vac} , and zero Standard Model momenta and fermions, the added stress tensor vanishes because $V(H_{\text{vac}}) = 0$. Thus this enlarged zero-matter point is still z_0 .

30. A small connection with zero semisimple centralizer

Lemma 30.1 (Finite curvature-centralizer seed). *On the trivial $SU(n)$ bundle, $n = 2, 3$, there is a smooth Lie-algebra-valued one-form $a^{(n)}$ such that, for every sufficiently small $\delta \neq 0$, the connection $A_\delta^{(n)} = \delta a^{(n)}$ has no nonzero covariantly constant adjoint section.*

Proof. Choose a basis $X_1, \dots, X_{n^2-1}$ of $\mathfrak{su}(n)$ and pairwise disjoint coordinate balls B_k . In B_k choose a compactly supported one-form b_k and point x_k such that $b_k(x_k) = 0$ and $db_k(x_k) \neq 0$. Put

$$a^{(n)} = \sum_k X_k b_k.$$

Then

$$F_{\delta a^{(n)}}(x_k) = \delta X_k db_k(x_k)$$

because every $b_j(x_k)$ vanishes except possibly $b_k(x_k) = 0$. If $D_{A_\delta} \alpha = 0$, the integrability identity gives $[F_{A_\delta}, \alpha] = 0$. Parallel-transport the finitely many curvature values to one base point. After division by δ , they converge as $\delta \rightarrow 0$ to nonzero multiples of the basis X_k . The linear map

$$X \mapsto ([X_1, X], \dots, [X_{n^2-1}, X])$$

is injective because the centre of $\mathfrak{su}(n)$ is zero. Injectivity of this finite-dimensional map is open, so the transported curvature values have zero common centralizer for all sufficiently small nonzero δ . Hence $\alpha = 0$. $\square$

Choose the two forms given by lemma 30.1, and also a smooth real one-form a^Y with $da^Y \neq 0$. Let f be a nonconstant smooth real function and $\eta \in \mathbb{C}^2$. Define

$$\mathcal{A}_\delta = (\delta a^{(3)}, \delta a^{(2)}, \delta a^Y), \quad H_\delta = H_{\text{vac}} + \delta f \eta. \quad (107)$$

After decreasing δ_0 , H_δ is nowhere zero and nonconstant for $0 < |\delta| < \delta_0$, while each selected nonabelian curvature and the hypercharge curvature are nonzero.

Lemma 30.2 (Trivial joint internal stabilizer). *For $0 < |\delta| < \delta_0$, the only infinitesimal internal parameter $\alpha = (\alpha_3, \alpha_2, \alpha_Y)$ satisfying*

$$D_{\mathcal{A}_\delta} \alpha = 0, \quad \alpha \cdot H_\delta = 0$$

is $\alpha = 0$.

Proof. Lemma 30.1 gives $\alpha_3 = \alpha_2 = 0$. The adjoint connection on $\mathfrak{u}(1)$ is trivial, so α_Y is constant. The remaining action is $ig_1 Y_H \alpha_Y H_\delta$. Since $g_1 > 0$, $Y_H = 1/2$, and H_δ is nowhere zero, it vanishes only when $\alpha_Y = 0$. $\square$

31. Exact Gauss and Lorentz constraints

Set

$$E_\delta = 0, \quad \Pi_{H,\delta} = \Pi_{H^\dagger,\delta} = 0, \quad \Psi_\delta = 0. \quad (108)$$

Every charge density in (80)–(82) is then zero, and $D_i E^i = 0$. Thus all twelve internal Gauss components vanish exactly. The fermionic primary conditions hold on the zero section, while their matrix (78) remains invertible.

For any corrected metric momentum π^{ij} , choose the symmetric coframe lift

$$P^i{}_I = 2\pi^{ij} e_{jI}.$$

At zero fermion its Lorentz moment map is exactly zero by lemma 6.1. The matter spatial momentum also vanishes because every term in (84), (86), and (88) contains E, Π_H , or Ψ .

32. Exact even parametric correction

Use the accepted shifted scales

$$\begin{aligned} E_q^{4A} &= H^{q+2}(S^2T^*) \oplus H^{q+1}(S^2T^*) \oplus H^{q+1} \oplus H^q, \\ F_q &= H^q(\text{Dens}) \oplus H^q(T^*\Sigma \otimes \text{Dens}), \\ \mathcal{P}_q^{\text{GH}} &= H^{q+1}(T^*\Sigma \otimes \mathfrak{g}_{\text{SM}}) \oplus H^{q+1}(\mathbb{C}^2). \end{aligned}$$

Let $\mathcal{X}_0 = \text{im } R_{z_0} \subset E^{4A}$, so $D\Phi_{4A}(z_0)|_{\mathcal{X}_0}$ is an isomorphism with inverse R_{z_0} . For $u \in \mathcal{X}_0$, define

$$\mathcal{F}(u, \delta) = \Phi_{4A}(z_0 + u) + \mathcal{S}_{\text{GH}}(z_0 + u; \mathcal{A}_\delta, H_\delta), \quad (109)$$

where $\mathcal{S}_{\text{GH}}$ is the gauge–Higgs contribution to (89)–(90) with (108). Then

$$\mathcal{F}(0, 0) = 0, \quad D_u \mathcal{F}(0, 0) = D\Phi_{4A}(z_0)|_{\mathcal{X}_0}.$$

The source is quadratic at the vacuum:

$$\|\mathcal{S}_{\text{GH}}(z; \mathcal{A}_\delta, H_\delta)\|_{F_q} \leq C_q(1 + \|z\|_{E_{q+2}^{4A}})\|(\mathcal{A}_\delta, H_\delta - H_{\text{vac}})\|_{\mathcal{P}_{q+1}}^2.$$

Its parameter derivatives satisfy one-high-norm Moser estimates with the same fixed coefficient loss two as R_z . Hence the parametric smooth-tame inverse theorem applies to (109) [19]. There are $\delta_1 > 0$ and a unique smooth-tame curve $u_\delta \in \mathcal{X}_0$, $u_0 = 0$, such that

$$\mathcal{F}(u_\delta, \delta) = 0, \quad |\delta| < \delta_1. \quad (110)$$

This is an exact equation at every Sobolev level, not an asymptotic series. Moreover $u_\delta = O(\delta^2)$ in every fixed lower norm.

Remark 32.1 (No odd analytic direction). The only parameter in (109) is the even pair $(\mathcal{A}_\delta, H_\delta)$. Fermions remain at the invariant zero section. The graded carrier and algebra were proved separately and are not inputs to a Banach or tame inverse theorem over Grassmann variables.

33. Licensed covariant history

The corrected smooth datum consists of

$$z_\delta = z_0 + u_\delta, \quad (\mathcal{A}_\delta, E_\delta = 0), \quad (H_\delta, \Pi_{H,\delta} = 0), \quad \Psi_\delta = 0,$$

together with the symmetric coframe momentum lift. It satisfies every classical constraint exactly.

In harmonic gauge for the metric and a hyperbolic gauge for the internal connections, the Einstein–Yang–Mills–Higgs–scalar equations are a quasilinear hyperbolic system with smooth lower-order potential terms [20–22]. Constraint propagation gives a local geometrically unique development. The fermion equations are homogeneous and linear in the fermions (with Higgs-dependent Yukawa mixing), so zero initial fermions give the invariant zero solution. The standard maximal-development construction then supplies a maximal globally hyperbolic representative, unique up to an initial-surface-fixing isometry [28]. No geodesic completeness or global clock conclusion follows.

Theorem W4B.9 (Explicit nonzero Standard Model bosonic occupation). For every sufficiently small $0 < |\delta| < \delta_1$, the datum above is an exact zero of all primary, Lorentz, internal Gauss, spatial, and normal constraints. Its gauge curvature is nonzero, its Higgs field is nonzero and nonconstant, and its joint internal infinitesimal stabilizer is trivial. It has a licensed source-attached torsion-free Einstein–Yang–Mills–Higgs–scalar development with invariant zero fermions.

Proof. Lemmas 30.1 and 30.2 prove the nonzero and stabilizer properties. Equation (108) proves the internal Gauss, matter-momentum, and fermionic-primary statements. The symmetric lift proves the Lorentz constraint. Equation (110) proves the remaining Hamiltonian–momentum constraints exactly. The hyperbolic specialization licenses the covariant history and preserves the zero fermion section. $\square$

Part XX

WDW-4B: The SMReg4B split and local quotient

34. The full even constraint map

At the zero-fermion witness define

$$\Phi_{4B} = (\Phi_{\text{tot}}, \mathcal{G}_{3,2,Y}, \mathcal{J}).$$

Use the row space

$$\mathcal{Y}_q = F_q \oplus H^q(\Sigma, \mathfrak{g}_{\text{SM}}^*) \oplus H^q(\Sigma, \mathfrak{so}(1,3)^*)$$

and the selected column space

$$\mathcal{X}_q = E_q^{4A} \oplus \underbrace{\left[H^{q+1}(T\Sigma \otimes \mathfrak{g}_{\text{SM}}^*) \oplus H^q(\mathcal{H}^*) \right]}_{\mathcal{U}_q^G} \oplus \underbrace{H^q(T\Sigma \otimes \mathbb{R}^{1,3})}_{\mathcal{U}_q^J}.$$

The second Gauss column is the Higgs-momentum variation, and the Lorentz column is the vertical coframe-momentum variation. No odd column belongs to this analytic map.

35. The internal Gauss normal inverse

At the pair $(\mathcal{A}_\delta, H_\delta)$, let

$$K_\delta(\dot{E}, \dot{\Pi}) = D_{\mathcal{A}_\delta, i} \dot{E}^i + \mu_H(H_\delta, \dot{\Pi})$$

be the Gauss linearization in the selected momentum columns. With fixed invariant fibre metrics, its formal adjoint is

$$K_\delta^\# \alpha = (-D_{\mathcal{A}_\delta} \alpha, \rho_H(\alpha) H_\delta).$$

Thus

$$\langle \alpha, K_\delta K_\delta^\# \alpha \rangle = \|D_{\mathcal{A}_\delta} \alpha\|_{L^2}^2 + \|\rho_H(\alpha) H_\delta\|_{L^2}^2. \quad (111)$$

By lemma 30.2, the kernel is zero. The normal operator

$$L_\delta^G = K_\delta K_\delta^\# : H^{q+2}(\mathfrak{g}_{\text{SM}}) \longrightarrow H^q(\mathfrak{g}_{\text{SM}}^*)$$

is positive elliptic, self-adjoint, Fredholm of index zero on compact Σ , and hence invertible [29]. The compatible right inverse is

$$Q_\delta^G = K_\delta^\# (L_\delta^G)^{-1}, \quad K_\delta Q_\delta^G = 1. \quad (112)$$

It is one fixed distributional expression at every Sobolev level, so elliptic regularity gives level compatibility.

On a small coefficient neighborhood,

$$\|Q_w^G f\|_{\mathcal{U}_q^G} \leq C_q \left[\|f\|_{H^q} + (1 + \|w\|_{\mathcal{X}_{q+2}}) \|f\|_{H^{q_0}} \right]. \quad (113)$$

Every parameter derivative obeys the corresponding one-high-norm estimate, obtained by differentiating $(L_w^G)^{-1}$ as $-(L_w^G)^{-1}(DL_w^G)(L_w^G)^{-1}$.

36. The block-triangular right inverse

At the exact witness, the Hamiltonian is quadratic in (E, Π_H) , whereas the spatial constraint is linear in those momenta. Define

$$A_\delta = D\Phi_{\text{tot}}(w_\delta)|_{E^{4A}}, \quad J_{e_\delta} = D\mathcal{J}(w_\delta)|_{\text{vert}},$$

and let $B_\delta : \mathcal{U}_q^G \rightarrow F_q$ be the order-zero momentum column

$$\begin{aligned} B_\delta(\dot{E}, \dot{\Pi}) &= (0, b_\delta(\dot{E}, \dot{\Pi})_i), \\ b_\delta(\dot{E}, \dot{\Pi})_i &= \sum_{s=3,2,Y} \dot{E}_s^j \cdot F_{s,ij} + \dot{\Pi}_H \mathcal{D}_i H_\delta + (\mathcal{D}_i H_\delta)^\dagger \dot{\Pi}_H^\dagger. \end{aligned} \quad (114)$$

The zero in the first component of B_δ is the Hamiltonian row: its derivative in (E, Π_H) vanishes at $E = \Pi_H = 0$. The second component is generally nonzero because $F_{\mathcal{A}_\delta} \neq 0$ and H_δ is nonconstant. The Gauss law is linear in these momenta, and the horizontal coframe coordinates are separated from vertical momentum. Therefore, on the selected columns, the exact base linearization is upper triangular:

$$D\Phi_{4B}(w_\delta) = \begin{pmatrix} A_\delta & B_\delta & 0 \\ 0 & K_\delta & 0 \\ 0 & 0 & J_{e_\delta} \end{pmatrix}. \quad (115)$$

The first block is a small matter perturbation of the WDW-4A block and has a compatible right inverse R_δ by the same low-norm Neumann argument used in equation (109). The other inverses are (112) and S_e from (68). Set

$$Q_\delta^\Delta = \begin{pmatrix} R_\delta & -R_\delta B_\delta Q_\delta^G & 0 \\ 0 & Q_\delta^G & 0 \\ 0 & 0 & S_{e_\delta} \end{pmatrix}. \quad (116)$$

Because

$$A_\delta R_\delta = 1, \quad K_\delta Q_\delta^G = 1, \quad J_{e_\delta} S_{e_\delta} = 1,$$

direct block multiplication gives

$$D\Phi_{4B}(w_\delta) Q_\delta^\Delta = I. \quad (117)$$

In particular, the top-middle product is $-A_\delta R_\delta B_\delta Q_\delta^G + B_\delta Q_\delta^G = 0$; no matter column is discarded.

The typing and loss are explicit. Since $\mathcal{A}_\delta, H_\delta \in H^{q+1}$,

$$F_{\mathcal{A}_\delta}, \mathcal{D}H_\delta \in H^q.$$

The Sobolev multiplication theorem at $q_0 > \frac{3}{2} + 4$ gives

$$\begin{aligned} \|B_\delta u\|_{F_q} &\leq C_q (\|(F_{\mathcal{A}_\delta}, \mathcal{D}H_\delta)\|_{H^{q_0}} \|u\|_{\mathcal{U}_q^G} \\ &\quad + \|(F_{\mathcal{A}_\delta}, \mathcal{D}H_\delta)\|_{H^q} \|u\|_{\mathcal{U}_{q_0}^G}). \end{aligned} \quad (118)$$

Combining (118) with the compatible loss-two estimates for R_δ and Q_δ^G yields

$$\|R_\delta B_\delta Q_\delta^G f\|_{E_q^{4A}} \leq C_q [\|f\|_{H^q} + (1 + \|w_\delta\|_{q+2}) \|f\|_{H^{q_0}}]. \quad (119)$$

The same Moser product rule gives one-high-norm estimates after every parameter derivative. Thus Q_δ^Δ is a compatible loss-two tame right inverse at the exact witness.

At a nearby point w , retain the same horizontal/vertical splitting and write

$$D\Phi_{4B}(w) Q_\delta^\Delta = I + \mathcal{E}_w, \quad \mathcal{E}_w = (D\Phi_{4B}(w) - D\Phi_{4B}(w_\delta)) Q_\delta^\Delta.$$

After shrinking the neighborhood, $\|\mathcal{E}_w\|_{q_0} < 1/2$. Define

$$Q_w^{4B} = Q_\delta^\Delta (I + \mathcal{E}_w)^{-1} = Q_\delta^\Delta \sum_{k=0}^{\infty} (-\mathcal{E}_w)^k. \quad (120)$$

Then

$$D\Phi_{4B}(w) Q_w^{4B} = 1$$

and, with the common coefficient loss $\mu_{4B} = 2$,

$$\|Q_w^{4B} f\|_{\mathcal{X}_q} \leq C_q [\|f\|_{\mathcal{Y}_q} + (1 + \|w\|_{\mathcal{X}_{q+2}}) \|f\|_{\mathcal{Y}_{q_0}}]. \quad (121)$$

Differentiating (120) gives one-high-norm tame estimates for every parameter derivative. The fixed distributional operators and the same Neumann series on nested levels make the family compatible under Sobolev inclusion.

37. Joint generalized KIDs and the local quotient

Surjectivity of $D\Phi_{4B}(w)$ implies

$$\ker D\Phi_{4B}(w)^\# = 0.$$

At the witness this also has the transparent sequential reading: the WDW-4A no-KID block kills lapse and shift, the Gauss energy (111) kills the internal parameter, and the algebraic Lorentz split kills the Lorentz parameter. Openness of (120) preserves this on the local chart.

Let

$$\mathcal{R}_0 = \text{Diff}_0(\Sigma) \ltimes (\mathcal{L}_0 \times \mathcal{G}_{0,\text{SM}})$$

be the connected group of proper spatial, Lorentz, and internal redescrptions on the fixed bundle and spin branch. Trivial infinitesimal stabilizer and the tame split give a local slice to its action on the regular constraint locus. Therefore

$$\mathcal{C}_{4B}^{\text{reg}}/\mathcal{R}_0$$

is a nonempty local smooth-tame quotient. Outside this chart, stabilizers are retained as singular data. The quotient does not identify different global forms, spin structures, bundles, line sectors, or boundary conditions.

Lemma 37.1 (Clock openness on the occupied chart). *After decreasing the matter-amplitude interval, the corrected development has a nonempty neighborhood on which*

$$g^{-1}(d\phi, d\phi) < 0, \quad n(\phi) > 0.$$

The inequalities and transversality persist on a local quotient neighborhood.

Proof. The accepted WDW-4A inequalities are strict at z_0 . Equation (110) gives $u_\delta = O(\delta^2)$, and the hyperbolic development depends continuously on the smooth initial data. Timelike covectors and positive normal contraction form open conditions. Scalar covariance makes the conditions invariant under the proper redescription group, so they descend to the local slice quotient. $\square$

38. Clause-by-clause certificate

Certificate 38.1 (SMReg4B). For a classical WDW-4B datum w , SMReg4B(w) is the conjunction of the twelve rows below.

Clause	Coordinate	Typed witness
SMR4B-01	Complete joint constraint zero	Theorem W4B.9
SMR4B-02	Nondegenerate tetrad and fixed spin lift	Theorem W4B.2
SMR4B-03	Invertible fermionic second-class matrix	(78)
SMR4B-04	Trivial internal stabilizer	Lemma 30.2
SMR4B-05	Surjective internal Gauss linearization	(112)
SMR4B-06	Surjective local Lorentz linearization	(68)
SMR4B-07	No joint generalized KID	$\ker D\Phi_{4B}^\# = 0$
SMR4B-08	Compatible tame right inverse	(120)–(121)
SMR4B-09	Source-attached covariant history	Hyperbolic development in Theorem W4B.9
SMR4B-10	Nonempty occupied local chart	Split locus around w_δ
SMR4B-11	Nonzero SM bosonic content	(107)
SMR4B-12	Persisting scalar-clock patch	Lemma 37.1

Theorem W4B.10 (SMReg4B occupancy and proper local quotient). For every sufficiently small nonzero δ , SMReg4B(w_δ) holds. The full even constraint map has the compatible smooth-tame right inverse (120) with coefficient loss two, its joint adjoint kernel is zero, and its zero locus admits a nonempty local quotient by $\mathcal{R}_0$. Stabilizers and distinct global data remain singular or separate rather than being erased.

Proof. The table supplies a theorem, equation, or construction for each clause. Lemma 37.1 proves SMR4B-12 directly from W4B.9 and does not use the W4B.10 conclusion. Equations (68), (112), (114), (116), (117) and (120) prove the three inverse coordinates while retaining the nonzero matter column. Equations (118), (119) and (121) prove the common loss-two tame bound. The tame submersion and slice theorems applied to the free proper local action give the stated locus and quotient. $\square$

Part XXI

WDW-4B: Clock retention, separation, and classical handoff

39. Persistence of the WDW-4A local clock

At z_0 , the accepted scalar development satisfies, on an open spacetime neighborhood $\mathcal{O}_0$,

$$g^{-1}(d\phi, d\phi) < 0, \quad n(\phi) > 0.$$

Lemma 37.1 proves that the correction and hyperbolic development preserve the strict timelike and orientation conditions. Hence there is a common local neighborhood $\mathcal{O}_\delta$ on which the corrected scalar remains a time-oriented timelike-gradient function.

Define

$$V_\phi = \frac{\nabla\phi}{g^{-1}(d\phi, d\phi)}. \quad (122)$$

With signature $(-+++)$, $n(\phi) > 0$ makes $\nabla\phi$ past-directed; the denominator in (122) is negative. Thus V_ϕ is future-directed and

$$V_\phi(\phi) = 1.$$

Its flow is transverse to the $\phi = \text{constant}$ leaves and gives a unit-rate local relational readout. Because ϕ is a scalar, (122) descends through proper spatial, Lorentz, and internal gauge redescrptions on the regular quotient.

The clock certificate fails, rather than silently continuing, if any of the following occurs:

$$\begin{aligned} & d\phi = 0 \quad \text{or} \quad g^{-1}(d\phi, d\phi) \geq 0, \\ & \text{loss of time orientation or of the regular quotient chart,} \\ & \text{or exit from the licensed development.} \end{aligned}$$

Caustics, topology change, nonproper group action, or a global recollapse can also prevent continuation. No claim of a global or preferred time is made.

40. No-backflow and matter-target separation

The scalar branch and Standard Model branch are different theorem objects:

$$\mathfrak{D}_{\text{WDW}}^\phi \neq \mathfrak{D}_{\text{WDW}}^{\text{SM}}.$$

WDW-4B enriches the latter with coframe, spin, gauge, Higgs, and graded fermion coordinates. It neither revises WDW-4A nor retroactively changes the meaning of its scalar capstone. If a WDW-4B-only anomaly, topology, or regularity condition fails outside this local branch, the accepted scalar theorem remains unchanged.

41. Exact WDW-5 handoff

The permitted classical Standard-Model handoff consists of:

1. the reduced graded phase carrier and Dirac bracket;
2. the complete Lorentz, internal Gauss, improved spatial, and normal constraints with the algebra of Theorem W4B.7;
3. the source-attached covariant history class;
4. the nonempty SMReg4B quotient chart and all singular guards;
5. the fixed spin, global-form, bundle, line, topology, parameter, and boundary coordinates;
6. the anomaly-status table of Theorem W4B.8; and
7. the local failure-typed scalar-clock class.

WDW-5 must separately freeze its functional representation, regularization, factor ordering, common operator domain, constraint implementation, state space, inner product, and observable question. Nothing here supplies those quantum objects.

Theorem W4B.11 (Clock retention and branch-typed handoff). For all sufficiently small occupied WDW-4B data, the accepted WDW-4A timelike-gradient clock persists on a nonempty local quotient neighborhood. Its continuation vector is (122), is future-directed, and satisfies $V_\phi(\phi) = 1$. The clock remains local, nonpreferred, and failure typed. The Standard Model branch remains separate from WDW-4A and supplies exactly the seven-coordinate classical handoff above, with no quantum promotion.

Proof. Lemma 37.1 preserves the strict timelike and orientation inequalities. The contraction and time orientation of (122) are direct. Scalar covariance gives quotient descent. The dependency graph contains no edge from WDW-4B to WDW-4A, while the handoff list contains no functional or quantum object. $\square$

Part XXII

WDW-4B: Capstone and audit appendices

42. Capstone assembly

Theorem W4B.12 (Torsion-Free Tetrad/Spin/Gauge Standard Model Classical Occupancy and Joint-Constraint Closure). On the fixed compact, connected, boundaryless, oriented, spin, topologically trivial branch, second-order torsion-free tetrad gravity coupled to the accepted WDW-4A scalar clock and the declared three-generation minimal chiral Standard Model has:

1. a complete classical graded canonical carrier;
2. all local Lorentz, internal Gauss, spatial, and normal constraints on one Dirac-reduced domain;
3. the exact joint first-class algebra of Theorem W4B.7;
4. a nonempty source-attached regular local class with nonzero gauge curvature and nonconstant Higgs data;
5. all twelve SMReg4B clauses and a compatible tame right inverse;
6. a local quotient by proper spatial, Lorentz, and internal gauge redescrptions;
7. the retained WDW-4A local scalar clock and its failure boundary; and

8. the exact target-indexed anomaly-status ledger of Theorem W4B.8.

The result is classical, local, torsion-free, trivial-bundle, compact-boundaryless, and uses a zero-fermion occupancy witness within a fully present graded fermion carrier. It supplies only the branch-typed classical input package of Theorem W4B.11.

Proof. Theorem W4B.1 freezes the target and dependency order. Theorems W4B.2 and W4B.3 construct its geometric and representation carriers. Theorem W4B.4 attaches the covariant action and source. Theorems W4B.5 to W4B.7 establish the graded canonical carrier, complete constraints, and exact algebra. Theorem W4B.8 supplies the bounded anomaly theorem. Theorems W4B.9 and W4B.10 construct and regularize one coherent nonzero bosonic occupant. Theorem W4B.11 retains the clock and fixes the handoff. No conclusion uses a later gate to prove an earlier premise. $\square$

43. Fourteen exact nonclaims

NC4B-01:

WDW-4B does not derive the Standard Model action from the AASC kernel.

NC4B-02:

It does not derive numerical gauge couplings, Yukawa matrices, Higgs parameters, masses, mixing angles, or CP phases.

NC4B-03:

It does not derive the $\mathbb{Z}_6$ global form from local fields.

NC4B-04:

It does not classify or occupy every nontrivial bundle, magnetic, monopole, or line-operator sector.

NC4B-05:

It does not introduce an independent spin connection, torsion, nonmetricity, or Einstein–Cartan four-fermion interaction.

NC4B-06:

It does not claim equivalence with a minimally coupled first-order Palatini or Einstein–Cartan target with fermions.

NC4B-07:

It does not add right-handed neutrinos, Majorana masses, the Weinberg operator, or sterile sectors.

NC4B-08:

It does not claim a nonzero classical fermion excitation at the explicit witness.

NC4B-09:

It does not apply a Banach implicit-function theorem in untyped Grassmann directions.

NC4B-10:

It does not infer complete global or dynamical-gravity anomaly freedom from local cancellation.

NC4B-11:

It does not transport flat-space Poincare or CPT closure into dynamical gravity.

NC4B-12:

It does not supply a Wheeler–DeWitt functional derivative, regularization, ordering, common operator domain, physical inner product, or quantum observable.

NC4B-13:

It does not promote the local scalar clock to a global or preferred time.

NC4B-14:

It does not rule out richer Higgs, BSM, torsionful, nontrivial-topology, boundary, or alternative-global-form targets.

44. Constraint and canonical audit tables

Sector	Final canonical role	Status
gravity	Tetrad lift reducing to accepted ADM pair	PROVED
metric projection	Accepted WDW-4A metric phase carrier	PROVED
clock	Retained local material-clock subsystem	PROVED
color gauge	Color connection and electric density	PROVED
weak gauge	Weak connection and electric density	PROVED
hypercharge gauge	Hypercharge connection and electric density	PROVED
higgs	Bridge/fixation doublet and canonical momentum	PROVED
left quark	Three-generation chiral matter carrier	PROVED
right up	Three-generation chiral matter carrier	PROVED
right down	Three-generation chiral matter carrier	PROVED
left lepton	Three-generation chiral matter carrier	PROVED
right electron	Three-generation chiral matter carrier	PROVED
fermion reduction	Canonical reduced graded carrier	PROVED
lapse shift	Multipliers for Hamiltonian and momentum constraints	PROVED
lorentz multiplier	Multiplier for local Lorentz constraint only; not an independent spatial connection	PROVED
internal multipliers	Multipliers for internal Gauss constraints	PROVED

ID	Stage	Class	Required result	Status
C4B-01	primary	first class	Enumerate and preserve multiplier role	PROVED
C4B-02	primary	first class	Enumerate all SU(3), SU(2), and U(1) components	PROVED
C4B-03	primary	first class	Keep distinct from an independent spin-connection theory	PROVED
C4B-04	primary	second class	Display full graded Dirac matrix with density, species, generation, and internal indices	PROVED
C4B-05	secondary	first class	Include tetrad and fermion spin contributions and show reduction to the ADM metric pair	PROVED
C4B-06	secondary	first class	Include quark color current	PROVED
C4B-07	secondary	first class	Include Higgs and left-doublet weak currents	PROVED
C4B-08	secondary	first class	Include every hypercharge current with one fixed normalization	PROVED
C4B-09	secondary	first class	Give raw and gauge/Lorentz-improved forms separately	PROVED
C4B-10	secondary	first class	Trace every density to the covariant action and source projections	PROVED
C4B-11	derived	first class	Prove exact Lie derivative action on all reduced variables	PROVED
C4B-12	derived	mixed	Write as a linear combination of every first-class constraint plus the second-class multipliers before reduction	PROVED
C4B-13	derived	certificate	Count even and odd components before and after first- and second-class reduction	PROVED
C4B-14	derived	certificate	Show no surface completion is needed on the compact boundaryless branch and retain the target-update guard	PROVED

45. Import-fidelity ledger

ID	Exact payload	Guard	Status
IMP-ARC	Separate D_WDW^SM target; WDW-4B burden list; matter-target separation theorem	Provisional outline is not proof	VERIFIED
IMP-W4A	Accepted scalar/clock phase carrier, total constraints, ScalarReg3 occupant, tame right inverse, local refoliation chart, local clock, branch guards	Does not contain tetrad, SM gauge, fermion, or anomaly closure	VERIFIED
IMP-W3	Exact scalar hypersurface-deformation algebra and regular local same-arrow theorem	Does not prove SM matter algebra or internal gauge reduction	VERIFIED TRANSITIVE
IMP-W2	Proper spatial-redescription quotient and source-fibre descent	Does not automatically quotient internal gauge or local Lorentz sectors	VERIFIED TRANSITIVE
IMP-COV	Finite-rank associated-bundle connection normal form and minimal natural metric Levi-Civita closure	Does not derive spin lift, infinite-dimensional functional derivative, or an independent spin connection	VERIFIED
IMP-MA	Attachment-safe symmetric source, Ward/divergence interface, composite exchange cancellation	Does not supply canonical SM densities by itself	VERIFIED TRANSITIVE
IMP-SM-SPINE	Sigma_SM structural tuple, minimal chiral representation-response skeleton, H_br/fix, Yukawa-availability slots, charge/anomaly/vacuum ledgers, three-generation record, parameter nonclaims	Does not supply curved-space spin/tetrad realization, canonical constraints, or numerical parameters	VERIFIED
IMP-SM-ARC	Carrier/skin grammar; gauge and matter carriers; charge/RG status; flavor/mixing status; carrier exhaustion; ClosedGenCert(n) iff n=3	Does not furnish a gravitationally coupled action or Dirac-Bergmann analysis	VERIFIED
IMP-HIGGS	H_br/fix carrier, identity, no primitive scale, vacuum-route, extension, and domain endpoint	Does not derive the Higgs potential, vev, pole mass, self-coupling, or canonical action	VERIFIED
IMP-SYM	Symmetry-status calculus; gauge architecture; target-update guard for dynamical gravity and tetrads; anomaly taxonomy; discrete-symmetry ledger	Does not close quantum-gravity anomaly or carry Poincare/CPT unchanged into WDW-4B	VERIFIED
EXT-SPIN	Spin lift, coframe redundancy, local Lorentz moment map, derived Levi-Civita spin connection	Must be specialized to the second-order torsion-free target	VERIFIED SPECIALIZED
EXT-SM-ACTION	Exact gauge, Higgs, chiral fermion, Yukawa, and stress-tensor formulas	Action is declared physical input, not kernel output	VERIFIED SPECIALIZED
EXT-DIRAC	Fermionic primary constraints, graded Dirac matrix, half-density reduction	First-order Einstein-Cartan results cannot be imported as if identical to the second-order target	VERIFIED SPECIALIZED
EXT-BOS-CAN	Gauss, momentum, Hamiltonian densities and first-class bracket structure	Must include the actual target conventions and all matter currents	VERIFIED SPECIALIZED
EXT-ANOM	Local anomaly sums, Witten parity, global-form and bordism distinctions	Global and dynamical-gravity anomalies remain target indexed	VERIFIED SPECIALIZED

ID	Exact payload	Guard	Status
EXT-IFT	Small even matter parameter produces corrected gravity/scalar constraint zero	No application in odd Grassmann directions	VERIFIED SPECIALIZED
EXT-EYMH	Licensed covariant history for the bosonic witness	Does not by itself construct a nonzero fermion development	VERIFIED SPECIALIZED

46. Acceptance-condition adjudication

The following sixty-eight rows are synchronization surfaces. Their PASS labels assert that the cited proof location supplies the required theorem act; the labels are not substitutes for those proofs.

ID	Condition	Status	Proof
AC4B-01	The final WDW-4A publication interface and WDW-4B handoff are explicit	PASS	Theorem W4B.1
AC4B-02	WDW-1 through WDW-3 are used only through their accepted interfaces	PASS	Theorem W4B.1
AC4B-03	Every SM-1 through SM-6 and generation-cardinality import has an exact theorem/status boundary	PASS	Theorem W4B.3
AC4B-04	All six Higgs papers are imported at structural carrier/status scope only	PASS	Theorem W4B.3
AC4B-05	SYM target-update and anomaly interfaces are specialized to the exact WDW-4B target	PASS	Theorem W4B.3
AC4B-06	Every conventional external theorem has source, hypotheses, specialization, and failure mode	PASS	Theorem W4B.1
AC4B-07	No downstream canonical success authorizes the AASC kernel	PASS	Theorem W4B.1
AC4B-08	No flat-space action, Poincare theorem, or CPT theorem is transported without an explicit bridge	PASS	Theorem W4B.1
AC4B-09	The compact boundaryless spatial manifold is inherited unchanged	PASS	Theorem W4B.1
AC4B-10	Orientation and one spin structure are fixed	PASS	Theorem W4B.2
AC4B-11	The tetrad is a lift of the metric branch rather than a new gravitational theory	PASS	Theorem W4B.2
AC4B-12	The spin connection is $\omega_{LC}(e)$ and not an independent phase variable	PASS	Theorem W4B.2
AC4B-13	Torsion, nonmetricity, Palatini, Einstein-Cartan, and Holst variants are explicit target updates	PASS	Theorem W4B.2
AC4B-14	$G_{SM}^{(6)}$ and the Z6 quotient convention are stated exactly	PASS	Theorem W4B.3
AC4B-15	The trivial principal-bundle sector and no-line-insertion branch are explicit	PASS	Theorem W4B.3
AC4B-16	Alternative global forms and nontrivial bundle sectors remain registered distinct targets	PASS	Theorem W4B.3
AC4B-17	Three generations and the minimal chiral matter inventory are fixed without a right-handed neutrino	PASS	Theorem W4B.3
AC4B-18	Gauge couplings, Yukawa matrices, Higgs parameters, and topological angles have truthful declared statuses	PASS	Theorem W4B.3
AC4B-19	One complete covariant action contains gravity, the WDW-4A clock, gauge, Higgs, fermion, and Yukawa sectors	PASS	Theorem W4B.4
AC4B-20	Every covariant derivative has a declared Levi-Civita, spin, or internal-gauge connection ancestor	PASS	Theorem W4B.4

ID	Condition	Status	Proof
AC4B-21	Every Standard Model representation and hypercharge appears in one fixed convention	PASS	Theorem W4B.4
AC4B-22	The Higgs doublet is explicitly related to H_{br}/fix without promoting its representative skin to primitive identity	PASS	Theorem W4B.4
AC4B-23	Metric/tetrad variation produces the complete symmetric source tensor	PASS	Theorem W4B.4
AC4B-24	Gauge, Higgs, fermion, and Yukawa exchange terms cancel in the total Ward identity	PASS	Theorem W4B.4
AC4B-25	Module-A attachment and source descent coordinates are supplied	PASS	Theorem W4B.4
AC4B-26	The covariant source and canonical constraint densities are traced term by term	PASS	Theorem W4B.4
AC4B-27	The unreduced even and odd canonical carrier is displayed	PASS	Theorem W4B.5
AC4B-28	All lapse, shift, Lorentz, and internal temporal multipliers and their primary momenta are enumerated	PASS	Theorem W4B.5
AC4B-29	All fermionic canonical momenta and primary constraints are displayed with indices and density weights	PASS	Theorem W4B.5
AC4B-30	The graded second-class matrix is computed and invertible on the nondegenerate tetrad branch	PASS	Theorem W4B.5
AC4B-31	The reduced half-density fermion bracket is derived rather than assumed	PASS	Theorem W4B.5
AC4B-32	The local Lorentz moment map contains both tetrad and fermion spin terms	PASS	Theorem W4B.6
AC4B-33	The $SU(3)$, $SU(2)$, and $U(1)$ Gauss laws contain all Higgs and fermion currents	PASS	Theorem W4B.6
AC4B-34	The total spatial momentum density contains every matter contribution	PASS	Theorem W4B.6
AC4B-35	The total Hamiltonian density contains every gauge, Higgs, fermion, Yukawa, clock, and gravity contribution	PASS	Theorem W4B.6
AC4B-36	Raw and gauge/Lorentz-improved spatial generators are not conflated	PASS	Theorem W4B.6
AC4B-37	The complete Dirac Hamiltonian is a sum of all constraints with multipliers	PASS	Theorem W4B.6
AC4B-38	An even/odd degree-of-freedom count agrees before and after reduction	PASS	Theorem W4B.6
AC4B-39	The internal Gauss constraints close with the exact Lie-algebra structure	PASS	Theorem W4B.7
AC4B-40	The local Lorentz constraints close and act correctly on tetrads and spinors	PASS	Theorem W4B.7
AC4B-41	The improved spatial generator acts by the correct Lie derivative on every reduced field	PASS	Theorem W4B.7
AC4B-42	Gauss-Lorentz and spatial cross brackets are displayed	PASS	Theorem W4B.7
AC4B-43	The Hamiltonian is invariant under internal gauge and local Lorentz generators	PASS	Theorem W4B.7
AC4B-44	The spatial-spatial and spatial-normal brackets close on the common Dirac-reduced domain	PASS	Theorem W4B.7
AC4B-45	The normal-normal bracket yields the improved spatial generator with the inherited structure function	PASS	Theorem W4B.7; exact HHraw formula and weak HHweak restriction
AC4B-46	Any Gauss or Lorentz term in a raw-generator bracket is shown explicitly	PASS	Theorem W4B.7

ID	Condition	Status	Proof
AC4B-47	Boundary terms vanish for the declared compact boundaryless target	PASS	Theorem W4B.7
AC4B-48	Classical covariance is not substituted for the bracket calculation	PASS	Theorem W4B.7
AC4B-49	All local perturbative anomaly sums are displayed in one left-handed convention	PASS	Theorem W4B.8
AC4B-50	The Witten SU(2) parity test counts color multiplicity and all generations	PASS	Theorem W4B.8
AC4B-51	The chosen spin/tangential and Z6 global-form data are included in the anomaly target	PASS	Theorem W4B.8
AC4B-52	Local cancellation is not promoted to complete global anomaly closure	PASS	Theorem W4B.8
AC4B-53	Quantum gravitational-constraint anomaly freedom remains a WDW-6 burden	PASS	Theorem W4B.8
AC4B-54	No anomaly, inflow, or CPT result changes the classical target retrospectively	PASS	Theorem W4B.8
AC4B-55	The seed gauge/Higgs configuration has F_A nonzero, H nonzero/nonconstant, and exact internal Gauss law	PASS	Theorem W4B.9
AC4B-56	The joint gauge/Higgs pair has trivial infinitesimal internal stabilizer	PASS	Theorem W4B.9
AC4B-57	Zero fermions are shown to satisfy the fermionic equations while the carrier and second-class block remain present	PASS	Theorem W4B.9
AC4B-58	The WDW-4A tame right inverse is applied only to the even gravity/scalar constraint residual	PASS	Theorem W4B.9
AC4B-59	The parametric correction solves the full Hamiltonian and momentum constraints exactly	PASS	Theorem W4B.9
AC4B-60	The local Lorentz constraints are solved by the symmetric coframe-momentum lift	PASS	Theorem W4B.9
AC4B-61	The full even constraint linearization has a compatible block-triangular tame right inverse	PASS	Theorem W4B.10; explicit B_delta column and triangular inverse
AC4B-62	All SMReg4B clauses and a licensed bosonic covariant history are occupied on a nonempty local class	PASS	Theorem W4B.10; SMReg4B certificate after triangular repair
AC4B-63	The WDW-4A timelike-gradient clock patch survives for sufficiently small matter data	PASS	Theorem W4B.11
AC4B-64	No global or preferred clock is inferred	PASS	Theorem W4B.11
AC4B-65	WDW-4A and WDW-4B remain distinct theorem objects with no backward dependency	PASS	Theorem W4B.11
AC4B-66	The classical SM branch handoff includes all constraints, quotient, source, regularity, anomaly-status, and global-form guards	PASS	Theorem W4B.11; repaired algebra and regular quotient handoff
AC4B-67	No functional differential, quantum-domain, ordering, regulator, inner-product, or quantum-observable claim appears	PASS	Theorem W4B.11
AC4B-68	The capstone reports only torsion-free, trivial-bundle, compact-boundaryless, local-classical occupancy	PASS	Theorem W4B.12

47. Adversarial-map gate

Every registered attack is tested below against the exact closure supplied in the manuscript. A reader must trace the cited closure backward before reopening a row.

ID	Attack	Actual closure	Status
ADV4B-01	AASC is being asked to generate the Standard Model action	Declare the covariant action as the independently specified physical occupant and use AASC only for necessary governance and target fidelity	CLOSED
ADV4B-02	WDW-4A already proved the full matter target	Point to the WDW-4A branch ledger and prove the new tetrad, gauge, fermion, and anomaly objects independently	CLOSED
ADV4B-03	Flat-space Standard Model closure is silently transported into dynamical gravity	Construct the tetrad/spin/gauge covariant action and canonical bridge at the new target	CLOSED
ADV4B-04	The Z6 quotient is claimed as experimentally or structurally forced	Label Z6 as the frozen global-form branch and retain $\Gamma=1, Z_2, Z_3$ as distinct target variants	CLOSED
ADV4B-05	The trivial bundle sector is mistaken for complete global completion	State the local-branch scope and register non-trivial bundles, monopoles, lines, and theta sectors	CLOSED
ADV4B-06	Spin structure is assumed without target registration	Freeze one spin structure and include it in every fermion and anomaly statement	CLOSED
ADV4B-07	Torsion-free Einstein-Dirac is conflated with minimally coupled Einstein-Cartan theory	Use the second-order metric branch and explicitly exclude independent connection/torsion imports	CLOSED
ADV4B-08	An independent spin connection has been smuggled into the minimal branch	Define $\omega_{LC}(e)$ and show that only its temporal Lorentz multiplier role appears canonically	CLOSED
ADV4B-09	The tetrad introduces extra gravitational degrees of freedom	Exhibit the local Lorentz moment map and symplectic reduction to the accepted ADM pair	CLOSED
ADV4B-10	Local Lorentz constraints are omitted or merged with internal Gauss laws	Enumerate J_{IJ} separately and compute its own bracket algebra	CLOSED
ADV4B-11	Fermions are treated as ordinary commuting canonical fields	Use a Z_2 -graded carrier and graded Poisson/Dirac brackets	CLOSED
ADV4B-12	The fermionic first-order kinetic term is declared canonical without second-class reduction	Compute every primary pair and the invertible Dirac matrix	CLOSED
ADV4B-13	Zero fermions mean that the full fermionic target is absent	Distinguish carrier occupancy from nonzero excitation and show the graded carrier and second-class block exist at the zero section	CLOSED
ADV4B-14	A right-handed neutrino or Majorana sector is silently imported	Freeze the minimal no_nu_R branch and route every neutrino-mass option to target update	CLOSED
ADV4B-15	Numerical couplings or Yukawa values are reported as derived	Keep them as fixed target parameters and preserve the SM/Higgs nonclaim ledgers	CLOSED
ADV4B-16	The Higgs ARC is used as if it supplied a potential or vev value	Import only H_{br}/fix and declare the doublet action/potential parameters independently	CLOSED

ID	Attack	Actual closure	Status
ADV4B-17	The displayed bosonic seed does not satisfy Gauss law	Choose $E=Pi_H=psi=0$ and verify all charge densities vanish before the gravitational correction	CLOSED
ADV4B-18	The $U(1)$ factor leaves a continuous stabilizer and destroys regularity	Choose a joint gauge/Higgs pair with trivial common infinitesimal stabilizer and prove the adjoint-kernel test	CLOSED
ADV4B-19	A nonzero connection is called irreducible without proof	Construct or import a small joint pair and verify its stabilizer directly	CLOSED
ADV4B-20	Classical covariance is substituted for the full constraint algebra	Calculate the complete Dirac-bracket table on one common reduced domain	CLOSED
ADV4B-21	Raw and improved momentum generators are conflated	Display both and trace every additional Gauss/Lorentz term	CLOSED
ADV4B-22	The Hamiltonian-Hamiltonian bracket hides internal-gauge terms	State the bracket in raw and improved conventions and prove equivalence	CLOSED
ADV4B-23	Local anomaly cancellation is reported as all-anomaly closure	Separate polynomial, Witten, global-form, bordism, and quantum-gravity rows	CLOSED
ADV4B-24	The Witten $SU(2)$ anomaly is ignored	Count four doublets per generation after color multiplicity and verify even parity	CLOSED
ADV4B-25	A dynamical-gravity or large-diffeomorphism anomaly theorem is imported from fixed-background SYM	Keep that row open for WDW-6 under the exact spin/global-form target	CLOSED
ADV4B-26	Flat-space CPT closure is used to validate the curved canonical theory	Use no CPT premise in W4B.1-W4B.12 and defer target-matched recovery	CLOSED
ADV4B-27	The occupancy argument is only perturbative rhetoric	Use the parametric implicit-function theorem with the accepted WDW-4A right inverse and prove an exact zero	CLOSED
ADV4B-28	The implicit-function theorem is illegally applied in Grassmann directions	Parameterize only even gauge/Higgs data and keep fermions at the invariant zero section	CLOSED
ADV4B-29	Matter columns destroy the WDW-4A right inverse	Use the gravity/scalar columns as a fixed surjective block and control the new terms parametrically	CLOSED
ADV4B-30	The internal Gauss and Lorentz blocks lack compatible right inverses	Construct the Gauss normal inverse from trivial adjoint kernel and the Lorentz algebraic split	CLOSED
ADV4B-31	No covariant history corresponds to the corrected initial data	Import and specialize local Einstein-Yang-Mills-Higgs-scalar development with zero fermions	CLOSED
ADV4B-32	The no-KID property is lost after adding matter	Use openness of the WDW-4A inverse and the joint-stabilizer argument	CLOSED
ADV4B-33	The scalar clock becomes a preferred foliation for the SM branch	Retain only the WDW-4A local quotient clock class and its failure boundary	CLOSED
ADV4B-34	Boundary charges are deleted by choosing a closed manifold	State that they are absent only in the frozen branch and are target updates elsewhere	CLOSED
ADV4B-35	The classical closure already determines the SM-coupled WDW operator	Keep functional calculus, regulator, ordering, domains, quantum anomalies, and inner product in WDW-5/6	CLOSED

ID	Attack	Actual closure	Status
ADV4B-36	The endpoint excludes BSM, torsionful, alternative-global-form, or nontrivial-topology theories	State the exact same-target scope and explicit richer-scope exits	CLOSED

48. Publication verification

The integrated edition preserves the complete WDW-4B theorem order, interface typing, acceptance conditions, adversarial checks, and scope boundary. Assembly does not amend a WDW-4B criterion or replace its final module verdict.

Part XXIII

Integrated WDW-4 capstone and classical handoff

49. The integrated certificate

Certificate 49.1 (Integrated WDW-4 regularity and occupancy). For a WDW-4A datum z_0 , an interface $\mathfrak{J}_{4A \rightarrow 4B}$, a sufficiently small nonzero even parameter δ , and a WDW-4B datum $w_\delta = \mathcal{L}_\delta(\mathfrak{J}_{4A \rightarrow 4B})$, the statement $W4\text{Reg}(z_0, w_\delta)$ is the conjunction of the eight rows below.

Clause	Coordinate	Typed witness
W4R-01	Occupied pure scalar/clock antecedent	ScalarReg3(z_0), Certificate 32.1 and Theorem W4.4
W4R-02	Exact typed 4A-to-4B interface	$\mathfrak{J}_{4A \rightarrow 4B}$, (1)
W4R-03	Exact total-constraint lift, without projection conflation	$\mathcal{L}_\delta$, (2), Theorem W4B.9
W4R-04	Occupied full Standard Model regularity certificate	SMReg4B(w_δ), Certificate 38.1 and Theorem W4B.10
W4R-05	Compatible joint tame split with common loss two	equations (116), (120) and (121)
W4R-06	Compatible local scalar-clock continuation and failure boundary	Theorem W4.6, lemma 37.1, and Theorem W4B.11
W4R-07	Target, topology, anomaly, singular, and quantum guards retained	Theorems W4B.1, W4B.8 and W4B.12 and equations (124) and (125)
W4R-08	Acyclic module dependency and separately checkable theorem objects	Theorem W4.1 and Theorem W4B.1 and <code>audit/integration_dependency.csv</code>

50. Integrated occupancy theorem

Theorem W4.I.1 (Integrated classical matter occupancy and joint-constraint closure). On the fixed compact, connected, boundaryless, oriented, spin branch, there exist:

1. a nonvacuum Einstein–Klein–Gordon datum z_0 occupying every clause of ScalarReg3;
2. its typed interface $\mathfrak{J}_{4A \rightarrow 4B}$;
3. a sufficiently small nonzero gauge–Higgs parameter δ ; and

4. an exact torsion-free tetrad/spin/gauge Standard Model datum $w_\delta = \mathcal{L}_\delta(\mathfrak{I}_{4A \rightarrow 4B})$

such that $\text{W4Reg}(z_0, w_\delta)$ holds.

The first datum is a zero of the pure gravity–scalar constraint map. The second is a zero of the full Standard-Model-coupled constraint map and occupies all twelve clauses of SMReg4B . Their relation is the exact parametric lift (2), not equality of their constraint maps or zero loci. The local scalar clock is future-directed, satisfies $V_\phi(\phi) = 1$, descends through the proper local quotient, and retains its failure boundary. The datum w_δ also determines the source-attached coupled covariant-history class $\mathcal{H}_{4B}^{\text{cov}}(w_\delta)$ defined below. The complete target-indexed anomaly ledger and all global-form, bundle, topology, singular-stratum, boundary, and quantum guards remain attached.

Proof. Theorems W4.3 and W4.4 construct z_0 , eliminate generalized KIDs, and populate ScalarReg3 ; Theorems W4.5, W4.6 and W4.8 supply its regular quotient, local clock, and scalar capstone. These results establish W4R-01 and the data listed in (1).

Theorems W4B.1 to W4B.3 freeze the downstream target and construct its coframe, spin, global-form, and representation carriers. Theorems W4B.4 to W4B.7 construct the covariant action, graded canonical carrier, complete constraint system, and exact common-domain algebra. Theorem W4B.9 applies the accepted 4A split to the even parametric equation (2) and produces the exact full constraint zero w_δ with nonzero gauge curvature and nonconstant Higgs data. It never asserts that the corrected gravity–scalar projection separately solves the pure 4A equations.

Theorem W4B.10 proves $\text{SMReg4B}(w_\delta)$, including the full upper-triangular inverse and the local proper-redescription quotient. Theorems W4B.8, W4B.11 and W4B.12 supply the bounded anomaly theorem, clock persistence, guards, source-attached handoff, and Standard Model capstone. These conclusions establish W4R-02–W4R-07. The no-backflow theorems Theorem W4.1 and Theorem W4B.1 and the displayed integration dependency establish W4R-08. Hence every clause of W4Reg is occupied. $\square$

51. The unified classical WDW-5 input

The source coordinate imported from W4B.9 and W4B.11 is the pointed development class

$$\mathcal{H}_{4B}^{\text{cov}}(w_\delta) := [\text{Dev}_{\text{EYMH}\phi}(w_\delta; \Psi_\delta = 0)]_{\cong_{\iota_\delta}}. \quad (123)$$

Here $\iota_\delta : \Sigma \hookrightarrow \mathcal{M}_\delta$ is the distinguished initial embedding, and $\cong_{\iota_\delta}$ is initial-surface-fixing isometry together with the proper Lorentz and internal gauge redescriptions of the frozen local branch. The class is nonempty by Theorem W4B.9; it is a covariant-history coordinate, not the canonical datum w_δ or the lifting map $\mathcal{L}_\delta$.

The fixed classical target guard is the finite ordered status ledger

$$\begin{aligned} \mathfrak{G}_{\text{target}} := & ((\Sigma, \text{compact, connected, } \partial\Sigma = \emptyset, \text{oriented, } \mathfrak{s}), \\ & (\text{second-order metric GR, } e, \omega_{\text{LC}}(e), \text{torsion} = 0, \text{nonmetricity} = 0), \\ & (\text{accepted WDW-4A scalar/clock subsystem, } \mathcal{U}_{\text{clk}}, \mathfrak{F}_{\text{clk}}), \\ & (G_{\text{SM}}^{(6)}, P_0 \simeq \Sigma \times G_{\text{SM}}^{(6)}, \text{no line insertion}), \\ & (N_{\text{gen}} = 3, \mathcal{R}_{\text{SM}}, \nu_R \text{ absent, } H \in (\mathbf{1}, \mathbf{2})_{1/2}), \\ & (g_3, g_2, g_1, Y_u, Y_d, Y_e \text{ fixed; } V_H = \lambda(H^\dagger H - v^2/2)^2 \text{ fixed}), \\ & (\theta \text{ registered at the frozen local value}), \\ & (\text{proper spatial/Lorentz/internal redescriptions, singular stabilizers retained}), \\ & (\text{Sobolev smoothness above multiplication/elliptic thresholds}), \\ & (\mathbb{Z}_2\text{-graded Dirac-Bergmann carrier, bosonic-body analytic regularity}), \\ & (\text{compact boundaryless sector, nontrivial bundles/topologies/boundaries excluded})). \end{aligned} \quad (124)$$

This is a grouped display of all eighteen nonquantum declared and inherited coordinates in `audit/4b/`

`target_freeze.csv`; it does not derive a target choice from local field data. The nineteenth row of that ledger is the separate quantum-scope boundary defined next.

The still-open quantum guard is the finite status ledger

$$\begin{aligned} \mathfrak{G}_{\text{quantum}} := & ((\text{functional differential calculus, OPEN-WDW-5}), \\ & (\text{regularized constraints/common domain, OPEN-WDW-5}), \\ & (\text{ordering/renormalization, OPEN-WDW-5}), \\ & (\text{quantum constraint anomaly, OPEN-WDW-6}), \\ & (\text{determinant line/bordism/large diffeomorphisms, OPEN-WDW-6}), \\ & (\text{state space/inner product/rigging map, OPEN-WDW-5}), \\ & (\text{quantum observables/clock relation, OPEN-WDW-5}), \\ & (\text{boundary/inflow data, INACTIVE WITH TARGET GUARD})). \end{aligned} \quad (125)$$

These are absence-and-continuation statuses, not quantum objects supplied by WDW-4.

Let $\mathfrak{A}_{4B}^{\text{con}}$ denote the exact first-class Dirac-bracket object on the Lorentz, internal Gauss, improved-spatial, and normal constraint family, with every structure function and weak restriction fixed by Theorem W4B.7.

Define the exact seven-coordinate WDW-4B package by

$$\begin{aligned} \mathfrak{H}_{\text{W4B}}^{\text{cl}}(w_\delta) := & ((\mathcal{P}_{4B}^{\text{red}}, \{\cdot, \cdot\}_D), \\ & (\Phi_{4B}, \mathfrak{A}_{4B}^{\text{con}}), \\ & \mathcal{H}_{4B}^{\text{cov}}(w_\delta), \\ & (\mathcal{C}_{4B}^{\text{reg}}/\mathcal{R}_0, \text{all singular guards}), \\ & \mathfrak{G}_{\text{target}}, \mathcal{A}_{\text{status}}, \\ & (\mathcal{U}_{\text{clk}}, \mathfrak{F}_{\text{clk}}, \mathfrak{G}_{\text{quantum}})). \end{aligned} \quad (126)$$

The line breaks group, in order, precisely the seven coordinates of Theorem W4B.11. The quantum ledger is attached to the seventh coordinate only as an explicit boundary.

The complete integrated classical handoff is

$$\mathfrak{H}_{\text{W4}}^{\text{cl}} = (\mathfrak{H}_{\text{W4A}}, \mathfrak{H}_{\text{W4B}}^{\text{cl}}(w_\delta), \mathfrak{I}_{4A \rightarrow 4B}, \mathcal{L}_\delta). \quad (127)$$

Here $\mathfrak{H}_{\text{W4A}}$ is the exact 4A scalar package of equation (64); the named middle coordinate is the complete WDW-4B package (126); and the last two coordinates retain the typed 4A-to-4B interface and its exact lift. Thus the integrated tuple contains the coupled history (123) and both fully enumerated guards (124)–(125).

WDW-5 may use (127) as its classical input. It must independently construct any functional differential calculus, regularized constraint operator, common operator domain, ordering, renormalization prescription, quantum anomaly theorem, state space, physical inner product, rigging map, or quantum observable algebra.

52. Integrated nonclaims

The integrated result does not:

1. identify the pure WDW-4A and full WDW-4B constraint maps;
2. assert that the gravity–scalar projection of w_δ is a separate WDW-4A constraint zero;
3. derive the Standard Model action or its numerical parameters from AASC;
4. derive the $\mathbb{Z}_6$ global form from local representation data;

5. classify nontrivial bundles, monopoles, line insertions, or boundary charge sectors;
6. introduce an independent spin connection, torsion, nonmetricity, or an Einstein–Cartan interaction;
7. add right-handed neutrinos, Majorana masses, sterile sectors, or beyond-Standard-Model fields;
8. claim a nonzero classical fermion at the explicit occupancy witness;
9. infer determinant-line, bordism, large-diffeomorphism, or dynamical-gravity anomaly freedom from the displayed finite sums;
10. promote the scalar clock to a global or preferred time;
11. construct a Wheeler–DeWitt operator or any other quantum object; or
12. use integration to alter an accepted module hypothesis, conclusion, failure boundary, or proof dependency.

53. Integration acceptance conditions

The following rows audit only the act of integration. They supplement, and do not replace, the 40 WDW-4A and 68 WDW-4B acceptance rows already printed in their respective modules.

ID	Acceptance condition	Controlling surface	Status
ACI-01	Final WDW-4A theorem set retained	Integrated overview	PASS
ACI-02	Final WDW-4B theorem set retained	Integrated overview	PASS
ACI-03	WDW-4A interface and scope boundary retained	Module 4A	PASS
ACI-04	WDW-4B interface and scope boundary retained	Module 4B	PASS
ACI-05	W4.1–W4.8 labels and order preserved	Module 4A	PASS
ACI-06	W4B.1–W4B.12 labels and order preserved	Module 4B	PASS
ACI-07	ScalarReg3 remains scoped to the pure map	Certificate 32.1	PASS
ACI-08	SMReg4B remains scoped to the full map	Certificate 38.1	PASS
ACI-09	Interface typed with the existing 4A branch guard	(1)	PASS
ACI-10	Parametric lift explicitly typed	(2)	PASS
ACI-11	Corrected projection not called a pure 4A zero	Theorem W4.I.1	PASS
ACI-12	Nonzero gauge–Higgs occupant retained	Theorem W4B.9	PASS
ACI-13	Clock sign and orientation remain correct	Theorem W4.6 and Theorem W4B.11	PASS
ACI-14	Exact off-shell bracket retained	Theorem W4B.7	PASS
ACI-15	Nonzero momentum column retained	(114)	PASS
ACI-16	Upper-triangular inverse remains exact	(116)	PASS
ACI-17	Common coefficient loss two remains explicit	(121)	PASS
ACI-18	Anomaly boundaries remain unchanged	Theorem W4B.8	PASS
ACI-19	AASC/conventional proof labor remains separated	Integrated overview	PASS
ACI-20	Module no-backflow theorems remain intact	Theorem W4.1 and Theorem W4B.1	PASS
ACI-21	Unified handoff includes the coupled history and typed guards	equations (123) to (125) and (127)	PASS
ACI-22	Quantum and global nonclaims remain explicit	Integrated nonclaims	PASS
ACI-23	Bibliography keys are unique and resolved	Final compilation	PASS
ACI-24	Publication edition passes presentation QA	Publication verification	PASS

54. Integrated adversarial map

The registered integration-level attacks test theorem-import fidelity, dependency regressions, notation alignment, interface typing, and publication consistency. Any mathematical objection to a module theorem must still be traced through that module’s full adversarial map.

ID	Attack	Exact closure	Status
ADVI-01	Integration changes an accepted theorem	Original sources and bounded diff preserve theorem content	CLOSED
ADVI-02	The constraint maps are identified	Interface and W4.I.1 explicitly deny identification	CLOSED
ADVI-03	The corrected projection is called a pure 4A zero	(2) retains the gauge–Higgs source	CLOSED
ADVI-04	WDW-4B repairs WDW-4A	Theorem W4.1 and Theorem W4B.1 preserve one-way dependency	CLOSED
ADVI-05	WDW-4A is treated as already Standard Model complete	The module boundary and interface retain the richer-carrier burden	CLOSED
ADVI-06	The clock-sign defect reappears	Both clock vectors satisfy $V_\phi(\phi) = 1$	CLOSED
ADVI-07	The WDW-4B bracket defect reappears	Exact off-shell closure retains $\mathcal{D} + \mathcal{G} + \mathcal{J}$	CLOSED
ADVI-08	The false diagonalization reappears	B_δ and the triangular inverse remain explicit	CLOSED
ADVI-09	ScalarReg3 is assumed at the corrected projection	W4Reg distinguishes z_0 from w_δ	CLOSED
ADVI-10	AASC generates the positive SM construction	Proof-labor discipline assigns it to conventional mathematics	CLOSED
ADVI-11	The $\mathbb{Z}_6$ form is locally derived	Declared-target status and nonclaim retained	CLOSED
ADVI-12	Local anomaly cancellation becomes quantum anomaly freedom	Anomaly ledger retains every open level	CLOSED
ADVI-13	The scalar clock becomes global or preferred	Failure ledger and integrated nonclaim remain explicit	CLOSED
ADVI-14	Zero fermions erase the graded carrier	Full second-class matrix and reduced bracket remain present	CLOSED
ADVI-15	Duplicate BibTeX keys corrupt references	WDW-4B duplicates are namespaced and resolved	CLOSED
ADVI-16	The integrated edition is only a presentation wrapper	The unified theorem chain and complete typed interfaces are explicit	CLOSED
ADVI-17	The interface contains an unbound renamed guard	$\mathfrak{G}_{\text{branch}}$ is imported without alias from (64)	CLOSED
ADVI-18	The unified handoff drops a history or leaves guards unbound	equations (123) to (126) define every coordinate before assembly	CLOSED

55. Conclusion

WDW-4 is now one publication manuscript with two separately checkable classical constructions. Module 4A closes the scalar occupancy and local clock antecedent. Module 4B uses that accepted analytic interface to construct and occupy the larger torsion-free Standard Model constraint system. The integrated capstone records the exact lift between them, preserves the nonidentity of their constraint maps, and hands WDW-5 a single source-attached classical package with its coupled history and every branch, target, failure, and open-quantum coordinate visible.

The endpoint is therefore neither a juxtaposition of unrelated papers nor a collapse of two theorem objects into one undifferentiated claim. It is an acyclic, interface-complete WDW-4 classical matter closure.

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